Quantum Computing Syllabus Breakdown

i want to prepare my semister subjects with the help of you so what ever i ask in this chat you have to answer with me in a semister point of view so you have to give in detail and comprehsive knowledge about the concept. ok this is the syallbus at first breakdown the concepts first unite wise subject name " GENERAQUANTUM COMPUTING FOR ENGINEERS " "UNIT-I 9 Lectures
FUNDAMENTALS OF QUANTUM COMPUTING : Classical vs Quantum Computing, Basic Concepts: Qubits, Bloch sphere, measurement, Quantum Mechanics Basics: Dirac notation, entanglement, Quantum Hardware Overview, Quantum Programming Platforms.
Learning Outcomes:
1. Understand the differences between classical and quantum computing, including core concepts such as qubits, superposition, and measurement.
2. Apply basic quantum mechanics principles (Dirac notation, entanglement) and identify major quantum hardware and programming platforms.
UNIT – II 9 Lectures
QUANTUM GATES AND CIRCUITS: Quantum Logic Gates: Pauli, Hadamard, Phase, Rotation, Multi-Qubit Gates: CNOT, Toffoli, SWAP, Quantum Circuits and Reversible Computation, Quantum Phenomena: Interference, Parallelism, Noise and Error Basics
Learning Outcomes:
1. Analyze and implement fundamental quantum gates (Pauli, Hadamard, Phase, Rotation) and multi-qubit gates like CNOT and Toffoli.
2. Design simple quantum circuits and explain key quantum phenomena such as interference, parallelism, and the impact of noise.
44
UNIT –III 9 Lectures
QUANTUM SEARCH ALGORITHMS: Unstructured Search Problem, Grover's Algorithm: Oracle, amplitude amplification, Applications: Database search, optimization, Variants: Quantum counting, multiple target search.
Learning Outcomes :
1. Explain the unstructured search problem and the working principles of Grover's algorithm, including oracle and amplitude amplification.
2. Evaluate applications and variants of quantum search algorithms such as quantum counting and multiple-target search.
UNIT-IV 9 Lectures
QUANTUM ALGORITHMS FOR IMAGE PROCESSING: Quantum Image Representation: Encoding pixels, Models: FRQI, NEQR, Techniques: Edge detection.
Learning Outcomes :
1. Describe quantum image representation models (FRQI, NEQR) and methods for encoding image data into quantum states.
2. Apply quantum techniques such as edge detection to basic image processing problems.
UNIT-V 9 Lectures
APPLICATION-ORIENTED QUANTUM ALGORITHMS: Quantum Optimization: VQE, QAOA, Applications: Logistics, scheduling, Quantum Machine Learning: Classifiers, neural networks.
Learning Outcomes :
1. Analyze quantum optimization algorithms (VQE, QAOA) and their applications in realworld problems like logistics and scheduling.
2. Explain foundational concepts of quantum machine learning, including classifiers and

quantum neural networks.
TEXT BOOKS:
1. Thomas G. Wong, Introduction to Classical and Quantum Computing, 1st Edition, Rooted Grove, 2022.(Unit – I,II,III)
2. Jack D. Hidary, Quantum Computing: An Applied Approach, 2nd Edition, Springer, 2021.(Unit-IV,V)"
Show moreShow less

Perfect. From now on in this chat, I'll teach your semester subject "Quantum Computing for Engineers" in an exam-oriented + concept-oriented + engineering-oriented way.
I will give:

- Unit-wise breakdown of all concepts
- Detailed explanations from basics to advanced
- Derivations and formulas where needed
- Circuit diagrams / examples when useful
- Important university exam points
- Short notes, 2-mark, 5-mark, and 10-mark preparation
- Connections between topics for better understanding

Subject Roadmap: Quantum Computing for Engineers

UNIT I
Fundamentals of Quantum Computing
9 lectures • Foundation for all other units
UNIT II
Quantum Gates and Circuits
9 lectures • The operational heart of quantum computing
UNIT III
Quantum Search Algorithms
9 lectures • Grover's algorithm and amplitude amplification
UNIT IV
Quantum Algorithms for Image Processing
9 lectures • FRQI, NEQR, and edge detection
UNIT V
Application-Oriented Quantum Algorithms
9 lectures • VQE, QAOA, logistics, scheduling, QML

UNIT–I: Fundamentals of Quantum Computing (Most Important Foundation)

This unit is theory-heavy and very important for 10-mark questions.

1. Classical vs Quantum Computing

Concepts to study

Classical Computing

- Bit (0 or 1)
- Deterministic computation
- Boolean logic
- Sequential/parallel processing

Quantum Computing

- Qubit
- Superposition
- Entanglement
- Probabilistic measurement
- Quantum parallelism

Exam focus

- Difference between bit and qubit
- Why quantum computing is powerful
- Advantages and limitations

Expected 10-mark question

“Compare classical and quantum computing with suitable examples.”

2. Qubits

Subtopics

- Definition of qubit
- Basis states: $|0\rangle$ and $|1\rangle$
- General qubit state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where

$$|\alpha|^2 + |\beta|^2 = 1$$

Must understand

- Probability amplitudes
- Difference between amplitude and probability
- Measurement outcomes

Numerical possibility

If

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

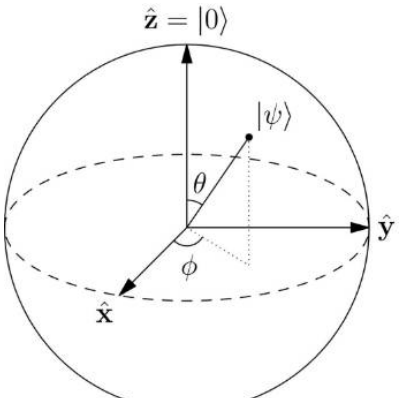
Then

- P(0) = 1/2
- P(1) = 1/2

3. Bloch Sphere

Subtopics

- 3D representation of a qubit
- X, Y, Z axes
- North pole = $|0\rangle$
- South pole = $|1\rangle$
- Superposition states on the surface



Exam focus

- Draw Bloch sphere
- Explain how quantum gates rotate qubits

4. Quantum Measurement

Subtopics

- Measurement postulate
- State collapse
- Probabilities from amplitudes
- Repeated measurements

Key idea

Before measurement:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After measuring 0:

$$|\psi\rangle \rightarrow |0\rangle$$

Frequently asked

“Explain quantum measurement and state collapse.”

Quantum Mechanics Basics

This is the mathematical language of quantum computing.

5. Dirac Notation (Bra–Ket Notation)

Subtopics

Ket

$$|\psi\rangle$$

Bra

$$\langle\psi|$$

Inner product

$$\langle\phi|\psi\rangle$$

Outer product

$$|\psi\rangle\langle\phi|$$

Why important?

Used in:

- Gates
- Measurements
- Entanglement
- Algorithms

Exam tip

Practice converting between:

- Vector form
- Dirac notation

Example:

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$$

6. Entanglement

Core concept

Two qubits become correlated in a way impossible classically.

Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Properties

- Measurement of one affects the other
- Cannot be separated into individual qubits
- Resource for quantum communication and computing

10-mark question

“Explain quantum entanglement with Bell states and applications.”

7. Quantum Hardware Overview

Engineering-oriented topic.

Hardware types

Superconducting Qubits
IBM, Google

Trapped Ions
IonQ

Photonic Qubits
Xanadu

Spin / Quantum Dots
Research platforms

For each hardware know

- Physical realization
- Advantages
- Challenges

Short note

“Superconducting qubits vs trapped-ion qubits.”

8. Quantum Programming Platforms

Important platforms

Qiskit
IBM
Cirq
Google
PennyLane
QML
Q#
Microsoft

What to know

- Creating qubits
- Applying gates
- Running circuits
- Simulation vs real hardware

Very important for viva

“Which platform is most widely used for learning quantum computing?” → Qiskit

UNIT–I Concept Flow (How topics are connected)

Classical vs Quantum
↓
Qubits
↓
Superposition
↓
Bloch Sphere
↓
Measurement
↓
Dirac Notation
↓

Entanglement

↓

Hardware & Platforms

UNIT–II: Quantum Gates and Circuits

This unit is numerical + circuit-oriented.

1. Single-Qubit Gates

Pauli Gates

Gate	Function
X	Bit flip
Y	Bit + phase flip
Z	Phase flip

Hadamard Gate

Creates superposition.

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Phase Gates

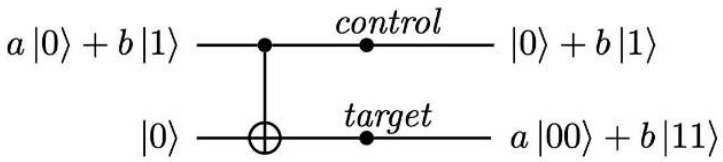
- S gate
- T gate

Rotation Gates

- Rx(θ)
- Ry(θ)
- Rz(θ)

2. Multi-Qubit Gates

CNOT



Control	Target	Result
$\|0\rangle$	$\|0\rangle$	$\|00\rangle$
$\|0\rangle$	$\|1\rangle$	$\|01\rangle$
$\|1\rangle$	$\|0\rangle$	$\|11\rangle$

Toffoli (CCNOT)

Used in reversible computation.

SWAP

Exchanges qubit states.

3. Quantum Circuits

Concepts

- Circuit model
- Wires and gates
- Measurement
- Reversibility

4. Reversible Computation

Important theory question.

- Classical irreversible gates lose information.
- Quantum gates must be unitary and reversible.

5. Quantum Phenomena

Interference

Constructive and destructive.

Quantum Parallelism

Evaluate many states simultaneously.

Noise and Errors

- Bit-flip error
- Phase-flip error
- Decoherence

UNIT–III: Quantum Search Algorithms

This unit is centered on Grover's Algorithm.

1. Unstructured Search

Classical complexity:

$$O(N)$$

Quantum complexity:

$$O(\sqrt{N})$$

2. Grover's Algorithm

Steps

Initialize

↓

Create Superposition

↓

Apply Oracle

↓

Amplitude Amplification

↓

Measure

3. Oracle

Marks the target state by phase inversion.

4. Amplitude Amplification

Increases probability of the correct answer.

5. Applications

- Database search
- Optimization
- Cryptanalysis

6. Variants

- Quantum counting

- Multiple target search

UNIT–IV: Quantum Algorithms for Image Processing

Mostly theory + architecture.

1. Quantum Image Representation

FRQI

Flexible Representation of Quantum Images.

NEQR

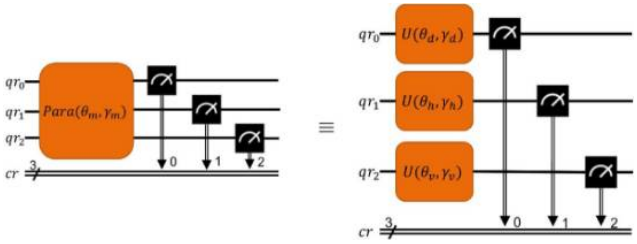
Novel Enhanced Quantum Representation.

2. Pixel Encoding

- Position qubits
- Color/intensity qubits

3. Edge Detection

Quantum version of detecting boundaries.



Exam focus

- FRQI vs NEQR comparison
- Advantages of quantum image processing

UNIT–V: Application-Oriented Quantum Algorithms

Modern and application-based unit.

1. VQE (Variational Quantum Eigensolver)

Components

- Parameterized circuit
- Measurement
- Classical optimizer

Used for chemistry and optimization.

2. QAOA

Quantum Approximate Optimization Algorithm.

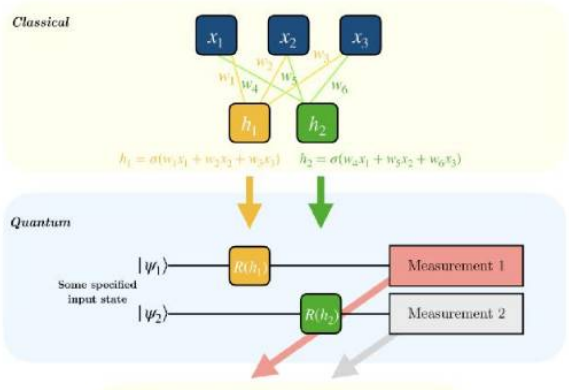
Applications

- Scheduling
- Routing
- Logistics
- Resource allocation

3. Quantum Machine Learning

Topics

- Quantum classifiers
- Quantum kernels
- Quantum neural networks



Important note

Current QML is hybrid quantum-classical.

Most Important Topics for University Exams

VERY IMPORTANT
Qubits, superposition, measurement
UNIT I • Frequently asked 10-mark
VERY IMPORTANT
Bloch sphere and Dirac notation
UNIT I • Theory + diagram
VERY IMPORTANT
Entanglement and Bell states
UNIT I • High-weightage concept
VERY IMPORTANT
Hadamard, Pauli, CNOT gates
UNIT II • Numerical + circuit
VERY IMPORTANT
Reversible computation
UNIT II • Theory question
VERY IMPORTANT

Grover's algorithm with circuit and complexity
UNIT III • Core algorithm
IMPORTANT
FRQI vs NEQR
UNIT IV • Comparison question
IMPORTANT
VQE and QAOA
UNIT V • Application oriented
IMPORTANT
Quantum machine learning basics
UNIT V • Short/medium answer

How I Suggest We Study

Step 1
UNIT I complete mastery
Foundation for every other unit
Step 2
UNIT II with gate practice
Learn circuits and calculations
Step 3
UNIT III Grover thoroughly
Most important algorithm
Step 4
UNIT IV theory and comparisons
Easy scoring unit
Step 5
UNIT V applications and modern concepts
Viva and future-oriented understanding

Your Next Step

We should start UNIT–I from the first concept.
Type:

- "Start Unit 1 – Classical vs Quantum Computing" → full detailed lecture
- "Teach me Qubits in detail"
- "Explain Bloch sphere for exams"
- "Explain Dirac notation from basics"
- "Explain Entanglement with examples"

I'll teach each topic as if you are preparing for a B.Tech semester examination and viva, with detailed notes, diagrams, formulas, examples, and important questions.

> ok next

Great! We'll start **UNIT–I** in detail.

UNIT–I: Fundamentals of Quantum Computing

Topic 1: Classical vs Quantum Computing

This is the **first and most fundamental topic** of quantum computing. Understanding it well will make the remaining topics much easier.

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1. Introduction

For decades, computers have been based on **classical computing**, where information is processed using **bits** (0 or 1). As technology advanced, scientists realized that at very small (atomic and subatomic) scales, nature follows the laws of **quantum mechanics** rather than classical physics.

Quantum computing uses these quantum mechanical principles to perform computations in ways that are impossible for classical computers.

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2. What is Classical Computing?

A **classical computer** is an electronic machine that processes information using **binary digits (bits)**.

A bit can have only one value at a time:

- 0 (OFF)
- 1 (ON)

Examples of classical computers include:

- Desktop computers
- Laptops
- Smartphones
- Supercomputers
- Servers

Working Principle

A classical computer performs calculations using:

- Logic gates (AND, OR, NOT, XOR)
- Transistors
- CPU
- Memory
- Algorithms

It follows the rules of **Boolean Algebra**.

---

3. What is Quantum Computing?

A **Quantum Computer** is a computing system that uses **quantum bits (qubits)** instead of classical bits.

Unlike a classical bit, a qubit can exist in:

- State $|0\rangle$
- State $|1\rangle$
- A combination of both states simultaneously (called **superposition**)

Quantum computers exploit quantum mechanical properties such as:

- Superposition
- Entanglement
- Interference

These properties allow them to solve certain problems much faster than classical computers.

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4. Why Do We Need Quantum Computing?

Some computational problems become extremely difficult as the input size grows.

Examples include:

- Breaking modern encryption
- Molecular simulation
- Drug discovery
- Weather prediction
- Financial optimization
- Artificial Intelligence
- Logistics optimization

A classical computer may require **millions of years** for some of these problems, while a sufficiently powerful quantum computer could solve them much faster.

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5. Classical Bit vs Quantum Bit (Qubit)

Classical Bit

A classical bit stores exactly one value.

```
Bit = 0

or

Bit = 1
```

Only one state is possible at a time.

Quantum Bit (Qubit)

A qubit can be:

```
|0⟩
|1⟩
or
α|0⟩ + β|1⟩
```

where:

- α and β are probability amplitudes.
- Their squared magnitudes determine the probabilities of measuring 0 or 1.
- They satisfy:

$$|\alpha|^2 + |\beta|^2 = 1$$

This simultaneous existence is called **superposition**.

---

6. Working Example

Imagine a light switch.

Classical Computer

The switch is either:

```
OFF
or
ON
```

It cannot be both simultaneously.

Quantum Computer

Before measurement, the qubit behaves like it is in a combination of:

```
OFF
and
ON
```

After measurement, it collapses to either OFF or ON.

Important Note: This does **not** mean a qubit literally stores two definite values at once. It exists in a quantum state that allows both outcomes with certain probabilities until measured.

7. Comparison Table

Feature	Classical Computing	Quantum Computing
Basic Unit	Bit	Qubit
Values	0 or 1	0, 1, or superposition
Physics	Classical Physics	Quantum Mechanics
Processing	Sequential/Parallel	Quantum Parallelism
Logic Gates	Boolean Gates	Quantum Gates
Copying Data	Possible	Cannot perfectly copy an unknown quantum state (No-Cloning Theorem)
Speed	Efficient for general tasks	Can be much faster for certain specialized problems
Measurement	Does not fundamentally change stored value	Measurement changes (collapses) the quantum state
Error Sensitivity	Relatively low	High (affected by noise and decoherence)

8. Advantages of Classical Computing

- Mature and reliable technology
- Easy to program
- Inexpensive hardware
- Suitable for everyday applications
- Well-developed software ecosystem

9. Limitations of Classical Computing

- Some optimization problems take an impractical amount of time.
- Difficult to accurately simulate quantum systems.
- Limited scalability for certain NP-hard problems.
- Some cryptographic tasks remain computationally expensive.

10. Advantages of Quantum Computing

- Faster for certain classes of problems.
- Excellent for optimization.

- Useful in molecular and chemical simulations.
- Can accelerate specific search algorithms (e.g., Grover's algorithm).
- Forms the basis for emerging quantum machine learning techniques.

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11. Limitations of Quantum Computing

- Hardware is still in early development.
- Very expensive to build and maintain.
- Requires specialized environments (often extremely low temperatures for some technologies).
- Highly sensitive to environmental noise.
- Limited number of stable qubits in current devices.
- Programming is more complex than classical computing.

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12. Real-Life Applications of Quantum Computing

Healthcare

- Drug discovery
- Protein folding
- Personalized medicine

Finance

- Portfolio optimization
- Risk analysis
- Fraud detection

Artificial Intelligence

- Quantum machine learning
- Pattern recognition
- Faster optimization

Cybersecurity

- Quantum cryptography
- Post-quantum cryptography research

Logistics

- Route optimization
- Traffic management

- Supply chain optimization

Scientific Research

- Material science
- Climate modeling
- Chemical simulations

---

13. Classical vs Quantum Example

Suppose we need to search for one specific record among **1,000,000** records.

Classical Search

A linear search may need to check nearly all records in the worst case.

Time complexity:

$$O(N)$$

Quantum Search

Using **Grover's Algorithm**, the search can be performed in approximately:

$$O(\sqrt{N})$$

For 1,000,000 records:

- Classical: up to **1,000,000** checks.
- Quantum: about **1,000** iterations (approximately √1,000,000).

This demonstrates a significant speedup for this specific problem.

---

14. Important Definitions

Bit

A bit is the smallest unit of information in a classical computer, having a value of either **0** or **1**.

Qubit

A qubit is the fundamental unit of information in a quantum computer. It can exist in the state **|0⟩**, **|1⟩**, or a **superposition** of both.

Quantum Computing

Quantum computing is a computing paradigm that uses the principles of quantum mechanics, such as **superposition**, **entanglement**, and **interference**, to process information.

15. Frequently Asked University Exam Questions

2-Mark Questions

1. Define a bit.
2. Define a qubit.
3. What is quantum computing?
4. State two differences between classical and quantum computing.
5. What is a superposition state?

5-Mark Questions

1. Explain the need for quantum computing.
2. Compare classical computing and quantum computing.
3. Explain the advantages and limitations of quantum computing.

10-Mark Questions

1. Explain the differences between classical and quantum computing with a neat comparison table.
2. Discuss the advantages, limitations, and applications of quantum computing.
3. Explain why quantum computing is considered the future of computation.

Key Takeaways

- Classical computers use **bits**, while quantum computers use **qubits**.
- A classical bit has only one value (**0 or 1**), whereas a qubit can exist in a **superposition** of states.
- Quantum computing is based on **quantum mechanics**, while classical computing relies on **classical physics**.
- Quantum computers are **not replacements for classical computers**; they are designed to solve certain specialized problems more efficiently.
- The main principles behind quantum computing are **superposition**, **entanglement**, and **interference**.

Next Topic (Topic 2)

The next topic is **Qubits**, where we'll cover:

1. What is a qubit?
2. Mathematical representation of a qubit
3. Probability amplitudes

4. Normalization condition
5. Superposition in detail
6. Basis states
7. Multiple qubits
8. Measurement of a qubit
9. Numerical examples
10. University exam questions and important derivations

This is one of the **most important topics** in Unit I and forms the mathematical foundation of quantum computing.

> ok next

UNIT–I: Topic 2 – Qubits (Quantum Bits)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> This is the **heart of quantum computing**. Almost every concept—quantum gates, circuits, algorithms (Grover, Shor), entanglement, and quantum machine learning—starts with understanding **qubits**.

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1. Introduction to Qubits

A **Qubit (Quantum Bit)** is the **basic unit of information** in a quantum computer, just as a **bit** is the basic unit in a classical computer.

- Classical Computer → **Bit**
- Quantum Computer → **Qubit**

A classical bit can only be **0** or **1**.

A qubit can be:

- **$|0\rangle$**
- **$|1\rangle$**
- **A superposition of $|0\rangle$ and $|1\rangle$**

This ability to exist in a superposition is what makes quantum computing fundamentally different from classical computing.

---

2. Classical Bit vs Qubit

Classical Bit	Qubit
Stores either 0 or 1	Can exist in a superposition of 0 and 1

Classical Bit	Qubit
Deterministic	Probabilistic until measured
Based on classical physics	Based on quantum mechanics
Represented using voltage levels	Represented using quantum states

3. Representation of a Qubit

In quantum computing, we use **Dirac (Bra-Ket) Notation**.

The two fundamental states are:

$$|0\rangle$$

and

$$|1\rangle$$

These are called **Computational Basis States**.

They are represented as vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These vectors form the basis of all quantum computations.

4. General Form of a Qubit

A qubit is generally written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where:

- **|ψ⟩** → Quantum state
- **α (alpha)** → Probability amplitude for $|0\rangle$
- **β (beta)** → Probability amplitude for $|1\rangle$

The values **α** and **β** can be **real or complex numbers**.

5. What are Probability Amplitudes?

One of the biggest differences between classical and quantum computing is the concept of **probability amplitudes**.

The amplitudes themselves are **not probabilities**.

The probabilities are obtained by squaring their magnitudes.

If

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

then

Probability of measuring **0**

$$P(0) = |\alpha|^2$$

Probability of measuring **1**

$$P(1) = |\beta|^2$$

Example

Suppose

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Then

$$P(0) = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

$$P(1) = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

So there is:

- 50% chance of measuring 0
- 50% chance of measuring 1

---

6. Normalization Condition

Every valid qubit must satisfy:

$$|\alpha|^2 + |\beta|^2 = 1$$

This is called the **Normalization Condition**.

It ensures that the total probability is always **100%**.

Example 1

$$|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$$

Check:

$$\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2$$
$$= \frac{9}{25} + \frac{16}{25}$$
$$= \frac{25}{25} = 1$$

Hence, it is a **valid qubit**.

Example 2

Suppose

$$|\psi\rangle = 2|0\rangle + 3|1\rangle$$

Check:

$$2^2 + 3^2 = 4 + 9 = 13$$

Since

$$13 \neq 1$$

this is **not a valid quantum state**.

---

7. Superposition

Superposition is one of the most important principles of quantum mechanics.

A qubit can exist in a combination of:

- $|0\rangle$
- $|1\rangle$

at the same time until it is measured.

Example:

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

This means the qubit has equal probability of collapsing to:

- $|0\rangle$
- $|1\rangle$

during measurement.

> **Important:** Superposition does **not** mean the qubit contains both classical values simultaneously. It means the quantum state allows both outcomes with certain probabilities until measurement.

---

8. Measurement of a Qubit

Measurement converts a quantum state into a classical result.

Suppose

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

When measured,

Possible outcomes:

- 0 (50%)
- 1 (50%)

If the outcome is **0**, then the state becomes

$$|0\rangle$$

If the outcome is **1**, then the state becomes

$$|1\rangle$$

This is known as **State Collapse**.

---

9. State Collapse

Before measurement:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After measuring:

- Result = 0

The state becomes

$$|0\rangle$$

OR

- Result = 1

The state becomes

$$|1\rangle$$

The original superposition no longer exists after measurement.

---

10. Visual Understanding

Imagine flipping a coin.

Classical Bit

The coin is placed on the table.

It is either:

- Heads
- Tails

Qubit

Imagine a spinning coin.

While spinning,

it is described by a quantum state with probabilities for heads and tails.

Once you stop and observe it,

you get only one result.

This analogy is useful for intuition, but remember that a qubit is **not literally a spinning coin**—it follows the laws of quantum mechanics.

---

11. Multiple Qubits

If one qubit has **2 possible basis states**, then:

Number of Qubits	Basis States
1	2
2	4
3	8
4	16
n	2^n

For example:

Two qubits:

$$|00\rangle$$

$$|01\rangle$$

$$|10\rangle$$

$$|11\rangle$$

A general two-qubit state is:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

with

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$$

---

12. Why Are Qubits Powerful?

As the number of qubits increases, the number of possible basis states grows **exponentially**.

Qubits	Basis States
1	2
2	4
5	32
10	1,024
20	1,048,576
30	1,073,741,824

This exponential state space is one reason quantum algorithms can outperform classical algorithms for certain problems.

---

13. Applications of Qubits

Qubits are used in:

- Quantum cryptography
- Quantum machine learning
- Quantum image processing
- Drug discovery
- Weather prediction
- Financial optimization
- Artificial intelligence
- Search algorithms (Grover's Algorithm)
- Quantum simulations

---

14. Advantages of Qubits

- Can represent superposition states.
- Enable quantum parallelism.
- Form the foundation of quantum algorithms.
- Allow entanglement between particles.
- Provide computational advantages for certain tasks.

---

15. Limitations of Qubits

- Extremely sensitive to environmental noise.
- Lose quantum information due to **decoherence**.
- Difficult to manufacture and control.
- Error correction is challenging.
- Measurements destroy the superposition state.

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16. Frequently Asked University Exam Questions

2-Mark Questions

1. Define a qubit.
2. What is a computational basis state?
3. What is superposition?
4. State the normalization condition.
5. What is state collapse?

5-Mark Questions

1. Explain the mathematical representation of a qubit.
2. Describe probability amplitudes with an example.
3. Explain measurement in quantum computing.
4. Differentiate between a bit and a qubit.

10-Mark Questions

1. Explain the concept of a qubit with mathematical representation.
2. Describe superposition and measurement with suitable examples.
3. Explain probability amplitudes, normalization, and state collapse.

Quick Revision (Exam Summary)

- A **qubit** is the fundamental unit of information in a quantum computer.

- Basis states are **|0⟩** and **|1⟩**.
- A general qubit is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- Probabilities are:

$$P(0) = |\alpha|^2, \qquad P(1) = |\beta|^2$$

- Normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

- Measurement causes the quantum state to **collapse** to one of the basis states.
- An **n-qubit** system has 2^n computational basis states.

Next Topic: Bloch Sphere

In the next lesson, we'll study:

1. What is the Bloch Sphere?
2. Geometry of a qubit
3. X, Y, and Z axes
4. Polar and azimuthal angles (θ and ϕ)
5. Representation of any single-qubit state on the Bloch Sphere
6. How quantum gates rotate qubits on the Bloch Sphere
7. Important diagrams and university exam questions

This topic is **very important** because it provides the geometric intuition behind single-qubit states and quantum gates.

> ok next

UNIT–I: Topic 3 – Bloch Sphere (Complete Detailed Notes)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> The **Bloch Sphere** is the standard graphical representation of a **single qubit**. It helps us visualize quantum states and understand how quantum gates affect them.

---

1. Introduction

In classical computing, a bit can only be either **0** or **1**, so representing its state is straightforward.

A qubit, however, can exist in infinitely many superposition states. To visualize these states, physicists use the **Bloch Sphere**.

The Bloch Sphere is a **three-dimensional unit sphere** where **every point on the surface represents a valid pure state of a single qubit**.

---

2. Why Do We Need the Bloch Sphere?

The mathematical expression

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

is accurate but difficult to visualize.

The Bloch Sphere helps us:

- Visualize a qubit's state.
- Understand superposition.
- See how quantum gates rotate qubits.
- Understand measurement probabilities geometrically.

> **Important:** The Bloch Sphere represents **only one qubit**. It cannot fully represent multi-qubit systems, especially entangled states.

---

3. Structure of the Bloch Sphere

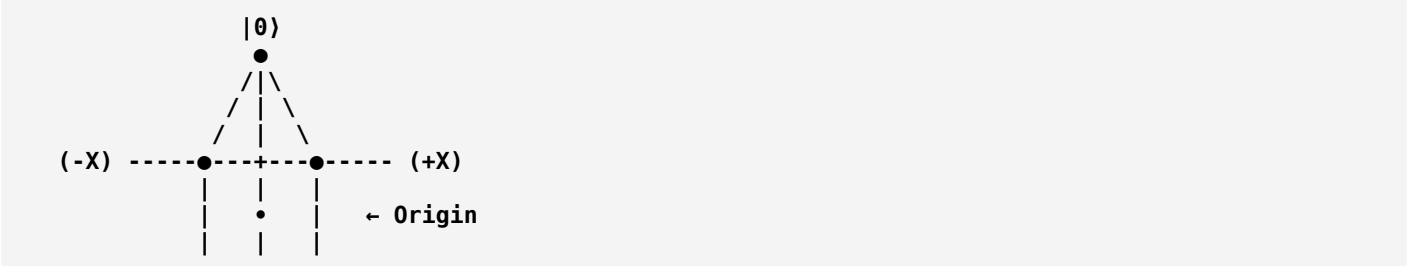
Imagine a sphere like the Earth.

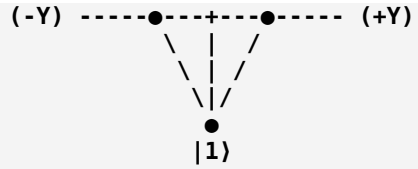
It has:

- North Pole
- South Pole
- Equator
- X-axis
- Y-axis
- Z-axis

The center represents the origin.

ASCII Diagram





4. Axes of the Bloch Sphere

The Bloch Sphere has three mutually perpendicular axes:

Z-axis

Represents the computational basis states.

- North Pole = $|0\rangle$
- South Pole = $|1\rangle$

This is the most important axis because measurements in the computational basis distinguish between these two states.

X-axis

Represents equal superpositions:

Positive X:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Negative X:

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These states are commonly created using the **Hadamard gate**.

Y-axis

Represents superpositions with a phase difference:

Positive Y:

$$\frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

Negative Y:

$$\frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

These are important when studying phase and rotation gates.

---

5. General Representation of a Qubit

Every pure single-qubit state can be written as:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

where:

- θ = Polar angle (0° to 180°)
- ϕ = Azimuthal angle (0° to 360°)

These two angles uniquely determine a point on the sphere.

---

6. Meaning of θ (Theta)

Theta controls how far the qubit is from the north pole.

Case 1

$$\theta = 0^\circ$$

State:

$$|0\rangle$$

The qubit is at the **North Pole**.

Case 2

$$\theta = 180^\circ$$

State:

$$|1\rangle$$

The qubit is at the **South Pole**.

Case 3

$$\theta = 90^\circ$$

The qubit lies on the **equator**, corresponding to an equal-weight superposition (the exact state depends on ϕ).

---

7. Meaning of φ (Phi)

Phi determines the direction around the equator (the phase of the state).

Changing ϕ:

- Does **not** change the total probability of measuring 0 or 1.
- **Does** change the relative phase, which affects interference and the action of later quantum gates.

This makes phase a uniquely quantum property.

---

8. Important States on the Bloch Sphere

State 1

$$|0\rangle$$

Location:

North Pole

State 2

$$|1\rangle$$

Location:

South Pole

State 3

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Location:

Positive X-axis

State 4

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Location:

Negative X-axis

State 5

$$\frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

Location:

Positive Y-axis

State 6

$$\frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

Location:

Negative Y-axis

---

9. Superposition on the Bloch Sphere

Every point on the surface (except the poles) represents a superposition of $|0\rangle$ and $|1\rangle$.

Examples:

- North Pole → Pure $|0\rangle$
- South Pole → Pure $|1\rangle$
- Equator → Equal superposition states (different phases)

Thus, the Bloch Sphere gives a geometric picture of superposition.

---

10. Quantum Gates as Rotations

One of the biggest advantages of the Bloch Sphere is that quantum gates can be visualized as **rotations**.

Examples:

- X Gate → Rotation by **180° about the X-axis**
- Y Gate → Rotation by **180° about the Y-axis**
- Z Gate → Rotation by **180° about the Z-axis**
- Rotation gates R_x, R_y, and R_z → Rotate the state by an arbitrary angle around the corresponding axis

This makes the Bloch Sphere extremely useful when studying **Unit II (Quantum Gates and Circuits)**.

---

11. Measurement on the Bloch Sphere

Suppose a qubit is somewhere on the sphere.

When measured in the computational basis:

- It collapses to $|0\rangle$ with probability $|\alpha|^2$.
- It collapses to $|1\rangle$ with probability $|\beta|^2$.

After measurement, the state is no longer the original superposition.

12. Applications of the Bloch Sphere

The Bloch Sphere is used in:

- Quantum computing
- Quantum communication
- Quantum cryptography
- Quantum gate design
- Quantum algorithms
- Quantum error correction
- Quantum control

13. Advantages

- Easy visualization of single-qubit states.
- Clearly illustrates superposition.
- Helps explain phase and rotations.
- Simplifies understanding of quantum gates.
- Widely used in teaching and research.

14. Limitations

- Represents **only one qubit**.
- Cannot fully represent entangled states.
- Does not visualize multi-qubit systems.

15. Bloch Sphere vs Classical Bit

Classical Bit	Bloch Sphere (Qubit)
Two possible states	Infinitely many pure states on the surface
Only 0 or 1	Any superposition of (
No phase information	Includes relative phase (ϕ)
No geometric representation needed	Represented as a point on a sphere

16. Frequently Asked University Exam Questions

2-Mark Questions

1. What is a Bloch Sphere?
2. Which states are located at the north and south poles?
3. What do the X, Y, and Z axes represent?
4. Why is the Bloch Sphere important?

5-Mark Questions

1. Explain the structure of the Bloch Sphere.
2. Describe the role of θ and ϕ.
3. Explain how quantum gates are represented as rotations.

10-Mark Questions

1. Explain the Bloch Sphere with a neat diagram and discuss its significance.
2. Describe the representation of a qubit using the Bloch Sphere.
3. Explain the geometric interpretation of quantum states and gates using the Bloch Sphere.

Key Takeaways (Exam Revision)

- The **Bloch Sphere** is a **3D unit sphere** used to represent the state of a **single qubit**.
- The **North Pole** corresponds to $|0\rangle$, and the **South Pole** corresponds to $|1\rangle$.
- Every point on the surface represents a valid **pure quantum state**.
- The angles θ and ϕ uniquely specify a qubit's state.
- Quantum gates can be viewed as **rotations** around the X, Y, and Z axes.
- The Bloch Sphere is limited to **single-qubit visualization** and cannot represent general multi-qubit entangled states.

Connection to the Next Topic

So far, we have learned:

1. ✅ Classical vs Quantum Computing
2. ✅ Qubits
3. ✅ Bloch Sphere

Now we know **what a qubit is** and **how to visualize it**.

The next topic is **Quantum Measurement**, where we'll answer questions such as:

- What happens when we measure a qubit?
- Why does the quantum state collapse?

- How are measurement probabilities calculated?
- What is the measurement postulate of quantum mechanics?
- How does measurement differ from classical observation?

Understanding quantum measurement is essential because **every quantum algorithm ultimately ends with a measurement to obtain a classical result**.

> ok next

UNIT–I: Topic 4 – Quantum Measurement (Complete Detailed Notes)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> **Quantum Measurement** is one of the most fundamental concepts in quantum computing. Every quantum algorithm eventually ends with a measurement because humans can only read **classical information (0s and 1s)** from a quantum computer.

---

1. Introduction

A **qubit** can exist in a **superposition** of states.

Example:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Before measurement:

- The qubit is in a quantum state.
- It is **not** simply 0 or 1.

When we **measure** the qubit:

- We obtain a **classical result** (either **0** or **1**).
- The quantum state **collapses** into the measured basis state.

This process is called **Quantum Measurement**.

---

2. What is Quantum Measurement?

Definition (Exam Definition):

> **Quantum Measurement** is the process of observing a quantum system, causing its quantum state to collapse into one of the possible basis states according to the probabilities determined by its probability amplitudes.

Unlike classical measurement, **quantum measurement changes the system being measured**.

3. Why is Measurement Needed?

Quantum computers process information using qubits.

However:

- Humans understand classical bits (0 or 1).
- Output devices also require classical data.

Therefore, at the end of every quantum computation, we **measure** the qubits to obtain classical outcomes.

Flow of a Quantum Program

```
Input
  |
  ▼
Initialize Qubits
  |
  ▼
Apply Quantum Gates
  |
  ▼
Create Superposition / Entanglement
  |
  ▼
Quantum Computation
  |
  ▼
Measurement
  |
  ▼
Classical Output (0s and 1s)
```

4. Measurement of a Single Qubit

Consider the quantum state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where:

- α = amplitude of $|0\rangle$
- β = amplitude of $|1\rangle$

The probabilities are:

$$P(0) = |\alpha|^2$$

$$P(1) = |\beta|^2$$

Since probabilities must sum to 1:

$$|\alpha|^2 + |\beta|^2 = 1$$

5. Example 1

Suppose

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Step 1: Calculate probabilities

$$P(0) = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

$$P(1) = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

Result

- Probability of getting **0 = 50%**
- Probability of getting **1 = 50%**

Each measurement gives only **one** outcome, chosen according to these probabilities.

---

6. Example 2

Suppose

$$|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$$

Probability of 0

$$P(0) = \left(\frac{3}{5}\right)^2 = \frac{9}{25} = 0.36$$

Probability of 1

$$P(1) = \left(\frac{4}{5}\right)^2 = \frac{16}{25} = 0.64$$

So:

- 36% chance of measuring **0**
- 64% chance of measuring **1**

---

7. State Collapse

This is one of the most important ideas in quantum mechanics.

Before Measurement

The qubit is in a superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After Measurement

If the result is **0**:

$$|\psi\rangle \rightarrow |0\rangle$$

If the result is **1**:

$$|\psi\rangle \rightarrow |1\rangle$$

The original superposition is destroyed.

This phenomenon is called **State Collapse** or **Wave Function Collapse**.

---

8. Visual Representation

Before Measurement

```
|\psi) = α|0) + β|1)
```

The qubit has probabilities for both outcomes.

↓

Measure

↓

After Measurement

```
Result = 0

State becomes

|0)
```

OR

```
Result = 1

State becomes

|1)
```

9. Why Does the State Collapse?

Quantum mechanics tells us that a measurement forces the system into one of the measurement basis states.

Before measurement:

- Multiple outcomes are possible with different probabilities.

After measurement:

- Only one outcome remains.
- The system is now definitely in that state.

This behavior has **no direct classical equivalent**.

---

10. Repeated Measurements

Suppose the first measurement gives:

$$|0\rangle$$

If you immediately measure again (without applying any gates), the result will be:

$$|0\rangle$$

with **100% probability**.

Similarly, if the first measurement gives:

$$|1\rangle$$

the next measurement will also give:

$$|1\rangle$$

with **100% probability**.

This happens because the state has already collapsed to a basis state.

---

11. Measurement in the Computational Basis

Most quantum computers measure qubits in the **computational basis**:

$$|0\rangle$$

and

$$|1\rangle$$

These are also called the **Z-basis states** because they correspond to the north and south poles of the Bloch Sphere.

> Advanced Note: Other measurement bases (such as the X-basis and Y-basis) are also used in quantum information, but for this course, the computational basis is the primary focus unless

explicitly mentioned.

12. Classical Measurement vs Quantum Measurement

Classical Measurement	Quantum Measurement
Reads data without fundamentally changing it	Changes the quantum state (state collapse)
Gives a deterministic value	Gives a probabilistic outcome
Object already has a definite value	Outcome depends on the quantum state and measurement basis
Repeat measurements give the same value because the object was already in that state	Repeat measurements give the same value because the first measurement collapsed the state

13. Real-Life Analogy (With a Limitation)

Imagine a **sealed lottery box**.

Before opening it:

- There are several possible outcomes, each with a known probability.

When you open the box:

- You see one winning number.

Similarly, a quantum measurement yields one outcome based on probabilities.

> **Important:** This is only an analogy. A quantum state is **not** literally an unopened lottery ticket. Quantum systems exhibit interference and other effects that have no classical counterpart.

14. Importance of Quantum Measurement

Measurement is essential because:

- It converts quantum information into classical information.
- It provides the final answer from a quantum algorithm.
- It determines the success probability of many quantum algorithms.
- It is used in quantum communication and cryptography.

Without measurement, we cannot read the result of a quantum computation.

15. Applications

Quantum measurement is used in:

- Quantum computing
- Quantum communication
- Quantum cryptography
- Quantum teleportation
- Quantum error correction
- Quantum sensing

---

16. Frequently Asked University Exam Questions

2-Mark Questions

1. Define quantum measurement.
2. What is state collapse?
3. Write the probability formulas for measuring a qubit.
4. What are computational basis states?

5-Mark Questions

1. Explain quantum measurement with an example.
2. Describe state collapse in quantum mechanics.
3. Differentiate between classical and quantum measurement.

10-Mark Questions

1. Explain the process of quantum measurement with suitable examples.
2. Describe state collapse and the role of probability amplitudes in quantum measurement.
3. Compare classical measurement and quantum measurement.

Quick Revision (Exam Summary)

- A qubit is measured to obtain a **classical bit (0 or 1)**.
- If

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

then

$$P(0) = |\alpha|^2, \qquad P(1) = |\beta|^2$$

- Measurement causes the quantum state to **collapse** into the measured basis state.
- After collapse, repeated measurements (without further operations) produce the same result with **100% probability**.

- Quantum measurement is **probabilistic**, unlike classical measurement.

Formula Box (Very Important for Exams)

General Qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Probability of Measuring 0

$$P(0) = |\alpha|^2$$

Probability of Measuring 1

$$P(1) = |\beta|^2$$

Normalization Condition

$$|\alpha|^2 + |\beta|^2 = 1$$

Connection to the Next Topic

So far, we have completed:

1. ✅ Classical vs Quantum Computing
2. ✅ Qubits
3. ✅ Bloch Sphere
4. ✅ Quantum Measurement

The next topic is **Quantum Mechanics Basics – Dirac Notation (Bra–Ket Notation)**.

This is the **mathematical language of quantum computing**. Every quantum gate, circuit, algorithm, and proof is written using Dirac notation. Once you understand it, reading quantum computing literature and solving problems becomes much easier.

> ok next

UNIT–I: Topic 5 – Dirac Notation (Bra–Ket Notation) (Complete Detailed Notes)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> **Dirac Notation** is the **mathematical language of quantum mechanics and quantum computing**. It was introduced by the physicist **Paul Dirac**. Almost every quantum algorithm, quantum circuit, and quantum gate is expressed using this notation.

---

1. Introduction

In classical computing, we represent data using:

- Numbers
- Binary digits (0 and 1)
- Matrices
- Vectors

In quantum computing, we represent quantum states using **Dirac Notation**.

Instead of writing long vectors repeatedly, we use symbols like:

$$|0\rangle, \quad |1\rangle, \quad |\psi\rangle$$

This notation makes quantum mathematics **compact, consistent, and easier to manipulate**.

---

2. Why Do We Need Dirac Notation?

Quantum systems are represented mathematically by **vectors** in a complex vector space called a **Hilbert Space**.

Writing vectors repeatedly becomes cumbersome.

For example,

Instead of writing

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

we simply write

$$|0\rangle$$

Similarly,

Instead of

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

we write

$$|1\rangle$$

This greatly simplifies calculations.

---

3. Components of Dirac Notation

Dirac notation has four important components:

1. Ket Vector
2. Bra Vector
3. Inner Product
4. Outer Product

These are the foundation of quantum mechanics.

---

4. Ket Notation (| ⟩)

A **Ket** represents a **column vector** describing a quantum state.

General notation:

$$|\psi\rangle$$

Read as:

> "Ket Psi"

Examples:

Basis State Zero

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Basis State One

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

General Qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

This represents a qubit in superposition.

---

5. Bra Notation (⟨ |)

A **Bra** is the **conjugate transpose (Hermitian transpose)** of a ket.

General notation:

$$\langle\psi|$$

Read as:

> "Bra Psi"

If

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

then

$$\langle\psi| = \begin{bmatrix} a^* & b^* \end{bmatrix}$$

where:

- a^* and b^* are the **complex conjugates** of a and b.

  Note: If the amplitudes are real numbers (no imaginary part), the complex conjugates are the same as the original values.

Example

If

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

then

$$\langle 0| = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Similarly,

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

becomes

$$\langle 1| = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

---

6. Difference Between Bra and Ket

Ket	Bra
Column vector	Row vector (conjugate transpose of ket)
Written as (	\psi\rangle)
Represents a quantum state	Used to calculate probabilities and overlaps
Appears on the right	Appears on the left

---

7. Inner Product (⟨φ|ψ⟩)

The **Inner Product** combines a bra and a ket.

General form:

$$\langle \phi | \psi \rangle$$

The result is a **single number (scalar)**.

It tells us:

- How similar two quantum states are.
- Whether two states are orthogonal (perpendicular).
- Probability amplitudes in quantum mechanics.

Example 1

$$\langle 0 | 0 \rangle$$

Calculation:

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$$

Answer:

$$\boxed{\langle 0 | 0 \rangle = 1}$$

Example 2

$$\langle 1 | 1 \rangle$$

Result:

$$\boxed{1}$$

Example 3

$$\langle 0 | 1 \rangle$$

Calculation:

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

Therefore,

$$\boxed{\langle 0 | 1 \rangle = 0}$$

This means the states are **orthogonal**.

---

8. Orthogonal States

Two states are **orthogonal** if:

$$\langle \phi | \psi \rangle = 0$$

Examples:

$$\langle 0 | 1 \rangle = 0$$

$$\langle 1 | 0 \rangle = 0$$

Thus,

- $|0\rangle$
- $|1\rangle$

are orthogonal basis states.

---

9. Normalization

Every valid quantum state must satisfy:

$$\langle \psi | \psi \rangle = 1$$

This is equivalent to:

$$|\alpha|^2 + |\beta|^2 = 1$$

Thus, the inner product provides a compact way to express normalization.

---

10. Outer Product ($|\psi\rangle\langle\varphi|$)

The **Outer Product** is formed by multiplying a ket by a bra.

General form:

$$|\psi\rangle\langle\phi|$$

The result is a **matrix (operator)**, not a scalar.

It is widely used for:

- Quantum gates
- Projection operators
- Density matrices
- Quantum measurements

Example

$$|0\rangle\langle 0|$$

Calculation:

$$\begin{bmatrix}1\\0\end{bmatrix}\begin{bmatrix}1 & 0\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$$

This is called the **projector onto the state** $|0\rangle$.

---

11. Summary: Inner Product vs Outer Product

Inner Product	Outer Product
(\langle\phi	\psi\rangle)
Bra × Ket	Ket × Bra
Scalar	Matrix
Measures overlap/similarity	Represents an operator or projector

---

12. Basis States

The computational basis consists of:

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$$

and

$$|1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

Any single-qubit state can be expressed as a combination of these basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

---

13. Importance of Dirac Notation

Dirac notation is used in:

- Quantum gates
- Quantum circuits
- Quantum algorithms
- Quantum teleportation

- Quantum cryptography
- Quantum error correction
- Quantum machine learning
- Quantum image processing

Without Dirac notation, quantum computations become difficult to express clearly.

---

14. Advantages of Dirac Notation

- Compact representation.
- Easy to write complex quantum states.
- Simplifies mathematical derivations.
- Standard notation used worldwide.
- Suitable for multi-qubit systems.

---

15. Frequently Asked University Exam Questions

2-Mark Questions

1. What is Dirac notation?
2. Define a ket.
3. Define a bra.
4. What is an inner product?
5. What is an outer product?

5-Mark Questions

1. Explain bra and ket notation with examples.
2. Differentiate between inner and outer products.
3. Explain normalization using Dirac notation.

10-Mark Questions

1. Explain Dirac notation in detail with suitable examples.
2. Describe bra vectors, ket vectors, inner products, and outer products.
3. Explain the importance of Dirac notation in quantum computing.

Formula Box (Important for Exams)

Ket

$$|\psi\rangle$$

Bra

$$\langle\psi|$$

General Qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Inner Product

$$\langle\phi|\psi\rangle$$

Outer Product

$$|\psi\rangle\langle\phi|$$

Normalization

$$\langle\psi|\psi\rangle = 1$$

Quick Revision (Exam Summary)

- **Dirac notation** is the standard mathematical notation used in quantum mechanics.
- A **ket** $(|\psi\rangle)$ is a **column vector** representing a quantum state.
- A **bra** $(\langle\psi|)$ is the **conjugate transpose** of a ket.
- The **inner product** $(\langle\phi|\psi\rangle)$ gives a **scalar** and measures the overlap between two states.
- The **outer product** $(|\psi\rangle\langle\phi|)$ gives a **matrix (operator)** used in gates, projectors, and measurements.
- The computational basis states satisfy:
  - $\langle 0|0\rangle = 1$
  - $\langle 1|1\rangle = 1$
  - $\langle 0|1\rangle = 0$
  - $\langle 1|0\rangle = 0$

Connection to the Next Topic

We have now completed:

1. ✅ Classical vs Quantum Computing
2. ✅ Qubits
3. ✅ Bloch Sphere

4. ✅ Quantum Measurement
5. ✅ Dirac Notation

Next, we will study **Quantum Entanglement**, one of the most fascinating concepts in quantum computing. You will learn:

- What is entanglement?
- Why is it different from classical correlation?
- Bell states (EPR pairs)
- How entanglement is created using quantum gates
- Applications in quantum teleportation, superdense coding, quantum cryptography, and quantum algorithms

This is one of the **highest-weightage topics** in Unit I and is frequently asked in 5-mark and 10-mark university exam questions.

> ok next

UNIT–I: Topic 6 – Quantum Entanglement (Complete Detailed Notes)

> **Weightage:** ⭐⭐⭐⭐⭐ (Extremely Important)
> **Quantum Entanglement** is one of the most important and unique features of quantum mechanics. It has no true equivalent in classical computing and is the foundation of many quantum technologies such as **quantum teleportation, quantum cryptography, superdense coding, and quantum algorithms**.

---

1. Introduction

Suppose two people each have a coin.

- If one person gets **Heads**, the other getting **Heads** or **Tails** is independent (unless they agreed beforehand).
- This is **classical independence**.

Now imagine two quantum particles that are prepared in a special way.

Even when they are separated by a large distance, measuring one particle gives information about the other because they share a **single quantum state**.

This phenomenon is called **Quantum Entanglement**.

> **Important Clarification:** Entanglement creates strong correlations between measurement outcomes, but it **does not allow faster-than-light communication**. This is a common misconception.

---

2. Definition

Exam Definition

> **Quantum Entanglement** is a quantum phenomenon in which two or more qubits are described by a **shared quantum state**, so the state of one qubit cannot, in general, be described independently of the others.

---

3. What is an Entangled State?

Consider two qubits.

Normally, if they are independent:

$$|\psi\rangle = |0\rangle \otimes |1\rangle$$

This is simply written as:

$$|01\rangle$$

Here:

- First qubit = 0
- Second qubit = 1

Each qubit has its own individual state.

This is **not entanglement**.

Now consider

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This is an **entangled state**.

We cannot assign an independent state to each qubit. The pair must be treated as one combined quantum system.

---

4. Why is it Called "Entangled"?

Think of two dancers performing a perfectly synchronized routine.

You cannot completely describe one dancer's movement without also considering the other.

Similarly, entangled qubits are mathematically linked through a shared quantum state.

> This analogy helps build intuition, but unlike dancers, entangled qubits exhibit uniquely quantum correlations that cannot be explained by classical coordination alone.

---

5. Classical Correlation vs Quantum Entanglement

Classical Correlation	Quantum Entanglement
Can be explained using classical probability	Requires quantum mechanics
Objects have definite properties before measurement	Measurement outcomes are correlated according to the shared quantum state
Independent descriptions are possible	Individual qubits generally cannot be described independently
No superposition required	Superposition is essential

---

6. Mathematical Representation

The most famous entangled state is

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This is called a **Bell State** (or **EPR Pair**).

It means:

The system is in a superposition of

- both qubits being 0
- both qubits being 1

---

7. Bell States

There are four standard Bell States.

Bell State 1

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Bell State 2

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

Bell State 3

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

Bell State 4

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

These four Bell states form an important basis for two-qubit systems and are widely used in quantum communication protocols.

---

8. Understanding the Bell State

Take

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

The probabilities are:

$$P(00) = \frac{1}{2}$$

$$P(11) = \frac{1}{2}$$

Notice:

There is **no probability** of obtaining

- 01
- 10

If we measure:

First qubit = 0

Then the second qubit will also be found to be 0 in the computational basis.

If we measure:

First qubit = 1

Then the second qubit will also be found to be 1.

The outcomes are perfectly correlated for this Bell state.

---

9. Creating Entanglement

One of the simplest ways to create entanglement is by using:

- Hadamard Gate (H)
- Controlled-NOT Gate (CNOT)

Step 1

Initial state

$$|00\rangle$$

Step 2

Apply Hadamard to the first qubit.

The first qubit becomes

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Combined state:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

Step 3

Apply the CNOT gate.

The control qubit is the first qubit.

- If control = 0 → target remains unchanged.
- If control = 1 → target flips.

Result:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This is the Bell state $|\Phi^+\rangle$.

Circuit Diagram

```
|0) ──H────■────
           |
|0) ───────X────
```

Output:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This is one of the most important quantum circuits in introductory quantum computing.

10. Properties of Entanglement

1. Strong Quantum Correlation

Measurements on entangled qubits are correlated in ways that cannot generally be explained by classical probability alone.

---

2. Shared Quantum State

The qubits are described by one combined quantum state.

---

3. Cannot Be Factorized

A separable state can be written as

$$|\psi_A\rangle \otimes |\psi_B\rangle$$

An entangled state **cannot** be written in this form.

---

4. Measurement Affects the Shared State

Measuring one qubit changes the overall shared quantum state.

---

5. Useful Resource

Entanglement is considered a resource for many quantum information tasks.

---

11. Applications of Entanglement

A. Quantum Teleportation

Transfers an **unknown quantum state** from one location to another using:

- Entanglement
- Classical communication

  It does **not** teleport matter or energy.

B. Quantum Cryptography

Used in **Quantum Key Distribution (QKD)** to detect eavesdropping and establish secure keys.

C. Superdense Coding

Allows the transmission of **two classical bits** of information by sending **one qubit**, provided the communicating parties share an entangled pair.

D. Quantum Algorithms

Many quantum algorithms use entanglement to create useful correlations between qubits during computation.

E. Quantum Networks

Entanglement enables quantum repeaters and future quantum internet technologies.

12. Advantages

- Enables quantum communication protocols.
- Improves computational capabilities in many algorithms.
- Essential for quantum teleportation and superdense coding.
- Supports advanced quantum networking.

13. Limitations

- Difficult to create and maintain.
- Fragile due to **decoherence** and environmental noise.
- Current hardware supports only limited numbers of high-quality entangled qubits.
- Verification of entanglement can be experimentally challenging.

14. Frequently Asked University Exam Questions

2-Mark Questions

1. Define quantum entanglement.
2. What is a Bell state?
3. Name the four Bell states.
4. Which gates are commonly used to create entanglement?

5-Mark Questions

1. Explain quantum entanglement with an example.
2. Describe Bell states.
3. Differentiate between classical correlation and quantum entanglement.

10-Mark Questions

1. Explain quantum entanglement with Bell states and suitable diagrams.
2. Describe the generation of entanglement using Hadamard and CNOT gates.
3. Explain the applications of quantum entanglement in modern quantum technologies.

Formula Box (Important for Exams)

Bell State

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Bell State

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

Bell State

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

Bell State

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

Quick Revision (Exam Summary)

- **Quantum entanglement** is a phenomenon where two or more qubits share a single quantum state and generally cannot be described independently.
- The most common entangled states are the **Bell States**.
- A Bell state can be created using a **Hadamard gate followed by a CNOT gate**.
- Entanglement is essential for **quantum teleportation, quantum cryptography, superdense coding, quantum algorithms, and quantum networking**.
- Entanglement produces strong quantum correlations but **does not enable faster-than-light communication**.

Connection to the Next Topic

We have now completed:

- ✅ Classical vs Quantum Computing

- ✅ Qubits
- ✅ Bloch Sphere
- ✅ Quantum Measurement
- ✅ Dirac Notation
- ✅ Quantum Entanglement

The next topic is **Quantum Hardware Overview**, where you'll learn:

- How real quantum computers are built.
- Different physical implementations of qubits (Superconducting, Trapped Ions, Photonic, Spin-based).
- Their advantages and disadvantages.
- Current challenges in building large-scale quantum computers.
- Companies and organizations leading quantum hardware development.

This is an important theory topic and is often asked as a **5-mark or 10-mark question** in university exams.

> ok next

UNIT–I: Topic 7 – Quantum Hardware Overview (Complete Detailed Notes)

> **Weightage:** ⭐⭐⭐⭐☆ (Important)
> This topic explains **how real quantum computers are built**. Unlike classical computers, which use transistors and silicon chips, quantum computers use **physical systems that behave according to quantum mechanics**.

---

1. Introduction

A classical computer uses:

- Transistors
- Integrated Circuits (ICs)
- Silicon Chips

A quantum computer uses:

- Atoms
- Electrons
- Photons
- Superconducting circuits

- Trapped ions

These physical systems act as **qubits**.

The biggest challenge is keeping qubits stable long enough to perform computations.

---

2. What is Quantum Hardware?

Definition (Exam Definition)

> **Quantum Hardware** refers to the physical devices and technologies used to create, manipulate, and measure qubits for performing quantum computations.

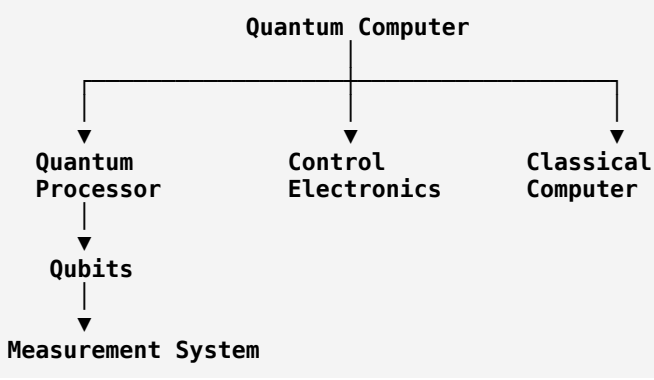
Unlike classical hardware, quantum hardware must preserve delicate quantum properties such as:

- Superposition
- Entanglement
- Quantum coherence

---

3. Components of a Quantum Computer

A complete quantum computer consists of several components.



Main Components

1. Quantum Processor (QPU)
2. Qubits
3. Control Electronics
4. Measurement System
5. Classical Computer
6. Cooling System (for some hardware types)

---

4. Quantum Processing Unit (QPU)

The **Quantum Processing Unit (QPU)** is the "brain" of a quantum computer.

It:

- Stores qubits
- Applies quantum gates
- Performs quantum operations
- Executes quantum algorithms

Just as a CPU processes classical bits, a **QPU processes qubits**.

---

5. Types of Quantum Hardware

There are several ways to physically implement qubits.

The most important technologies are:

1. Superconducting Qubits
2. Trapped Ion Qubits
3. Photonic Qubits
4. Spin Qubits (Quantum Dots)
5. Neutral Atom Qubits

Let's study each in detail.

---

6. Superconducting Qubits

Introduction

This is currently the **most widely used technology**.

Companies using it:

- IBM
- Google
- Rigetti

Working Principle

These qubits are made using superconducting electrical circuits.

At extremely low temperatures:

- Electrical resistance becomes almost zero.
- Quantum effects become observable.

- Current can behave like a quantum system.

Temperature Requirement

They operate at approximately

15 millikelvin (mK)

or about

−273°C

using a **dilution refrigerator**.

This is colder than outer space.

Advantages

- Fast quantum gate operations.
- Mature fabrication techniques.
- Easy integration with existing chip manufacturing.
- Supported by major technology companies.

Disadvantages

- Require extremely low temperatures.
- Sensitive to environmental noise.
- Relatively short coherence times compared to some other technologies.

---

7. Trapped Ion Qubits

Introduction

Instead of electrical circuits,

individual charged atoms (ions) are used as qubits.

Companies:

- IonQ
- Quantinuum

Working Principle

- Ions are trapped using electromagnetic fields.
- Lasers manipulate their quantum states.
- Lasers also perform quantum gates and measurements.

Advantages

- High gate accuracy (high fidelity).
- Long coherence times.
- Well-suited for precise quantum operations.

Disadvantages

- Slower gate operations than superconducting qubits.
- Scaling to very large systems is technically challenging.

---

8. Photonic Qubits

Introduction

Photonic quantum computers use **photons (particles of light)** as qubits.

Company:

- Xanadu

Working Principle

Information is encoded in properties of photons, such as:

- Polarization
- Path
- Phase

Photons are manipulated using optical components like beam splitters and phase shifters.

Advantages

- Operate at or near room temperature.
- Photons interact weakly with the environment, reducing decoherence.
- Excellent for long-distance quantum communication.

Disadvantages

- Two-photon interactions are difficult to implement.
- Large-scale photonic quantum computing remains an active research area.

---

9. Spin Qubits (Quantum Dots)

Introduction

A **spin qubit** uses the spin of an electron (or sometimes a nucleus) to represent quantum information.

Working Principle

Electron spin has two possible orientations:

- Spin Up (↑)
- Spin Down (↓)

These can represent:

- $|0\rangle$
- $|1\rangle$

Advantages

- Very small physical size.
- Potential compatibility with existing semiconductor manufacturing.
- Promising for future scalability.

Disadvantages

- Difficult to control individual spins precisely.
- Still largely in the research stage.

---

10. Neutral Atom Qubits

Introduction

Neutral atoms are trapped using laser light (optical tweezers).

Examples include atoms such as:

- Rubidium
- Cesium

Advantages

- Good scalability.
- Can naturally form large arrays of qubits.
- Strong potential for quantum simulation.

Disadvantages

- Laser systems are complex.
- Precision control is technically demanding.

---

11. Comparison of Quantum Hardware Technologies

Technology	Qubit Type	Advantages	Disadvantages	Examples
Superconducting	Electrical circuits	Fast gates, mature technology	Requires ultra-low temperatures	IBM, Google, Rigetti
Trapped Ion	Charged atoms	High fidelity, long coherence	Slower gates	IonQ, Quantinuum
Photonic	Photons	Room-temperature operation, good for communication	Difficult interactions	Xanadu
Spin Qubits	Electron spin	Small size, semiconductor compatibility	Difficult control	Research labs
Neutral Atom	Neutral atoms	Highly scalable	Complex laser systems	QuEra, research groups

---

12. Quantum Coherence

Definition

Quantum coherence is the ability of a qubit to maintain its quantum state over time.

The longer a qubit remains coherent:

- The more computations it can perform.
- The more reliable the quantum computer becomes.

Loss of coherence is called **decoherence**.

---

13. Decoherence

Definition

Decoherence is the loss of quantum information due to interactions with the surrounding environment.

Causes include:

- Heat
- Electromagnetic radiation
- Vibrations
- Material imperfections

Decoherence is one of the biggest challenges in quantum computing.

---

14. Quantum Error Correction

Since qubits are fragile:

Errors can occur because of:

- Noise
- Decoherence
- Imperfect quantum gates

Quantum Error Correction (QEC) uses **multiple physical qubits** to protect the information stored in a **logical qubit**.

Unlike classical error correction, QEC must preserve quantum information without simply copying an unknown quantum state (because of the **No-Cloning Theorem**).

---

15. Current Challenges in Quantum Hardware

Scientists are working to solve problems such as:

- Increasing the number of qubits.
- Reducing decoherence.
- Improving gate fidelity (accuracy).

- Reducing hardware costs.
- Developing better quantum error correction.
- Building fault-tolerant quantum computers.

16. Leading Companies in Quantum Hardware

Company	Hardware Technology
IBM	Superconducting Qubits
Google Quantum AI	Superconducting Qubits
Rigetti	Superconducting Qubits
IonQ	Trapped Ions
Quantinuum	Trapped Ions
Xanadu	Photonic Quantum Computing
QuEra	Neutral Atom Quantum Computing

17. Advantages of Quantum Hardware

- Can exploit quantum mechanical phenomena.
- Enables quantum algorithms.
- Useful for scientific simulations.
- Supports secure quantum communication.
- Potential to solve certain problems beyond the reach of classical computers.

18. Limitations

- Expensive to build and maintain.
- Very sensitive to environmental disturbances.
- Limited number of reliable qubits in current devices.
- Complex cooling and control systems (for many technologies).
- Error correction remains a major challenge.

19. Frequently Asked University Exam Questions

2-Mark Questions

1. What is quantum hardware?
2. What is a Quantum Processing Unit (QPU)?

3. What is decoherence?
4. Define quantum coherence.
5. Name any four quantum hardware technologies.

5-Mark Questions

1. Explain superconducting qubits.
2. Describe trapped-ion quantum computers.
3. Explain decoherence and its impact on quantum computing.
4. Compare different quantum hardware technologies.

10-Mark Questions

1. Explain various quantum hardware technologies with advantages and disadvantages.
2. Describe the architecture of a quantum computer.
3. Discuss the challenges in developing practical quantum computers.

Quick Revision (Exam Summary)

- **Quantum hardware** is the physical technology used to build quantum computers.
- The **Quantum Processing Unit (QPU)** performs quantum computations using qubits.
- Major hardware technologies include:
  - Superconducting Qubits
  - Trapped Ion Qubits
  - Photonic Qubits
  - Spin Qubits
  - Neutral Atom Qubits
- **Quantum coherence** is the ability of a qubit to maintain its quantum state.
- **Decoherence** is the loss of quantum information due to environmental interactions.
- **Quantum Error Correction (QEC)** protects quantum information from errors.
- Companies like **IBM, Google, IonQ, Xanadu, and QuEra** are actively developing quantum hardware.

Memory Trick for Hardware Technologies

Remember the acronym **"STPSN"**:

- **S** → Superconducting

- **T** → Trapped Ion
- **P** → Photonic
- **S** → Spin Qubits
- **N** → Neutral Atoms

This helps recall the major hardware approaches quickly during exams.

Connection to the Next Topic

We have now completed:

- ✅ Classical vs Quantum Computing
- ✅ Qubits
- ✅ Bloch Sphere
- ✅ Quantum Measurement
- ✅ Dirac Notation
- ✅ Quantum Entanglement
- ✅ Quantum Hardware Overview

The **final topic of Unit I** is **Quantum Programming Platforms**, where you'll learn:

- What quantum programming platforms are.
- Popular frameworks such as **Qiskit, Cirq, PennyLane, and Q#**.
- How quantum circuits are created and simulated.
- The difference between simulators and real quantum hardware.
- A simple example of creating a quantum circuit.

This is the practical bridge between quantum theory and writing quantum programs, and it is commonly asked as a **5-mark or 10-mark exam question**.

> ok next

UNIT–I: Topic 8 – Quantum Programming Platforms (Complete Detailed Notes)

> **Weightage:** ⭐⭐⭐⭐☆ (Important)
> This is the final topic of **UNIT–I**.
> After understanding qubits, gates, and hardware, we need software tools to **design, simulate, and execute quantum algorithms**. These tools are called **Quantum Programming Platforms**.

---

1. Introduction

A quantum computer requires specialized software to communicate with quantum hardware.

A **Quantum Programming Platform** provides:

- Programming languages
- Development frameworks
- Quantum circuit simulators
- Access to real quantum processors
- Libraries for quantum algorithms

It allows researchers and developers to create quantum programs without directly controlling physical qubits.

---

2. Definition

Exam Definition:

> **Quantum Programming Platforms are software frameworks and tools that allow users to design, simulate, execute, and analyze quantum algorithms on quantum simulators or real quantum computers.**

---

3. Need for Quantum Programming Platforms

Quantum hardware is extremely complex.

Programming platforms provide:

1. Quantum Circuit Design

Users can create circuits using quantum gates:

- Hadamard (H)
- Pauli gates
- CNOT
- Rotation gates

---

2. Simulation

Before running on real quantum hardware, programs can be tested on classical simulators.

Advantages:

- Low cost
- Easy debugging

- Faster experimentation

---

3. Hardware Access

Platforms provide cloud access to quantum computers.

Example:

A researcher can write a program on a laptop and execute it on an IBM quantum processor through the cloud.

---

4. Algorithm Development

They provide ready-made implementations of:

- Grover's Algorithm
- Quantum Fourier Transform
- Variational algorithms
- Quantum machine learning algorithms

---

4. Quantum Programming Workflow

A typical quantum programming process:

```
Problem Definition
        |
        ▼
Quantum Algorithm Design
        |
        ▼
Create Quantum Circuit
        |
        ▼
Run on Simulator
        |
        ▼
Optimize Circuit
        |
        ▼
Execute on Quantum Hardware
        |
        ▼
Measure Results
```

5. Major Quantum Programming Platforms

The most important platforms are:

1. IBM Qiskit
2. Google Cirq
3. Microsoft Q#
4. PennyLane

5. Amazon Braket

6. IBM Qiskit

Introduction

IBM developed **Qiskit**, one of the most popular open-source quantum computing frameworks.

It is written in:

- Python

It provides access to IBM quantum computers through the cloud.

Features of Qiskit

1. Quantum Circuit Creation

Users can create circuits using quantum gates.

Example:

```
Python
QuantumCircuit
```

2. Simulator Support

Qiskit provides simulators to test circuits before using real hardware.

3. Hardware Access

Users can execute programs on IBM quantum processors.

4. Libraries

Qiskit provides modules for:

- Quantum algorithms
- Optimization
- Machine learning
- Chemistry simulations

Qiskit Architecture

```
User Program
      │
      ▼
 Qiskit SDK
      │
      ▼
Quantum Circuit
      │
      ▼
Compiler / Transpiler
      │
      ▼
Simulator or Quantum Hardware
      │
      ▼
Measurement Results
```

7. Google Cirq

Introduction

Cirq is an open-source quantum computing framework developed by Google.

It focuses on:

- Designing quantum circuits
- Running experiments
- Research applications

Features

- Python-based framework
- Circuit optimization
- Quantum gate manipulation
- Integration with Google's quantum processors

Advantages

- Good control over quantum circuits.
- Suitable for quantum research.
- Efficient circuit representation.

---

8. Microsoft Q#

Introduction

Q# is a quantum programming language developed by Microsoft.

It is part of the Azure Quantum ecosystem.

Features

- Dedicated quantum programming language.
- Supports quantum algorithm development.
- Provides quantum simulators.
- Integrates with Microsoft development tools.

Advantages

- Designed specifically for quantum computing.
- Good support for large quantum algorithms.
- Strong mathematical foundation.

---

9. PennyLane

Introduction

PennyLane is a quantum machine learning framework developed by Xanadu.

It focuses mainly on:

- Quantum Machine Learning (QML)
- Hybrid quantum-classical algorithms

Features

Supports:

- Quantum neural networks
- Quantum gradients
- Variational algorithms

Advantages

- Easy integration with machine learning libraries:

  - PyTorch
  - TensorFlow
  - JAX
- Useful for AI researchers.

---

10. Amazon Braket

Introduction

Amazon Braket is a cloud-based quantum computing service provided by Amazon Web Services.

It allows users to access different quantum hardware technologies.

Features

Supports:

- Quantum circuit design
- Simulation
- Multiple quantum hardware providers

---

11. Comparison of Quantum Programming Platforms

Platform	Developer	Language	Main Focus
Qiskit	IBM	Python	General quantum computing
Cirq	Google	Python	Quantum circuit research
Q#	Microsoft	Q#	Quantum algorithm development
PennyLane	Xanadu	Python	Quantum machine learning
Braket	AWS	Python	Cloud quantum computing

---

12. Quantum Simulator vs Real Quantum Hardware

Quantum Simulator

A classical computer that imitates quantum behavior.

Examples:

- Qiskit Aer Simulator

- Cirq Simulator

Advantages

- Easy testing
- No hardware cost
- Good for learning

Limitations

- Limited by classical computing power
- Cannot efficiently simulate very large quantum systems

Real Quantum Hardware

Actual quantum processors containing physical qubits.

Examples:

- Superconducting quantum processors
- Trapped ion processors

Advantages

- Performs actual quantum computation
- Necessary for real experiments

Limitations

- Noise
- Errors
- Limited qubit count

---

13. Simple Quantum Programming Example

Creating a Superposition

Goal:

Create a qubit that has equal probability of 0 and 1.

Circuit:

```
|0⟩ ──H──M──
```

Steps:

1. Start with:

$$|0\rangle$$

2. Apply Hadamard gate:

$$H|0\rangle$$

Result:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

3. Measure.

Output:

- 0 → 50%
- 1 → 50%

---

14. Applications of Quantum Programming Platforms

Quantum Algorithm Development

- Grover's algorithm
- Shor's algorithm
- Quantum Fourier Transform

Quantum Machine Learning

- Quantum classifiers
- Quantum neural networks

Optimization

- QAOA
- VQE

Quantum Simulation

- Chemistry
- Physics
- Material science

---

15. Advantages of Quantum Programming Platforms

- Easy access to quantum technology.
- Reduce complexity of quantum programming.
- Allow testing before hardware execution.
- Support algorithm development.
- Provide educational resources.

---

16. Challenges

- Quantum programming is still evolving.
- Hardware limitations affect execution.
- Quantum errors must be handled.
- Requires knowledge of linear algebra and quantum mechanics.

---

17. Frequently Asked University Exam Questions

2-Mark Questions

1. Define a quantum programming platform.
2. What is Qiskit?
3. Name any four quantum programming platforms.
4. What is a quantum simulator?

5-Mark Questions

1. Explain Qiskit and its features.
2. Describe different quantum programming platforms.
3. Compare quantum simulators and real quantum hardware.

10-Mark Questions

1. Explain various quantum programming platforms with their features.
2. Describe the workflow of quantum programming.
3. Compare Qiskit, Cirq, Q#, and PennyLane.

UNIT–I Complete Revision Summary

We have completed all topics:

1. Classical vs Quantum Computing

- Bit vs Qubit
- Classical and quantum differences
- Applications

---

2. Qubits

- Basis states
- Superposition
- Probability amplitudes
- Normalization

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

---

3. Bloch Sphere

- Geometric representation of a qubit
- θ and φ angles
- Quantum rotations

---

4. Quantum Measurement

- Measurement probabilities
- State collapse

$$P(0) = |\alpha|^2$$

$$P(1) = |\beta|^2$$

---

5. Dirac Notation

- Ket
- Bra

- Inner product
- Outer product

---

6. Entanglement

- Bell states
- Quantum correlations
- Applications

---

7. Quantum Hardware

- Superconducting qubits
- Trapped ions
- Photonic qubits
- Spin qubits

---

8. Quantum Programming Platforms

- Qiskit
- Cirq
- Q#
- PennyLane
- Braket

UNIT–I Important 10-Mark Questions (Prepare These)

1. Explain classical computing and quantum computing with differences.
2. Explain qubits and superposition with mathematical representation.
3. Describe Bloch Sphere representation of a qubit.
4. Explain quantum measurement and state collapse.
5. Explain Dirac notation with examples.
6. Explain quantum entanglement and Bell states.
7. Discuss different quantum hardware technologies.
8. Explain quantum programming platforms.

✅ **UNIT–I is now completed.**

Next we will start:

UNIT–II: Quantum Gates and Circuits

First Topic:

Quantum Logic Gates

- Pauli Gates (X, Y, Z)
- Hadamard Gate
- Phase Gates (S, T)
- Rotation Gates (Rx, Ry, Rz)
- Matrix representation
- Circuit examples
- Applications

This unit is more **mathematical and circuit-oriented**, so we will study each gate with **matrix operations and examples**.

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 1 – Introduction to Quantum Logic Gates

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> Quantum gates are the **building blocks of quantum circuits**. Similar to classical computers using AND, OR, NOT gates, quantum computers use **quantum gates** to manipulate qubits.

---

1. Introduction to Quantum Gates

A **quantum gate** is a mathematical operation that changes the state of one or more qubits.

In classical computing:

```
Input Bits → Logic Gates → Output Bits
```

Example:

```
0 → NOT Gate → 1
```

In quantum computing:

```
Input Qubits → Quantum Gates → Modified Quantum State
```

Example:

$$|0\rangle \xrightarrow{H} \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

A Hadamard gate converts a definite state into a superposition state.

---

2. Definition

Exam Definition:

> **A quantum gate is a reversible unitary operation that transforms one quantum state into another while preserving the total probability of the system.**

---

3. Properties of Quantum Gates

Quantum gates have three important properties:

---

1. Unitary Operation

A quantum gate must satisfy:

$$U^\dagger U = UU^\dagger = I$$

where:

- U = quantum gate matrix
- $U^\dagger$ = conjugate transpose of U
- I = identity matrix

This ensures the operation is reversible.

---

2. Reversibility

Every quantum gate has an inverse operation.

If:

$$|\psi'\rangle = U|\psi\rangle$$

then:

$$|\psi\rangle = U^{-1}|\psi'\rangle$$

Example:

$$X^{-1} = X$$

The NOT gate can reverse itself.

---

3. Probability Preservation

Quantum gates must preserve:

$$|\alpha|^2 + |\beta|^2 = 1$$

The total probability remains 100%.

---

4. Matrix Representation of Quantum Gates

Quantum gates are represented using matrices.

A quantum operation is:

$$|\psi'\rangle = U|\psi\rangle$$

where:

- U = gate matrix
- $|\psi\rangle$ = input state
- $|\psi'\rangle$ = output state

Example

Input:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Apply X gate:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Calculation:

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Result:

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore:

$$X|0\rangle = |1\rangle$$

The qubit flips from 0 to 1.

---

5. Single-Qubit Gates

Single-qubit gates operate on only one qubit.

Important single-qubit gates:

1. Pauli-X Gate
2. Pauli-Y Gate
3. Pauli-Z Gate
4. Hadamard Gate
5. Phase Gates
6. Rotation Gates

---

6. Pauli Gates

Pauli gates are fundamental quantum gates named after physicist **Wolfgang Pauli**.

They correspond to rotations around the:

- X-axis
- Y-axis
- Z-axis

of the Bloch Sphere.

The three Pauli gates are:

- X Gate
- Y Gate
- Z Gate

---

7. Pauli-X Gate (Quantum NOT Gate)

Definition

The Pauli-X gate is the quantum equivalent of the classical NOT gate.

It flips:

$$|0\rangle \leftrightarrow |1\rangle$$

Matrix Representation

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Operation on $|0\rangle$

$$X|0\rangle$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Result:

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore:

$$X|0\rangle = |1\rangle$$

Operation on $|1\rangle$

$$X|1\rangle = |0\rangle$$

Circuit Symbol

```
|ψ⟩ ──X──
```

Bloch Sphere Effect

The X gate rotates the qubit by:

$$180^\circ$$

around the X-axis.

---

8. Pauli-Y Gate

Definition

The Y gate performs both:

- Bit flip
- Phase flip

Matrix Representation

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

where:

$$i = \sqrt{-1}$$

Operation

For $|0\rangle$:

$$Y|0\rangle = i|1\rangle$$

For $|1\rangle$:

$$Y|1\rangle = -i|0\rangle$$

The imaginary factor represents a phase change.

Bloch Sphere Effect

Rotation:

$$180^\circ$$

around Y-axis.

---

9. Pauli-Z Gate

Definition

The Z gate performs a phase flip.

It keeps:

$$|0\rangle$$

unchanged but changes the phase of:

$$|1\rangle$$

Matrix Representation

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Operation on $|0\rangle$

$$Z|0\rangle = |0\rangle$$

Operation on $|1\rangle$

$$Z|1\rangle = -|1\rangle$$

The negative sign indicates a phase change.

Circuit Symbol

```
|ψ⟩ ──Z──
```

10. Comparison of Pauli Gates

Gate	Function	Matrix	Rotation
X	Bit flip	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	X-axis
Y	Bit + phase flip	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	Y-axis
Z	Phase flip	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Z-axis

---

11. Importance of Pauli Gates

Pauli gates are used in:

- Quantum circuits

- Quantum error correction
- Quantum teleportation
- Quantum algorithms
- Quantum simulations

---

12. Exam Important Points

Remember:

X → Changes bit

$$|0\rangle \leftrightarrow |1\rangle$$

Y → Changes bit + phase

$$|0\rangle \rightarrow i|1\rangle$$

Z → Changes phase only

$$|1\rangle \rightarrow -|1\rangle$$

Frequently Asked University Questions

2-Mark Questions

1. Define a quantum gate.
2. What is a unitary operation?
3. Write the matrix of Pauli-X gate.
4. What is the function of Pauli-Z gate?

5-Mark Questions

1. Explain Pauli gates with matrices.
2. Describe the operation of X, Y, and Z gates.
3. Explain reversibility of quantum gates.

10-Mark Questions

1. Explain different Pauli gates with mathematical representation.
2. Describe quantum gates and their properties.

3. Compare classical logic gates and quantum gates.

Quick Revision

- Quantum gates manipulate qubit states.
- They are represented by **unitary matrices**.
- All quantum gates are reversible.
- Pauli gates are:
  - **X gate → Bit flip**
  - **Y gate → Bit + phase flip**
  - **Z gate → Phase flip**
- X, Y, Z gates correspond to rotations on the Bloch Sphere.

Next Topic

Hadamard Gate (H Gate)

We will study:

- Definition
- Matrix representation
- Creating superposition
- Operation on $|0\rangle$ and $|1\rangle$
- Bloch sphere interpretation
- Quantum circuit examples
- Importance in Grover's algorithm and quantum algorithms

The Hadamard gate is one of the **most important gates in quantum computing** and is asked very frequently in semester examinations.

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 2 – Hadamard Gate (H Gate)

Weightage: ⭐⭐⭐⭐⭐ (Extremely Important)
The **Hadamard Gate** is one of the most fundamental quantum gates. It is responsible for

creating **superposition**, which is the key property that gives quantum computers their computational power.

---

1. Introduction to Hadamard Gate

In classical computing:

A bit is either:

$$0$$

or

$$1$$

There is no intermediate state.

In quantum computing:

A qubit can be transformed into a combination of:

$$|0\rangle$$

and

$$|1\rangle$$

using the **Hadamard Gate**.

---

2. Definition

Exam Definition:

> **The Hadamard gate is a single-qubit quantum gate that creates an equal superposition of the computational basis states |0⟩ and |1⟩.**

It converts:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

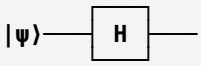
and

$$|1\rangle \rightarrow \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

---

3. Symbol of Hadamard Gate

Circuit representation:



Where:

H represents the Hadamard operation.

4. Matrix Representation

The Hadamard gate is represented by the matrix:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

or:

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

5. Applying Hadamard Gate on $|0\rangle$

Initial state:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Apply:

$$H|0\rangle$$

Calculation:

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Multiplication:

First element:

$$(1)(1) + (1)(0) = 1$$

Second element:

$$(1)(1) + (-1)(0) = 1$$

Therefore:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Hence:

$$\boxed{H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}}$$

6. Applying Hadamard Gate on $|1\rangle$

Initial state:

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Apply:

$$H|1\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Calculation:

First element:

$$(1)(0) + (1)(1) = 1$$

Second element:

$$(1)(0) + (-1)(1) = -1$$

Therefore:

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Hence:

$$\boxed{H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}}$$

7. Creating Superposition

The main purpose of the Hadamard gate is creating a superposition state.

Example:

Start with:

$$|0\rangle$$

Apply H:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Now the qubit has:

Probability of measuring 0:

$$P(0) = \frac{1}{2}$$

Probability of measuring 1:

$$P(1) = \frac{1}{2}$$

So:

- 50% chance of getting 0
- 50% chance of getting 1

---

8. Hadamard Gate and Measurement

Circuit:

```
|0⟩ ──H──M──
```

Steps:

Step 1:

Initial state:

$$|0\rangle$$

Step 2:

After H:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Step 3:

Measurement gives:

Either:

$$|0\rangle$$

or

$$|1\rangle$$

with equal probability.

9. Hadamard Gate on General Qubit

Consider:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Apply H:

$$H|\psi\rangle$$

Using:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

and

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Therefore:

$$H|\psi\rangle = \alpha\frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Rearranging:

$$= \frac{(\alpha+\beta)|0\rangle + (\alpha-\beta)|1\rangle}{\sqrt{2}}$$

---

10. Hadamard Gate Properties

1. Creates Superposition

Most important property.

Example:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

---

2. Reversible

Applying Hadamard twice returns the original state:

$$H^2 = I$$

Example:

$$|0\rangle \xrightarrow{H} \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Again:

$$\xrightarrow{H} |0\rangle$$

---

3. Unitary

The Hadamard matrix satisfies:

$$H^\dagger H = I$$

Therefore, it is a valid quantum gate.

---

11. Bloch Sphere Interpretation

The Hadamard gate performs a rotation on the Bloch Sphere.

It changes:

- Computational basis states
- Into superposition states

It maps:

$$|0\rangle \to |+\rangle$$

where:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

and:

$$|1\rangle \to |-\rangle$$

where:

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

---

12. Importance in Quantum Algorithms

The Hadamard gate is used extensively in:

---

1. Grover's Algorithm

Used to:

- Create equal superposition of search states.
- Enable amplitude amplification.

---

2. Quantum Fourier Transform

Hadamard gates are important building blocks.

---

3. Quantum Teleportation

Creates entanglement along with CNOT.

---

4. Quantum Random Number Generation

Creates equal probability outcomes.

---

13. Creating Entanglement Using H Gate

Hadamard is used with CNOT:

Circuit:

```
|0⟩ ──H──■──
        │
|0⟩ ─────X──
```

Steps:

Initial:

$$|00\rangle$$

After H:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

After CNOT:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

A Bell state is created.

---

14. Hadamard Gate vs Classical Gates

Classical Gate	Hadamard Gate
Works on bits	Works on qubits
Produces definite output	Produces superposition
Not reversible generally	Always reversible
Boolean operation	Unitary operation

15. Advantages

- Creates quantum superposition.
- Fundamental building block of quantum algorithms.
- Enables quantum parallelism.
- Helps create entanglement.
- Simple and reversible.

16. Limitations

- Works only on quantum states.
- Does not directly provide useful information without measurement.
- Measurement destroys the superposition.

17. Frequently Asked University Questions

2-Mark Questions

1. Define Hadamard gate.
2. Write the matrix representation of H gate.
3. What is the purpose of Hadamard gate?
4. What happens when H is applied to $|0\rangle$?

5-Mark Questions

1. Explain Hadamard gate with matrix representation.
2. Describe how Hadamard creates superposition.
3. Explain the reversibility property of H gate.

10-Mark Questions

1. Explain the Hadamard gate with mathematical derivation and circuit representation.
2. Describe the role of Hadamard gate in quantum algorithms.
3. Explain how Hadamard and CNOT gates create entanglement.

Formula Box (Very Important)

Hadamard Matrix

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Transformations

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Reversibility

$$H^2 = I$$

Quick Revision

- Hadamard is a **single-qubit gate**.
- It creates **equal superposition**.
- It transforms basis states into:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

and

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

- It is unitary and reversible.
- It is essential in:
  - Grover's Algorithm
  - Quantum Fourier Transform

- Quantum Teleportation

Next Topic

Phase Gates (S Gate and T Gate)

We will study:

- What is a phase shift?
- S gate matrix and operation
- T gate matrix and operation
- Difference between bit flip and phase flip
- Bloch sphere interpretation
- Applications in quantum circuits

Phase gates are important because they introduce **quantum phase changes**, which are responsible for quantum interference.

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 3 – Phase Gates (S Gate and T Gate)

> **Weightage:** ⭐⭐⭐⭐☆ (Important)
> Phase gates are fundamental single-qubit quantum gates that modify the **phase of a quantum state** without changing the probability of measuring the state. They play a major role in **quantum interference**, quantum algorithms, and circuit construction.

---

1. Introduction to Phase in Quantum Computing

A qubit is represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where:

- α and β are probability amplitudes.
- Their magnitudes determine measurement probabilities.
- Their phases affect interference.

The probability of measurement depends on:

$$|\alpha|^2$$

and

$$|\beta|^2$$

But the **relative phase** between states determines how quantum states combine.

---

2. What is a Phase Gate?

Definition:

> **A phase gate is a single-qubit quantum gate that changes the phase of the |1⟩ state while leaving the |0⟩ state unchanged.**

General phase gate:

$$P(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}$$

where:

- θ = phase rotation angle
- $e^{i\theta}$ = phase factor

---

3. Effect of Phase Gate

Consider:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Applying phase gate:

$$P(\theta)|\psi\rangle$$

Result:

$$= \alpha|0\rangle + \beta e^{i\theta}|1\rangle$$

The phase of |1⟩ changes, but measurement probabilities remain the same.

---

4. Types of Phase Gates

The important phase gates are:

1. S Gate
2. T Gate

---

5. S Gate (Phase Shift Gate)

Introduction

The S gate is also called:

- Phase gate
- √Z gate

because:

$$S^2 = Z$$

It introduces a phase shift of:

$$90^\circ$$

or:

$$\frac{\pi}{2}$$

---

6. S Gate Matrix Representation

The matrix of S gate is:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

where:

$$i = \sqrt{-1}$$

---

7. Operation of S Gate on $|0\rangle$

Input:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Apply S:

$$S|0\rangle$$

Calculation:

$$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Result:

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Therefore:

$$\boxed{S|0\rangle = |0\rangle}$$

The state is unchanged.

---

8. Operation of S Gate on $|1\rangle$

Input:

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Apply S:

$$S|1\rangle$$

Calculation:

$$= \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Result:

$$= \begin{bmatrix} 0 \\ i \end{bmatrix}$$

Therefore:

$$\boxed{S|1\rangle = i|1\rangle}$$

---

9. S Gate Effect on General Qubit

For:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After applying S:

$$S|\psi\rangle$$

Result:

$$\boxed{\alpha|0\rangle + i\beta|1\rangle}$$

---

10. T Gate (π/8 Gate)

Introduction

The T gate is a phase rotation gate.

It introduces a phase shift of:

$$45^\circ$$

or:

$$\frac{\pi}{4}$$

It is also called:

$$\pi/8 \text{ gate}$$

because its rotation angle is half of the S gate.

---

11. T Gate Matrix Representation

The matrix is:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Using Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

For:

$$\theta = \frac{\pi}{4}$$

$$e^{i\pi/4} = \frac{1+i}{\sqrt{2}}$$

Therefore:

$$\boxed{T = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix}}$$

---

12. Operation of T Gate

On $|0\rangle$

$$T|0\rangle = |0\rangle$$

No change.

On $|1\rangle$

$$T|1\rangle = e^{i\pi/4}|1\rangle$$

The phase changes by:

$$45^\circ$$

---

13. Relation Between Z, S, and T Gates

These gates are connected:

S Gate

$$S^2 = Z$$

Meaning:

Two S gates equal one Z gate.

T Gate

$$T^2 = S$$

Meaning:

Two T gates equal one S gate.

Therefore:

$$T^4 = Z$$

and:

$$T^8 = I$$

---

14. Difference Between Bit Flip and Phase Flip

Bit Flip

Changes the state:

$$|0\rangle \leftrightarrow |1\rangle$$

Example:

X Gate

$$X|0\rangle = |1\rangle$$

Phase Flip

Changes only phase:

$$|1\rangle \rightarrow -|1\rangle$$

Example:

Z Gate

$$Z|1\rangle = -|1\rangle$$

15. Comparison: X, Z, S, T Gates

Gate	Type	Operation	Matrix
X	Bit Flip	(	0\rangle \leftrightarrow
Z	Phase Flip	(	1\rangle\rightarrow-
S	Phase Rotation	90° phase shift	$\begin{bmatrix}1 & 0\\0 & i\end{bmatrix}$
T	Phase Rotation	45° phase shift	$\begin{bmatrix}1 & 0\\0 & e^{i\pi/4}\end{bmatrix}$

16. Bloch Sphere Interpretation

Phase gates rotate the qubit around the **Z-axis**.

S Gate:

Rotation:

$$\frac{\pi}{2}$$

around Z-axis.

T Gate:

Rotation:

$$\frac{\pi}{4}$$

around Z-axis.

17. Importance of Phase Gates

Phase gates are used in:

1. Quantum Fourier Transform

Phase rotations are essential components of QFT.

2. Quantum Algorithms

Used in:

- Shor's algorithm
- Grover's algorithm
- Quantum simulations

3. Quantum Error Correction

Phase correction requires phase gates.

4. Quantum Interference

Phase changes determine whether quantum states:

- Reinforce each other
- Cancel each other

18. Advantages

- Control quantum phase.
- Enable quantum interference.
- Essential for universal quantum computation.
- Used to construct complex quantum circuits.

19. Limitations

- Do not directly create superposition.
- Only modify phase information.
- Require accurate control to avoid errors.

---

20. Frequently Asked University Questions

2-Mark Questions

1. What is a phase gate?
2. Write the matrix of S gate.
3. Write the matrix of T gate.
4. What is the phase shift produced by S gate?

5-Mark Questions

1. Explain S gate with matrix representation.
2. Explain T gate and its operation.
3. Differentiate bit flip and phase flip.

10-Mark Questions

1. Explain phase gates S and T with mathematical representation.
2. Describe the importance of phase gates in quantum computing.
3. Compare Pauli gates and phase gates.

Formula Box (Important)

General Phase Gate

$$P(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}$$

S Gate

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

T Gate

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Relations

$$S^2 = Z$$

$$T^2 = S$$

$$T^4 = Z$$

Quick Revision

- Phase gates modify the **phase** of a qubit.
- They do not change measurement probabilities.
- **S gate** $\rightarrow 90^\circ$ phase rotation.
- **T gate** $\rightarrow 45^\circ$ phase rotation.
- Both rotate qubits around the **Z-axis of the Bloch sphere**.
- They are important for quantum interference and quantum algorithms.

Next Topic

Rotation Gates (Rx, Ry, Rz)

We will study:

- General rotation gates
- Rotation matrices
- X-axis, Y-axis, Z-axis rotations
- Bloch sphere interpretation
- Difference between Pauli gates and rotation gates
- Applications in quantum circuits

Rotation gates are important because they provide **continuous control over qubit states**, unlike fixed gates such as X, Y, Z, S, and T.

> ok Next

UNIT–II: Quantum Gates and Circuits

Topic 4 – Rotation Gates (Rx, Ry, Rz)

Weightage: ⭐⭐⭐⭐⭐ (Very Important)
Rotation gates are advanced single-qubit quantum gates that allow **continuous rotation of a qubit state** around the axes of the Bloch sphere. They are essential for designing flexible quantum circuits and implementing quantum algorithms.

---

1. Introduction to Rotation Gates

In previous topics, we studied fixed quantum gates:

- X gate → 180° rotation around X-axis
- Y gate → 180° rotation around Y-axis
- Z gate → 180° rotation around Z-axis

However, sometimes we need a rotation by an arbitrary angle.

For example:

- 30°
- 45°
- 90°
- 120°

Rotation gates provide this flexibility.

---

2. Definition

Exam Definition:

> **Rotation gates are single-qubit quantum gates that rotate the state vector of a qubit by a specified angle around the X, Y, or Z axis of the Bloch sphere.**

The three fundamental rotation gates are:

1. Rx gate
2. Ry gate
3. Rz gate

---

3. General Rotation Gate Concept

A rotation operation is represented as:

$$R(\theta)$$

where:

- R = rotation operator

- θ = angle of rotation

The rotation depends on the axis:

- X-axis → $R_x(\theta)$
- Y-axis → $R_y(\theta)$
- Z-axis → $R_z(\theta)$

---

4. Mathematical Representation

Rotation gates are derived from Pauli matrices.

General form:

$$R_n(\theta) = e^{-i\frac{\theta}{2}\sigma_n}$$

where:

- n represents the rotation axis.
- σ_n is the corresponding Pauli matrix.

---

5. Rx Gate (Rotation Around X-axis)

Definition

The Rx gate rotates a qubit state around the **X-axis** of the Bloch sphere.

Matrix Representation

$$R_x(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -i\sin(\frac{\theta}{2}) \\ -i\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix}$$

---

6. Operation of Rx Gate

Consider a qubit:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Applying Rx:

$$R_x(\theta)|\psi\rangle$$

The qubit rotates by angle:

$$\theta$$

around the X-axis.

---

7. Special Cases of Rx Gate

When:

$$\theta = \pi$$

Then:

$$R_x(\pi) = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}$$

This is related to the Pauli-X gate:

$$R_x(\pi) = -iX$$

The global phase $-i$ does not affect measurement results.

When:

$$\theta = 0$$

$$R_x(0) = I$$

No rotation occurs.

---

8. Ry Gate (Rotation Around Y-axis)

Definition

The Ry gate rotates a qubit around the **Y-axis** of the Bloch sphere.

Matrix Representation

$$R_y(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix}$$

---

9. Operation of Ry Gate

For:

$$|\psi\rangle$$

Applying:

$$R_y(\theta)|\psi\rangle$$

rotates the qubit state by angle:

$$\theta$$

around Y-axis.

---

10. Special Case of Ry Gate

When:

$$\theta = \pi$$

$$R_y(\pi) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

It is related to Pauli-Y:

$$R_y(\pi) = -iY$$

---

11. Rz Gate (Rotation Around Z-axis)

Definition

The Rz gate rotates a qubit around the **Z-axis** of the Bloch sphere.

Matrix Representation

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

---

12. Operation of Rz Gate

For a qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After Rz:

$$R_z(\theta)|\psi\rangle$$

Result:

$$= \alpha e^{-i\theta/2}|0\rangle + \beta e^{i\theta/2}|1\rangle$$

The relative phase between the two states changes.

---

13. Special Case of Rz Gate

When:

$$\theta = \pi$$

$$R_z(\pi) = -iZ$$

So Rz is related to the Pauli-Z gate.

---

14. Circuit Symbols

Rx Gate

```
|ψ⟩ ──Rx(θ)──
```

Ry Gate

```
|ψ⟩ ──Ry(θ)──
```

Rz Gate

```
|ψ⟩ ──Rz(θ)──
```

15. Bloch Sphere Interpretation

A qubit can be represented as a point on the Bloch sphere:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

Rotation gates move this point.

Rx Rotation

Rotation around:

$$X - axis$$

Ry Rotation

Rotation around:

$$Y - axis$$

Rz Rotation

Rotation around:

$$Z - axis$$

16. Relationship Between Rotation and Pauli Gates

Rotation gates generalize Pauli gates.

X Gate

$$R_x(\pi) = -iX$$

Y Gate

$$R_y(\pi) = -iY$$

Z Gate

$$R_z(\pi) = -iZ$$

The difference is that rotation gates allow **any angle**, while Pauli gates are fixed rotations of 180°.

17. Difference Between Phase Gates and Rotation Gates

Phase Gates	Rotation Gates
Fixed phase changes	Continuous rotations
Mainly rotate around Z-axis	Rotate around X, Y, Z axes

Phase Gates	Rotation Gates
Examples: S, T	Examples: Rx, Ry, Rz
Limited control	Complete qubit control

18. Importance of Rotation Gates

Rotation gates are used in:

1. Variational Quantum Algorithms

Examples:

- VQE
- QAOA

They use adjustable rotation angles to optimize solutions.

2. Quantum Machine Learning

Quantum neural networks use rotation gates as trainable parameters.

3. Quantum Circuit Optimization

They help create efficient quantum circuits.

4. Quantum Simulations

Used to simulate physical systems.

19. Universal Quantum Computation

A combination of:

- Single-qubit rotation gates
- Entangling gates (such as CNOT)

can create any quantum operation.

This property is called:

Universal Quantum Gate Set

Example:

$$\{Rx, Ry, Rz, CNOT\}$$

forms a universal gate set.

---

20. Advantages

- Provide precise control over qubit states.
- Allow arbitrary quantum transformations.
- Essential for quantum algorithms.
- Used in quantum machine learning.

---

21. Limitations

- Require accurate control of rotation angles.
- Sensitive to noise.
- Small errors can affect quantum computation.

---

22. Frequently Asked University Questions

2-Mark Questions

1. What are rotation gates?
2. Name the three rotation gates.
3. Write the matrix of Rx gate.
4. Around which axis does Rz rotate?

5-Mark Questions

1. Explain Rx, Ry, and Rz gates.
2. Describe the Bloch sphere interpretation of rotation gates.
3. Differentiate between Pauli gates and rotation gates.

10-Mark Questions

1. Explain rotation gates with mathematical representation.
2. Describe Rx, Ry, and Rz gates and their applications.
3. Explain the role of rotation gates in quantum algorithms.

Formula Box (Very Important)

Rx Gate

$$R_x(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$$

Ry Gate

$$R_y(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$$

Rz Gate

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

Quick Revision

- Rotation gates rotate qubits by arbitrary angles.
- Three main gates:
  - **Rx → X-axis rotation**
  - **Ry → Y-axis rotation**
  - **Rz → Z-axis rotation**
- They are generalized forms of Pauli gates.
- They are widely used in:
  - VQE
  - QAOA
  - Quantum Machine Learning
  - Quantum simulations
- Combination of rotation gates and CNOT provides universal quantum computation.

Next Topic

Multi-Qubit Gates

We will study:

1. **CNOT Gate**
    - Controlled operations
    - Matrix representation
    - Creating entanglement
2. **Toffoli Gate**
    - Three-qubit controlled-controlled-NOT gate
    - Applications
3. **SWAP Gate**
    - Exchange of qubit states
    - Circuit representation

These are very important for **quantum circuits and entanglement**, and frequently appear in semester exams.

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 5 – Multi-Qubit Gates (CNOT, Toffoli, SWAP Gates)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> Multi-qubit gates are quantum gates that operate on **two or more qubits simultaneously**. They are essential because they allow quantum computers to create **entanglement**, perform controlled operations, and implement complex quantum algorithms.

---

1. Introduction to Multi-Qubit Gates

A single qubit can be manipulated using:

- X gate
- Y gate
- Z gate
- Hadamard gate
- Rotation gates

However, quantum computation requires interactions between qubits.

For example:

- Creating entanglement

- Conditional operations
- Quantum communication

These operations are performed using **multi-qubit gates**.

---

2. Definition

Exam Definition:

> **Multi-qubit quantum gates are quantum operations that act on two or more qubits and create correlations between them by performing controlled transformations.**

---

3. Multi-Qubit State Representation

For two qubits, the combined state is represented using tensor product:

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$$

The computational basis states are:

$$|00\rangle$$

$$|01\rangle$$

$$|10\rangle$$

$$|11\rangle$$

---

4. Important Multi-Qubit Gates

The important multi-qubit gates are:

1. CNOT Gate
2. Toffoli Gate
3. SWAP Gate

PART A: CNOT Gate (Controlled-NOT Gate)

---

5. Introduction to CNOT Gate

The CNOT gate is one of the most important quantum gates.

It has:

- Two input qubits

- One control qubit
- One target qubit

The target qubit changes only when the control qubit is:

$$|1\rangle$$

---

6. Definition

Exam Definition:

> **The Controlled-NOT (CNOT) gate is a two-qubit quantum gate that applies an X (NOT) operation on the target qubit only when the control qubit is in the state $|1\rangle$.**

---

7. CNOT Gate Symbol

Circuit representation:

```
Control Qubit

|A) ──●──
       |
      X
       |
|B) ──⊕──

Target Qubit
```

The upper qubit is the control.

The lower qubit is the target.

---

8. Working of CNOT Gate

CNOT follows this rule:

Control	Target Before	Target After
0	0	0
0	1	1
1	0	1
1	1	0

When control = 1:

$$X$$

operation is applied.

---

9. Truth Table of CNOT

Input	Output
	00⟩
	01⟩
	10⟩
	11⟩

10. Matrix Representation of CNOT

The CNOT matrix is:

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The matrix operates on a two-qubit system.

11. Example of CNOT Operation

Input:

$$|10\rangle$$

Here:

First qubit:

$$|1\rangle$$

(Control)

Second qubit:

$$|0\rangle$$

(Target)

Since control = 1:

Target flips:

$$0 \to 1$$

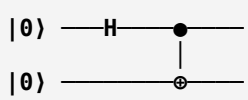
Output:

$$\boxed{|11\rangle}$$

12. Creating Entanglement Using CNOT

CNOT is commonly combined with Hadamard.

Circuit:



Step 1:

Initial:

$$|00\rangle$$

After Hadamard:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

After CNOT:

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

This creates a Bell state.

---

13. Applications of CNOT Gate

CNOT is used in:

1. Quantum Entanglement

Creates Bell states.

---

2. Quantum Teleportation

Essential gate in teleportation circuits.

---

3. Quantum Error Correction

Used to encode and protect quantum information.

---

4. Universal Quantum Computation

CNOT + single-qubit gates can implement any quantum operation.

PART B: Toffoli Gate (CCNOT Gate)

14. Introduction to Toffoli Gate

The Toffoli gate is a three-qubit gate.

It has:

- Two control qubits
- One target qubit

It is also called:

$$CCNOT$$

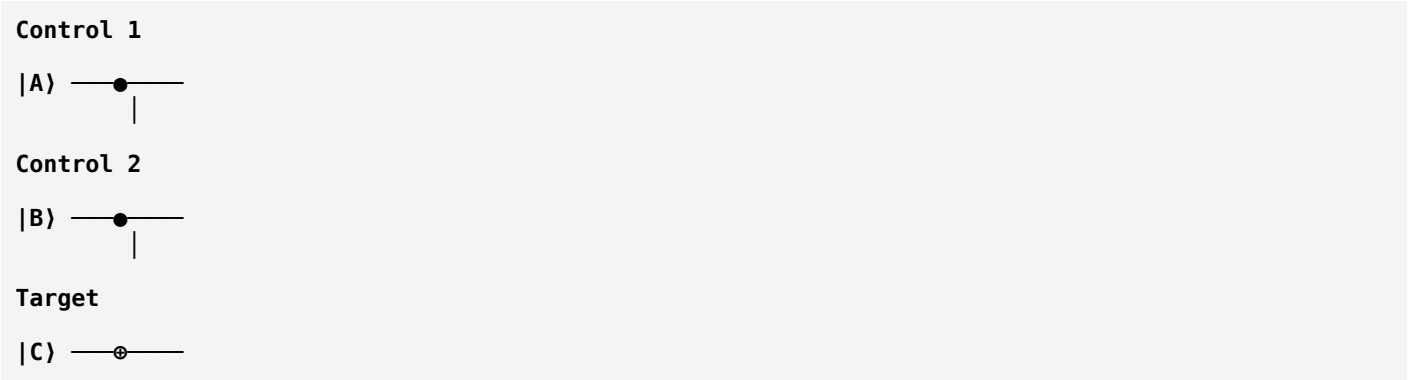
(Controlled-Controlled-NOT)

15. Definition

Exam Definition:

> **The Toffoli gate is a three-qubit quantum gate that applies an X operation to the target qubit only when both control qubits are in the state |1⟩.**

16. Toffoli Gate Symbol



17. Working of Toffoli Gate

Rule:

Target flips only when:

$$A = 1$$

and

$$B = 1$$

18. Truth Table of Toffoli Gate

A	B	C	Output
0	0	0	000
0	0	1	001
0	1	0	010
0	1	1	011
1	0	0	100
1	0	1	101
1	1	0	111
1	1	1	110

---

19. Matrix Representation

The Toffoli gate is an 8×8 matrix because it operates on three qubits.

It behaves like identity for all states except:

$$|110\rangle$$

and

$$|111\rangle$$

These two states are swapped.

---

20. Applications of Toffoli Gate

Used in:

- Quantum arithmetic
- Quantum error correction
- Reversible computation
- Quantum algorithms

PART C: SWAP Gate

---

21. Introduction to SWAP Gate

The SWAP gate exchanges the states of two qubits.

Example:

Input:

$$|01\rangle$$

Output:

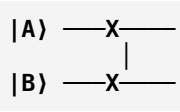
$$|10\rangle$$

22. Definition

Exam Definition:

The SWAP gate is a two-qubit quantum gate that exchanges the states of two qubits.

23. SWAP Gate Symbol



24. Working of SWAP Gate

Input	Output
00	00
01	10
10	01
11	11

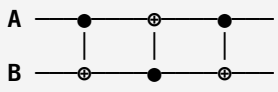
25. Matrix Representation

$$SWAP = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

26. Decomposition of SWAP Gate

A SWAP gate can be created using three CNOT gates:

Circuit:



Therefore:

$$SWAP = CNOT_{12}CNOT_{21}CNOT_{12}$$

27. Comparison of Multi-Qubit Gates

Gate	Number of Qubits	Function
CNOT	2	Controlled bit flip
Toffoli	3	Two-controlled bit flip
SWAP	2	Exchange qubit states

28. Importance of Multi-Qubit Gates

They enable:

- Quantum entanglement
- Quantum communication
- Quantum algorithms
- Quantum error correction
- Reversible computation

29. Frequently Asked University Questions

2-Mark Questions

1. Define CNOT gate.
2. What is a Toffoli gate?
3. What is the function of SWAP gate?
4. Which gate is used to create entanglement?

5-Mark Questions

1. Explain CNOT gate with truth table.
2. Describe Toffoli gate operation.
3. Explain SWAP gate with matrix representation.

10-Mark Questions

1. Explain CNOT, Toffoli, and SWAP gates with diagrams and applications.
2. Describe the role of multi-qubit gates in quantum computing.
3. Explain how CNOT creates entanglement.

Formula Box (Important)

CNOT Matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

SWAP Matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

SWAP Decomposition

$$SWAP = CNOT_{12}CNOT_{21}CNOT_{12}$$

Quick Revision

- Multi-qubit gates operate on two or more qubits.
- **CNOT**:
  - One control + one target
  - Creates entanglement
- **Toffoli**:
  - Two controls + one target
  - Used in quantum arithmetic
- **SWAP**:
  - Exchanges two qubit states
  - Can be built using three CNOT gates

Next Topic

Quantum Circuits and Reversible Computation

We will study:

- What is a quantum circuit?
- Circuit representation
- Circuit construction
- Reversible computation
- Difference between classical and quantum circuits
- Importance of reversibility in quantum computing

This topic is highly important because it connects **quantum gates into complete quantum algorithms**.

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 6 – Quantum Circuits and Reversible Computation

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> Quantum circuits are the **framework used to execute quantum algorithms**. They combine different quantum gates to manipulate qubits and perform computations.
> Reversible computation is a fundamental requirement because **all quantum operations must be reversible**.

---

1. Introduction to Quantum Circuits

A quantum computer does not execute instructions like a classical computer.

Instead, it performs a sequence of:

- Qubit initialization
- Quantum gate operations
- Measurements

This sequence is represented as a **quantum circuit**.

---

2. Definition of Quantum Circuit

Exam Definition:

A quantum circuit is a sequence of quantum gates applied to one or more qubits to transform an initial quantum state into a final state, followed by measurement.

---

3. Structure of a Quantum Circuit

A quantum circuit consists of:

1. Input qubits
2. Quantum gates
3. Measurement operations
4. Classical output

General structure:

```
Input Qubits
      |
      ▼
Quantum Gates
      |
      ▼
Quantum State Transformation
      |
      ▼
Measurement
      |
      ▼
Classical Output
```

4. Representation of Quantum Circuits

Quantum circuits are represented using:

- Horizontal lines → qubits
- Boxes → single-qubit gates
- Connected dots → controlled operations
- Measurement symbol → obtaining output

Example:

```
|0) ──H────●──M
           |
|0) ───────X──M
```

This circuit:

1. Creates superposition using H gate.
2. Creates entanglement using CNOT.
3. Measures the qubits.

---

5. Basic Components of Quantum Circuits

1. Qubit Lines

Each horizontal line represents one qubit.

Example:

```
|q0⟩ ──────────
|q1⟩ ──────────
```

2. Quantum Gates

Operations applied to qubits.

Examples:

Single-qubit gates:

- X
- Y
- Z
- H
- S
- T
- Rx, Ry, Rz

Multi-qubit gates:

- CNOT
- Toffoli
- SWAP

---

3. Measurement

Measurement converts quantum information into classical information.

Symbol:

```
|ψ⟩ ──M──
```

After measurement:

$$|\psi\rangle$$

collapses into:

$$|0\rangle$$

or

$$|1\rangle$$

---

6. Quantum Circuit Example: Superposition Circuit

Circuit:

```
|0⟩ ──H──M
```

Step 1:

Initial state:

$$|0\rangle$$

Step 2:

Apply Hadamard:

$$H|0\rangle$$

Output:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Step 3:

Measurement:

Results:

$$0 \text{ with probability } 50\%$$

$$1 \text{ with probability } 50\%$$

---

7. Quantum Circuit Example: Entanglement Circuit

Circuit:

```
|0⟩ ──H──●──
         |
|0⟩ ─────X──
```

Step 1:

Initial state:

$$|00\rangle$$

Step 2:

After Hadamard:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

Step 3:

After CNOT:

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

A Bell state is created.

8. Circuit Depth

The **depth** of a quantum circuit represents the number of sequential layers of operations.

Example:

```
Layer 1     Layer 2     Layer 3
 H            X           M
```

Circuit depth affects:

- Execution time
- Error rate
- Noise accumulation

Lower depth circuits are preferred.

9. Quantum Circuit Compilation

A quantum algorithm may not directly match hardware capabilities.

Compilation converts:

```
High-Level Quantum Program
          ↓
Optimized Quantum Circuit
```

```
          ↓
Hardware Instructions
```

The compiler performs:

- Gate optimization
- Mapping qubits
- Reducing circuit depth

---

10. Reversible Computation

Introduction

In classical computing, many operations are irreversible.

Example:

AND gate:

Input:

$$(0, 1)$$

Output:

$$0$$

The original input cannot be uniquely determined.

Definition

Exam Definition:

> **Reversible computation is a computation in which every output state uniquely determines the input state. The process can be reversed without loss of information.**

---

11. Why is Reversibility Important in Quantum Computing?

Quantum mechanics follows reversible evolution.

Quantum operations must satisfy:

$$U^\dagger U = I$$

Therefore:

- Quantum gates must be reversible.

- Information cannot be destroyed.

12. Classical Irreversible vs Quantum Reversible Computation

Classical Computation	Quantum Computation
Many operations are irreversible	All operations are reversible
Information can be lost	Information is preserved
Uses Boolean gates	Uses unitary gates
Example: AND, OR	Example: CNOT, Toffoli

13. Reversible Logic Gates

Important reversible gates:

1. NOT Gate
2. CNOT Gate
3. Toffoli Gate
4. Fredkin Gate

14. NOT Gate as Reversible Gate

Classical NOT:

$$0 \rightarrow 1$$

$$1 \rightarrow 0$$

The inverse of NOT is itself.

Therefore:

$$NOT^{-1} = NOT$$

15. CNOT as Reversible Gate

CNOT:

Input:

$$|10\rangle$$

Output:

$$|11\rangle$$

Applying CNOT again:

$$|11\rangle$$

returns:

$$|10\rangle$$

Therefore:

$$CNOT^2 = I$$

---

16. Toffoli Gate and Reversible Computation

The Toffoli gate is important because:

- It is reversible.
- It can implement classical reversible logic.
- It is universal for classical computation.

---

17. Landauer's Principle

An important concept related to reversible computing.

Landauer's Principle:

> Erasing one bit of information releases a minimum amount of heat into the environment.

Energy cost:

$$E = kT \ln 2$$

where:

- k = Boltzmann constant
- T = temperature

---

18. Advantages of Reversible Computation

1. Reduced Energy Loss

Less information destruction means less heat generation.

---

2. Required for Quantum Computing

Quantum mechanics requires reversible transformations.

---

3. Efficient Quantum Algorithms

Allows complex quantum operations.

---

19. Limitations of Reversible Computation

- More complex circuit design.
- Requires additional qubits.
- Difficult hardware implementation.
- More operations may increase error probability.

---

20. Applications of Quantum Circuits

Quantum circuits are used in:

1. Quantum Algorithms

Examples:

- Grover's algorithm
- Shor's algorithm

---

2. Quantum Simulation

Used for:

- Chemistry
- Physics
- Material science

---

3. Quantum Machine Learning

Used for:

- Quantum classifiers
- Quantum neural networks

---

4. Quantum Communication

Used in:

- Teleportation
- Cryptography

21. Frequently Asked University Questions

2-Mark Questions

1. Define a quantum circuit.
2. Why are quantum gates reversible?
3. What is circuit depth?
4. Define reversible computation.

5-Mark Questions

1. Explain the structure of a quantum circuit.
2. Describe reversible computation.
3. Explain the importance of CNOT and Toffoli gates.

10-Mark Questions

1. Explain quantum circuits with suitable examples.
2. Describe reversible computation and its importance in quantum computing.
3. Compare classical and quantum computation in terms of reversibility.

Formula Box (Important)

Quantum Evolution

$$|\psi'\rangle = U|\psi\rangle$$

Unitary Condition

$$U^\dagger U = I$$

Reversible Operation

$$U^{-1} = U^\dagger$$

Landauer Principle

$$E = kT \ln 2$$

Quick Revision

- A **quantum circuit** is a sequence of quantum gates acting on qubits.
- Circuit components:
  - Qubits
  - Gates
  - Measurement
- Quantum circuits must be reversible.
- Reversibility means:
  - Output uniquely determines input.
  - No information loss.
- Important reversible gates:
  - NOT
  - CNOT
  - Toffoli
- Quantum operations are represented by unitary matrices.

Next Topic

Quantum Phenomena in Computing

We will study:

1. **Quantum Interference**
2. **Quantum Parallelism**
3. **Quantum Noise and Error Basics**

These concepts explain **why quantum computers can outperform classical computers** and why building reliable quantum machines is challenging.

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 7 – Quantum Phenomena: Interference, Parallelism, Noise and Error Basics

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> Quantum phenomena are the fundamental principles that make quantum computers different from classical computers. The three important phenomena are:

1. Quantum Interference
2. Quantum Parallelism
3. Quantum Noise and Errors

PART A: Quantum Interference

---

1. Introduction to Quantum Interference

In classical computing, signals combine according to normal rules.

In quantum computing, quantum states combine through **interference**.

Quantum interference allows quantum computers to:

- Increase the probability of correct answers.
- Decrease the probability of incorrect answers.

---

2. Definition

Exam Definition:

> **Quantum interference is a phenomenon in which quantum probability amplitudes combine with each other, either increasing or decreasing the probability of quantum states.**

---

3. Understanding Probability Amplitudes

A quantum state is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where:

- α = amplitude of state $|0\rangle$
- β = amplitude of state $|1\rangle$

Measurement probabilities are:

$$P(0) = |\alpha|^2$$

$$P(1) = |\beta|^2$$

The amplitudes can interfere with each other.

---

4. Types of Quantum Interference

There are two types:

1. Constructive Interference
2. Destructive Interference

---

5. Constructive Interference

Definition:

When amplitudes combine in the same direction, they increase the probability of a state.

Example:

$$+\alpha + +\alpha = 2\alpha$$

Result:

Higher probability.

Example:

Two waves:

```
Wave 1:    /\      /\
          /  \    /  \

Wave 2:   /\      /\
         /  \    /  \

Combined:
          /\      /\
         /  \    /  \
        /    \  /    \
```

The amplitude increases.

---

6. Destructive Interference

Definition:

When amplitudes have opposite phases, they cancel each other.

Example:

$$+\alpha - \alpha = 0$$

Result:

Lower probability.

---

7. Role of Interference in Quantum Algorithms

Quantum algorithms are designed to:

- Amplify desired states.
- Cancel unwanted states.

Examples:

Grover's Algorithm

Uses amplitude amplification:

- Correct answer probability increases.
- Wrong answers decrease.

Quantum Fourier Transform

Uses interference patterns to extract information.

---

8. Example of Interference Using Hadamard Gate

Circuit:

```
|0⟩ ──H──H──
```

First H:

$$|0\rangle$$

becomes:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Second H:

Returns:

$$|0\rangle$$

The amplitudes interfere constructively to restore the original state.

PART B: Quantum Parallelism

---

9. Introduction to Quantum Parallelism

Quantum computers use superposition to process multiple possibilities simultaneously.

A classical computer processes:

```
One state at a time
```

A quantum computer can represent:

```
Multiple states simultaneously
```

10. Definition

Exam Definition:

> **Quantum parallelism is the ability of quantum computers to process multiple quantum states simultaneously by using superposition.**

---

11. Classical vs Quantum Parallelism

Classical Computer

A 2-bit system:

Possible states:

$$00, 01, 10, 11$$

But it stores only one state at a time.

Example:

```
00
```

Quantum Computer

Two qubits can exist as:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

All four states exist simultaneously.

---

12. N-Qubit Parallelism

For:

$$n$$

qubits:

Number of possible states:

$$2^n$$

Example:

1 qubit:

$$2^1 = 2$$

states.

---

10 qubits:

$$2^{10} = 1024$$

states.

---

50 qubits:

$$2^{50}$$

possible states.

---

13. Important Point About Quantum Parallelism

Quantum parallelism does NOT mean:

> "A quantum computer directly gives all answers at once."

Because measurement collapses the state.

The power comes from:

1. Superposition
2. Interference
3. Measurement strategy

---

14. Applications of Quantum Parallelism

Used in:

1. Quantum Search

Example:

Grover's algorithm.

2. Quantum Simulation

Simulating molecules and materials.

3. Optimization Problems

Finding best solutions.

4. Machine Learning

Processing high-dimensional data.

PART C: Quantum Noise and Errors

15. Introduction to Quantum Noise

Quantum systems are extremely sensitive.

Small environmental interactions can disturb qubit states.

This disturbance is called:

$$\textbf{Quantum Noise}$$

16. Definition

Exam Definition:

> **Quantum noise is unwanted disturbance in a quantum system caused by interaction with the environment, resulting in errors in quantum computation.**

17. Sources of Quantum Noise

Important sources:

1. Decoherence

The loss of quantum information due to interaction with surroundings.

Example:

A qubit loses its superposition state.

2. Thermal Noise

Heat affects quantum hardware.

3. Electromagnetic Noise

External electromagnetic fields disturb qubits.

4. Gate Errors

Quantum gates may not perform perfectly.

Example:

Required rotation:

$$90^\circ$$

Actual rotation:

$$89.9^\circ$$

5. Measurement Errors

Incorrect reading of qubit states.

18. Types of Quantum Errors

The main quantum errors are:

1. Bit Flip Error

Equivalent to classical NOT operation.

State changes:

$$|0\rangle \rightarrow |1\rangle$$

or:

$$|1\rangle \rightarrow |0\rangle$$

Represented by:

$$X$$

---

2. Phase Flip Error

Phase changes:

$$|0\rangle + |1\rangle$$

becomes:

$$|0\rangle - |1\rangle$$

Represented by:

$$Z$$

---

3. Bit-Phase Flip Error

Combination of:

- Bit flip
- Phase flip

Represented by:

$$Y$$

---

19. Decoherence

Definition:

> **Decoherence is the process by which a quantum system loses its quantum properties due to interaction with the environment.**

Before decoherence:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

After decoherence:

Classical mixture:

$$50\%|0\rangle$$

and

$$50\%|1\rangle$$

20. Quantum Error Correction

Because quantum errors cannot be avoided completely, quantum computers use:

$$\textbf{Quantum Error Correction (QEC)}$$

21. Principle of Quantum Error Correction

Quantum error correction:

- Detects errors.
- Corrects errors.
- Protects quantum information.

22. Quantum Error Correction Techniques

Examples:

1. Bit Flip Code

Protects against:

$$X$$

errors.

2. Phase Flip Code

Protects against:

$$Z$$

errors.

3. Shor Code

Protects against:

- Bit errors
- Phase errors

4. Surface Code

Most promising method for fault-tolerant quantum computing.

---

23. Noise Reduction Techniques

Methods used:

1. Better Hardware Design

- Improved qubit isolation.
- Better cooling systems.

---

2. Error Correction

Detect and fix errors.

---

3. Error Mitigation

Reduce impact without full correction.

---

4. Fault-Tolerant Quantum Computing

Build reliable quantum computers despite errors.

---

24. Comparison: Classical vs Quantum Computing

Feature	Classical	Quantum
Basic unit	Bit	Qubit
State	0 or 1	Superposition
Parallelism	Limited	Quantum parallelism
Errors	Bit errors	Bit + phase errors
Correction	Error correction	Quantum error correction

---

25. Importance of Quantum Phenomena

Quantum phenomena enable:

- Faster algorithms
- Secure communication
- Quantum simulations

- Optimization
- Machine learning applications

---

26. Frequently Asked University Questions

2-Mark Questions

1. Define quantum interference.
2. What is quantum parallelism?
3. Define quantum noise.
4. What is decoherence?
5. List quantum errors.

5-Mark Questions

1. Explain constructive and destructive interference.
2. Describe quantum parallelism with examples.
3. Explain sources of quantum noise.

10-Mark Questions

1. Explain quantum interference and its role in quantum algorithms.
2. Describe quantum parallelism and its applications.
3. Explain quantum noise, errors, and error correction techniques.

Formula Box (Important)

Number of States for n Qubits:

$$2^n$$

Quantum State:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Probability:

$$P = |\alpha|^2$$

Quick Revision

- **Quantum interference** → Controls probability using amplitude addition/cancellation.
- **Quantum parallelism** → Processes multiple states using superposition.
- **Quantum noise** → Disturbs quantum states.
- Main quantum errors:
  - Bit flip (X)
  - Phase flip (Z)
  - Bit-phase flip (Y)
- Decoherence is the biggest challenge in quantum computing.
- Quantum error correction is required for reliable quantum computers.

UNIT–II Completed ✅

Next: UNIT–III – Quantum Search Algorithms

First Topic:

Unstructured Search Problem

We will study:

- Classical search problem
- Why quantum search is needed
- Search complexity
- Database search model
- Introduction to Grover's Algorithm

This unit is highly important because **Grover's algorithm is one of the most famous quantum algorithms**.

> ok next

UNIT–III: Quantum Search Algorithms

Topic 1 – Unstructured Search Problem

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important)
> The unstructured search problem is the foundation for understanding **Grover's Algorithm**,

one of the most important quantum algorithms.
It explains how quantum computers can search faster than classical computers.

---

1. Introduction to Search Problems

Searching is one of the most common computational tasks.

Examples:

- Finding a name in a database
- Searching an item in a list
- Finding a solution among many possibilities
- Password searching
- Optimization problems

A search problem consists of:

- A set of possible solutions
- A method to check whether a solution is correct

---

2. Types of Search Problems

Search problems are mainly classified into:

1. Structured Search

The data has an organized pattern.

Examples:

- Sorted arrays
- Binary search trees
- Databases with indexes

Example:

Searching a number in a sorted list:

$$[2, 5, 8, 12, 20]$$

Binary search can find an element efficiently.

Complexity:

$$O(logN)$$

---

2. Unstructured Search

The data has no order or pattern.

Example:

Finding a person in an unsorted list.

$$[15, 3, 20, 8, 11]$$

No information is available about the location of the required element.

---

3. Definition of Unstructured Search Problem

Exam Definition:

> **An unstructured search problem is a problem where an item must be found from an unordered collection of N possible items without any additional information about their arrangement.**

---

4. Mathematical Representation

Suppose we have:

$$N$$

possible solutions:

$$x_0, x_1, x_2, \ldots, x_{N-1}$$

There exists a function:

$$f(x)$$

called an oracle function.

The function identifies the correct solution:

$$f(x) = \begin{cases} 1, & \text{if x is the correct solution} \\ 0, & \text{otherwise} \end{cases}$$

---

5. Classical Unstructured Search

In classical computing, the simplest method is:

Linear Search

The computer checks each item one by one.

Example:

Database:

Index	Value
0	5
1	12
2	8
3	20
4	15

Searching:

$$20$$

Steps:

1. Check 5 ❌
2. Check 12 ❌
3. Check 8 ❌
4. Check 20 ✅

---

6. Classical Search Complexity

For:

$$N$$

items:

Best Case:

The item is first:

$$O(1)$$

Worst Case:

The item is last:

$$O(N)$$

Average Case:

$$O(N/2)$$

Therefore:

$$\boxed{O(N)}$$

7. Why Classical Search is Slow?

For a very large database:

Example:

$$N = 1,000,000$$

A classical computer may need:

$$1,000,000$$

checks.

The search time increases linearly.

8. Quantum Approach to Search

Quantum computers use:

- Superposition
- Quantum interference
- Amplitude amplification

to improve searching.

Instead of checking one item at a time:

A quantum computer creates a superposition of all possibilities.

9. Quantum Representation of Search Space

Suppose:

$$N = 4$$

possible states.

The states are:

$$|00\rangle$$

$$|01\rangle$$

$$|10\rangle$$

$$|11\rangle$$

Using Hadamard gates:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

For N states:

$$|s\rangle = \frac{1}{\sqrt{N}}(|0\rangle + |1\rangle + ... + |N-1\rangle)$$

10. Quantum Search Model

Quantum search uses:

1. Oracle

A function that identifies the correct answer.

2. Amplitude Amplification

Increasing the probability of the correct answer.

3. Measurement

Obtaining the final result.

11. Oracle Concept

The oracle is a quantum operation that marks the correct solution.

It does not reveal the answer directly.

It only identifies it.

Example:

Searching:

$$x = 5$$

Oracle marks:

$$|5\rangle$$

by changing its phase.

12. Classical Search vs Quantum Search

Feature	Classical Search	Quantum Search
Data structure	Unordered list	Quantum state space
Method	Check one by one	Superposition
Technique	Linear search	Grover's algorithm
Complexity	O(N)	O(√N)

13. Need for Grover’s Algorithm

The unstructured search problem has:

$$O(N)$$

complexity classically.

Grover's algorithm reduces it to:

$$O(\sqrt{N})$$

Example:

For:

$$N = 1,000,000$$

Classical:

$$1,000,000$$

operations.

Quantum:

$$\sqrt{1,000,000} = 1000$$

operations.

Huge improvement.

14. Applications of Unstructured Search

1. Database Searching

Finding records in large databases.

2. Cryptography

Searching encryption keys.

---

3. Optimization Problems

Finding the best solution.

---

4. Artificial Intelligence

Searching solution spaces.

---

15. Limitations of Quantum Search

Although Grover's algorithm is faster:

- It provides quadratic speedup, not exponential.
- Requires quantum hardware.
- Large-scale implementation is difficult due to noise.

---

16. Important Concepts for Exams

Search Space

The complete collection of possible answers.

Example:

Password length 4:

Possible combinations:

$$10^4 = 10000$$

Target State

The correct solution state.

Oracle

Marks the target state.

Amplitude

Probability strength of a quantum state.

Amplitude Amplification

Increasing the probability of the correct state.

17. Frequently Asked University Questions

2-Mark Questions

1. Define unstructured search problem.
2. What is linear search complexity?
3. What is an oracle in quantum search?
4. What is the advantage of Grover's algorithm?

5-Mark Questions

1. Explain classical unstructured search.
2. Describe the quantum search approach.
3. Compare classical and quantum search algorithms.

10-Mark Questions

1. Explain the unstructured search problem and its limitations.
2. Describe how quantum computing improves search problems.
3. Explain the role of oracle in quantum search algorithms.

Formula Box (Important)

Classical Search Complexity:

$$O(N)$$

Grover Search Complexity:

$$O(\sqrt{N})$$

Quantum Superposition:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

Quick Revision

- Unstructured search means searching without any order.
- Classical solution:
  - Linear search
  - Complexity: **O(N)**
- Quantum solution:
  - Grover's Algorithm
  - Complexity: **O(√N)**
- Main components:
  - Oracle
  - Amplitude amplification
  - Measurement

Next Topic

Grover’s Algorithm

We will study:

1. Introduction and motivation
2. Quantum circuit of Grover's algorithm
3. Oracle operation
4. Diffusion operator
5. Amplitude amplification
6. Mathematical derivation
7. Applications and limitations

This is the **most important topic in UNIT–III** and frequently asked as a **10-mark question**.

ok next

UNIT–III: Quantum Search Algorithms

Topic 2 – Grover's Algorithm

> **Weightage:** ⭐⭐⭐⭐⭐ (Most Important Topic of UNIT–III)
> Grover's Algorithm is one of the most famous quantum algorithms that provides a **quadratic speedup** for searching an unstructured database.

---

1. Introduction to Grover's Algorithm

Searching is a fundamental computational problem.

Suppose we have:

$$N$$

unsorted items.

One item is the desired target.

Example:

A database contains:

$$1,000,000$$

records.

We need to find one specific record.

Classical Approach

Using linear search:

- Check each item one by one.
- Worst-case operations:

$$O(N)$$

Quantum Approach

Grover's algorithm searches in:

$$O(\sqrt{N})$$

operations.

This gives a quadratic speed improvement.

---

2. Definition

Exam Definition:

> **Grover's Algorithm is a quantum search algorithm that finds a marked item in an unstructured database using amplitude amplification with a complexity of O(√N), providing a quadratic speedup over classical search algorithms.**

---

3. Developed By

Grover's algorithm was proposed by:

$$\boxed{\text{Lov Grover in 1996}}$$

It is also called:

> Quantum Amplitude Amplification Algorithm

---

4. Problem Statement of Grover's Algorithm

Given:

- An unsorted database with N elements.
- One or more target elements.

Find the target element using minimum operations.

---

5. Components of Grover's Algorithm

Grover's algorithm consists of four major steps:

1. Initialization
2. Oracle Operation
3. Amplitude Amplification (Diffusion Operator)
4. Measurement

---

6. Overall Grover Circuit

Basic circuit:

```
Initial State
|0) ──H────Oracle────Diffusion────M
|0) ──H────Oracle────Diffusion────M
```

Step 1: Initialization

7. Preparing the Superposition State

Initially all qubits are:

$$|000...0\rangle$$

Hadamard gates are applied:

$$H^{\otimes n}$$

This creates an equal superposition:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

Example:

For 2 qubits:

Initial:

$$|00\rangle$$

After Hadamard:

$$\frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$$

All possible states have equal probability.

Step 2: Oracle Operation

8. What is an Oracle?

The oracle is a quantum operation that identifies the target state.

It does not directly give the answer.

It marks the correct state by changing its phase.

9. Oracle Function

Mathematically:

$$f(x)$$

where:

$$f(x) = \begin{cases} 1, & \text{target state} \\ 0, & \text{non-target state} \end{cases}$$

---

10. Oracle Phase Flip

The oracle performs:

$$|x\rangle \to (-1)^{f(x)}|x\rangle$$

For the target state:

$$|x\rangle \to -|x\rangle$$

For other states:

$$|x\rangle \to |x\rangle$$

Example:

Suppose:

Target:

$$|11\rangle$$

Before oracle:

$$\frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$$

After oracle:

$$\frac{|00\rangle + |01\rangle + |10\rangle - |11\rangle}{2}$$

The target state is marked.

Step 3: Amplitude Amplification

---

11. Need for Amplitude Amplification

After oracle operation:

The target state has negative amplitude.

However:

The probability is still small.

The diffusion operator increases the target probability.

---

12. Diffusion Operator

The diffusion operator is also called:

- Inversion about average operator
- Grover diffusion operator

It amplifies the amplitude of the target state.

---

13. Mathematical Representation

The diffusion operator:

$$D = 2|s\rangle\langle s| - I$$

where:

- |s⟩ = superposition state
- I = identity matrix

---

14. Working of Diffusion Operator

It performs two operations:

Step 1:

Invert amplitudes around zero.

Step 2:

Reflect amplitudes around their average value.

Example:

Before diffusion:

```
Target amplitude: Low
Other amplitudes: Higher
```

After diffusion:

```
Target amplitude: High
Other amplitudes: Lower
```

15. Grover Iteration

The combination of:

$$\text{Oracle}$$

and:

$$\text{Diffusion Operator}$$

is called:

$$\boxed{\text{Grover Iteration}}$$

One iteration:

$$G = D \times O$$

where:

- O = Oracle
- D = Diffusion operator

---

16. Number of Grover Iterations

The optimal number of iterations is:

$$r \approx \frac{\pi}{4}\sqrt{N}$$

where:

- N = number of elements

Example:

Database size:

$$N = 10000$$

Iterations:

$$r = \frac{\pi}{4}\sqrt{10000}$$

$$r \approx 78$$

Only around 78 searches are required.

---

17. Measurement

After enough Grover iterations:

The target state has maximum probability.

Measurement gives:

$$\boxed{\text{Desired Solution}}$$

---

18. Complete Working of Grover Algorithm

Step 1:

Initialize qubits:

$$|00...0\rangle$$

Step 2:

Apply Hadamard:

Create superposition.

$$|s\rangle$$

Step 3:

Apply Oracle:

Mark target state.

Step 4:

Apply Diffusion Operator:

Amplify target probability.

Step 5:

Repeat:

$$O(\sqrt{N})$$

times.

Step 6:

Measure:

Obtain answer.

19. Example of Grover Search

Suppose:

$$N = 4$$

States:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Target:

$$|11\rangle$$

Initial probability:

Each state:

$$25\%$$

After Grover iteration:

Target probability increases.

Measurement gives:

$$|11\rangle$$

with high probability.

20. Complexity Comparison

Algorithm	Complexity
Linear Search	$O(N)$
Binary Search	$O(logN)$ (sorted data)
Grover Search	$O(\sqrt{N})$

21. Applications of Grover's Algorithm

1. Database Search

Finding records in large databases.

---

2. Cryptography

Searching encryption keys.

Example:

Breaking symmetric encryption.

---

3. Optimization Problems

Finding minimum or maximum solutions.

---

4. Artificial Intelligence

Searching large solution spaces.

---

5. Constraint Satisfaction Problems

Finding solutions satisfying conditions.

---

22. Limitations of Grover's Algorithm

1. Requires Quantum Hardware

Needs reliable quantum computers.

---

2. Only Quadratic Speedup

It improves:

$$N$$

to:

$$\sqrt{N}$$

not exponential speedup.

---

3. Noise Sensitivity

Quantum errors affect results.

4. Large Memory Requirement

Requires quantum storage of search space.

23. Grover Algorithm vs Classical Search

Feature	Classical Search	Grover Search
Data	Unstructured database	Quantum state space
Method	Sequential checking	Amplitude amplification
Complexity	O(N)	O(√N)
Main Principle	Logic operations	Superposition + interference

24. Frequently Asked University Questions

2-Mark Questions

1. Who proposed Grover's algorithm?
2. What is the complexity of Grover's algorithm?
3. Define oracle operation.
4. What is amplitude amplification?

5-Mark Questions

1. Explain the working principle of Grover's algorithm.
2. Describe the role of oracle in Grover's algorithm.
3. Explain diffusion operator.

10-Mark Questions

1. Explain Grover’s algorithm with circuit and mathematical steps.
2. Derive the working of amplitude amplification in Grover’s algorithm.
3. Compare classical search with quantum search.

Formula Box (Very Important)

Grover Complexity

$$\boxed{O(\sqrt{N})}$$

Number of Iterations

$$\boxed{r \approx \frac{\pi}{4}\sqrt{N}}$$

Oracle Operation

$$|x\rangle \rightarrow (-1)^{f(x)}|x\rangle$$

Diffusion Operator

$$\boxed{D = 2|s\rangle\langle s| - I}$$

Quick Revision

- Grover's algorithm searches an unsorted database.
- Proposed by **Lov Grover (1996)**.
- Uses:
  - Superposition
  - Oracle
  - Amplitude amplification
  - Measurement
- Complexity:

$$O(\sqrt{N})$$

- Main idea:
  - Oracle marks the correct answer.
  - Diffusion increases its probability.

Next Topic

Grover’s Algorithm – Oracle and Amplitude Amplification (Detailed Mathematical Explanation)

We will study:

- Oracle construction
- Phase inversion
- Geometric interpretation
- Amplitude rotation
- Derivation of success probability
- Number of iterations

This topic is highly important for **10-mark semester questions**.

> ok next

UNIT–III: Quantum Search Algorithms

Topic 3 – Grover's Algorithm: Oracle and Amplitude Amplification (Detailed Explanation)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important for 10 Marks)
> This topic explains the **mathematical working mechanism** behind Grover's Algorithm.
> The two most important components are:
>
> 1. **Oracle**
> 2. **Amplitude Amplification (Diffusion Operator)**

---

1. Overview of Grover's Algorithm

Grover's algorithm solves the problem:

> Find a marked element from an unsorted database containing N elements.

It achieves this using:

$$O(\sqrt{N})$$

queries instead of:

$$O(N)$$

classical searches.

The algorithm works by repeatedly applying:

$$\boxed{G = D \cdot O}$$

where:

- O = Oracle operator

- D = Diffusion operator

PART A: Quantum Oracle

---

2. What is an Oracle?

In quantum computing, an oracle is a special quantum operation that identifies the correct solution.

It acts like a black box.

The oracle:

- Knows the solution.
- Marks the correct state.
- Does not reveal the answer directly.

---

3. Oracle Definition

Exam Definition:

> **A quantum oracle is a unitary operator that marks the correct solution state by changing its phase while leaving all other states unchanged.**

---

4. Oracle Mathematical Representation

The oracle operation is:

$$O|x\rangle = (-1)^{f(x)}|x\rangle$$

where:

$$f(x)$$

is a Boolean function.

The function:

$$f(x) = \begin{cases} 1, & \text{if x is the solution} \\ 0, & \text{otherwise} \end{cases}$$

---

5. Oracle Operation

For a non-target state:

$$f(x) = 0$$

Therefore:

$$(-1)^0 = 1$$

So:

$$|x\rangle \to |x\rangle$$

No change occurs.

For the target state:

$$f(x) = 1$$

Therefore:

$$(-1)^1 = -1$$

So:

$$|x\rangle \to -|x\rangle$$

The phase is flipped.

---

6. Example of Oracle Operation

Consider four possible states:

$$|00\rangle$$

$$|01\rangle$$

$$|10\rangle$$

$$|11\rangle$$

Suppose:

Target:

$$|11\rangle$$

Initial superposition:

$$|\psi\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$$

After oracle:

$$O|\psi\rangle = \frac{|00\rangle + |01\rangle + |10\rangle - |11\rangle}{2}$$

Only the target state changes phase.

---

7. Oracle Circuit Representation

Basic oracle circuit:

```
|x⟩ ——— Oracle ———
```

The internal design depends on the search problem.

---

8. Properties of Oracle

1. Unitary Operation

Oracle must satisfy:

$$O^\dagger O = I$$

---

2. Reversible

Applying oracle twice:

$$O^2 = I$$

returns the original state.

---

3. Phase Marking

It marks the correct state without measuring.

PART B: Amplitude Amplification

---

9. Introduction to Amplitude Amplification

After oracle operation:

The target state is marked, but its probability is still very small.

Example:

For:

$$N = 1,000,000$$

Initial probability:

$$\frac{1}{N}$$

$$= 0.000001$$

Amplitude amplification increases the target probability.

---

10. Definition

Exam Definition:

> **Amplitude amplification is a quantum technique that increases the probability amplitude of desired states while reducing the amplitude of undesired states using quantum interference.**

---

11. Diffusion Operator

Amplitude amplification is performed using the diffusion operator.

It is also called:

- Grover diffusion operator
- Inversion about average operator

---

12. Mathematical Form of Diffusion Operator

The diffusion operator is:

$$\boxed{D = 2|s\rangle\langle s| - I}$$

where:

- $|s\rangle$ = equal superposition state
- I = identity matrix

---

13. Working Principle of Diffusion Operator

The diffusion operator performs:

Step 1:

Calculate average amplitude.

Step 2:

Reflect every amplitude around the average.

Example:

Before diffusion:

State	Amplitude
Correct state	Low
Other states	Higher

After diffusion:

State	Amplitude
Correct state	Higher
Other states	Lower

---

14. Geometric Interpretation of Grover Search

Grover's algorithm can be understood as a rotation.

The quantum state rotates between:

1. Initial superposition state
2. Target state

The rotation angle is:

$$\theta$$

where:

$$\sin^2\theta = \frac{1}{N}$$

---

15. Amplitude Rotation

Each Grover iteration rotates the state by:

$$2\theta$$

After:

$$r$$

iterations:

$$(2r+1)\theta$$

rotation occurs.

---

16. Success Probability

After r iterations:

$$P_{success} = \sin^2((2r+1)\theta)$$

The maximum probability occurs when:

$$(2r+1)\theta \approx \frac{\pi}{2}$$

---

17. Optimal Number of Iterations

The number of required iterations:

$$\boxed{r \approx \frac{\pi}{4}\sqrt{N}}$$

---

18. Complete Grover Iteration

One complete iteration:

$$G = D \cdot O$$

Steps:

Step 1:

Prepare superposition:

$$|s\rangle$$

↓

Step 2:

Apply oracle:

$$O|s\rangle$$

↓

Step 3:

Apply diffusion:

$$D(O|s\rangle)$$

↓

Step 4:

Repeat:

$$\approx \frac{\pi}{4}\sqrt{N}$$

times.

---

19. Example Calculation

Suppose:

$$N = 16$$

Number of iterations:

$$r = \frac{\pi}{4}\sqrt{16}$$

$$= \frac{\pi}{4} \times 4$$

$$\approx 3$$

Only around 3 iterations are required.

---

20. Oracle vs Diffusion Operator

Oracle	Diffusion Operator
Marks target state	Amplifies target amplitude
Performs phase inversion	Performs amplitude reflection
First step	Second step
Depends on problem	General operation

---

21. Importance of Oracle and Amplitude Amplification

They enable:

- Faster database search
- Optimization problems
- Cryptographic analysis
- Machine learning search problems

22. Limitations

Oracle Construction

Creating an oracle may be difficult for complex problems.

Quantum Errors

Noise affects amplitude amplification.

Limited Speedup

Only quadratic improvement:

$$O(N) \rightarrow O(\sqrt{N})$$

23. Frequently Asked University Questions

2-Mark Questions

1. Define quantum oracle.
2. What is amplitude amplification?
3. Write diffusion operator formula.
4. What is the purpose of oracle?

5-Mark Questions

1. Explain oracle operation in Grover's algorithm.
2. Describe the diffusion operator.
3. Explain amplitude amplification.

10-Mark Questions

1. Explain the mathematical working of Grover's algorithm.
2. Describe oracle and diffusion operators with equations.

3. Derive the amplitude amplification process.

Formula Box (Very Important)

Oracle:

$$\boxed{O|x\rangle = (-1)^{f(x)}|x\rangle}$$

Diffusion Operator:

$$\boxed{D = 2|s\rangle\langle s| - I}$$

Grover Operator:

$$\boxed{G = D \cdot O}$$

Iterations:

$$\boxed{r \approx \frac{\pi}{4}\sqrt{N}}$$

Success Probability:

$$\boxed{P = \sin^2((2r+1)\theta)}$$

Quick Revision

- **Oracle** marks the correct answer by phase inversion.
- **Diffusion operator** increases the probability of the correct answer.
- One Grover iteration:

$$G = D \times O$$

- Repeating the iteration produces amplitude amplification.
- Optimal searches require:

$$\frac{\pi}{4}\sqrt{N}$$

iterations.

Next Topic

Applications of Grover's Algorithm

We will study:

1. Database search
2. Optimization problems
3. Cryptography applications
4. Quantum counting
5. Multiple target search variants

This completes the remaining concepts of **UNIT–III Quantum Search Algorithms**.

> ok next

UNIT–III: Quantum Search Algorithms

Topic 4 – Applications of Grover's Algorithm

> **Weightage:** ⭐⭐⭐⭐☆ (Important)
> Grover's Algorithm is not limited to database searching. Its main idea, **amplitude amplification**, can be applied to many problems where we need to find a solution from a large set of possibilities.

---

1. Introduction

Grover's algorithm provides a method to search an **unstructured search space** faster than classical methods.

The key improvement:

Classical search:

$$O(N)$$

Quantum search:

$$O(\sqrt{N})$$

The applications are based on converting a problem into:

1. Define possible solutions.
2. Build an oracle to identify correct solutions.
3. Use amplitude amplification.
4. Measure the final answer.

---

2. Application 1: Database Search

2.1 Introduction

One of the most direct applications of Grover's algorithm is searching large databases.

A database may contain:

$$N$$

records.

The goal is to find a specific record.

Classical Database Search

A classical computer checks records sequentially.

Example:

Database:

Index	Record
0	A
1	B
2	C
3	D

Finding D:

Checks:

A ❌

B ❌

C ❌

D ✅

Complexity:

$$O(N)$$

2.2 Quantum Database Search

A quantum computer represents all records as a superposition:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

The oracle marks the required record.

Amplitude amplification increases its probability.

2.3 Example

Suppose a database has:

$$N = 1,000,000$$

records.

Classical Search:

Maximum checks:

$$1,000,000$$

Grover Search:

Required operations:

$$\sqrt{1,000,000}$$

$$= 1000$$

Quantum improvement:

$$1000000 \rightarrow 1000$$

2.4 Real-World Examples

Applications include:

- Searching large data repositories.
- Finding specific information in cloud databases.
- Searching large scientific datasets.

3. Application 2: Cryptography

3.1 Introduction

Cryptography protects information using encryption keys.

A common attack method is:

> Brute-force key search

Example:

A password is protected by a key.

The attacker tries:

$$K_1, K_2, K_3, ..., K_N$$

until the correct key is found.

3.2 Classical Brute Force Attack

For a key space of size:

$$N$$

Classical complexity:

$$O(N)$$

3.3 Quantum Brute Force Attack

Grover's algorithm reduces:

$$O(N)$$

to:

$$O(\sqrt{N})$$

Example:

A 128-bit key:

Classical search:

$$2^{128}$$

operations.

Quantum search:

$$\sqrt{2^{128}}$$

$$= 2^{64}$$

operations.

3.4 Impact on Cryptography

Grover’s algorithm affects:

- Symmetric encryption
- Hash functions
- Password security

3.5 Security Improvement

To maintain security against quantum attacks:

Increase key size.

Example:

AES-128 → AES-256

---

4. Application 3: Optimization Problems

4.1 Introduction

Optimization means finding the best solution among many possibilities.

Examples:

- Minimum cost
- Maximum profit
- Shortest path
- Best schedule

4.2 Classical Optimization

Many optimization problems require checking:

$$N$$

possible solutions.

This becomes expensive for large problems.

4.3 Grover-Based Optimization

Grover's algorithm can speed up searching for:

- Minimum value
- Maximum value
- Best candidate

4.4 Example: Finding Minimum Value

Given:

$$[15, 8, 20, 3, 10]$$

Need to find minimum:

Classical method:

Check all values.

Quantum method:

Use Grover search repeatedly to find better candidates.

4.5 Applications

Optimization problems include:

Logistics

Finding:

- Best delivery routes.
- Minimum transportation cost.

Scheduling

Finding:

- Best timetable.
- Resource allocation.

Finance

Finding:

- Best investment strategy.

---

5. Application 4: Artificial Intelligence and Machine Learning

---

5.1 Introduction

Machine learning involves searching through large solution spaces.

Examples:

- Parameter optimization.
- Feature selection.
- Model searching.

5.2 Grover's Role in AI

Grover’s algorithm can improve:

- Searching training parameters.
- Finding optimal solutions.
- Optimization subroutines.

5.3 Example

Finding the best parameters:

Classical:

Test many combinations.

Quantum:

Use amplitude amplification to search promising parameters.

---

6. Application 5: Constraint Satisfaction Problems

6.1 Introduction

Constraint problems involve finding solutions satisfying certain conditions.

Examples:

- Sudoku
- Scheduling
- Graph coloring

6.2 Classical Approach

Try possible solutions:

$$S_1, S_2, S_3, ..., S_N$$

until valid solution found.

6.3 Quantum Approach

Oracle checks whether a solution satisfies constraints.

Grover amplifies valid solutions.

---

7. Application 6: Quantum Counting

Quantum counting is an extension of Grover's algorithm.

It estimates:

- Number of solutions.
- Size of marked states.

7.1 Need for Quantum Counting

Sometimes we do not know:

How many solutions exist?

Quantum counting helps estimate them.

7.2 Principle

Quantum counting combines:

- Grover operator
- Quantum Fourier Transform (QFT)

---

8. Application 7: Multiple Target Search

8.1 Introduction

Sometimes more than one solution exists.

Example:

Database contains:

$$M$$

correct records.

8.2 Complexity

For M solutions:

Grover complexity becomes:

$$O(\sqrt{\frac{N}{M}})$$

where:

- N = total items
- M = number of solutions

8.3 Example

Database:

$$N = 1,000,000$$

Solutions:

$$M = 100$$

Operations:

$$\sqrt{\frac{1000000}{100}}$$

$$= \sqrt{10000}$$

$$= 100$$

Only around 100 operations required.

---

9. Advantages of Grover Applications

1. Faster Searching

Quadratic speedup.

---

2. General Algorithm

Can solve many search-based problems.

---

3. Reduced Computational Cost

Requires fewer operations.

4. Useful for Large Search Spaces

Effective when:

$$N$$

is extremely large.

10. Limitations

1. Requires Quantum Computer

Not practical on classical machines.

2. Oracle Design Difficulty

Creating efficient oracles can be challenging.

3. Noise Effects

Quantum errors reduce accuracy.

4. Limited Speedup

Only quadratic improvement.

11. Comparison of Applications

Application	Purpose
Database Search	Find records faster
Cryptography	Search encryption keys
Optimization	Find best solutions
AI/ML	Search parameters
Constraint Problems	Find valid solutions
Quantum Counting	Estimate solutions
Multiple Search	Find multiple targets

12. Frequently Asked University Questions

2-Mark Questions

1. Mention two applications of Grover's algorithm.
2. How does Grover affect cryptography?
3. What is quantum counting?
4. Define multiple target search.

5-Mark Questions

1. Explain database search using Grover's algorithm.
2. Describe the application of Grover's algorithm in cryptography.
3. Explain Grover's algorithm for optimization problems.

10-Mark Questions

1. Discuss various applications of Grover’s algorithm.
2. Explain quantum counting and multiple target search.
3. Describe the impact of Grover's algorithm on modern cryptography.

Formula Box (Important)

Standard Grover Search:

$$O(\sqrt{N})$$

Multiple Target Search:

$$\boxed{O(\sqrt{\frac{N}{M}})}$$

Classical Search:

$$O(N)$$

Quick Revision

- Grover's algorithm is useful wherever searching is required.
- Major applications:
  - Database search
  - Cryptography
  - Optimization
  - AI/ML
  - Constraint problems
- Quantum counting estimates number of solutions.
- Multiple target search improves complexity:

$$O(\sqrt{N/M})$$

Next Topic

UNIT–III Final Topics:

Quantum Counting and Multiple Target Search (Detailed Working)

We will cover:

- Quantum counting algorithm
- Relationship with Grover iteration
- Quantum Fourier Transform role
- Multiple solution cases
- Mathematical explanation

This will complete **UNIT–III Quantum Search Algorithms**.

> ok next

UNIT–III: Quantum Search Algorithms

Topic 5 – Quantum Counting Algorithm and Multiple Target Search

Weightage: ⭐⭐⭐⭐☆ (Important)
Quantum counting and multiple target search are advanced extensions of Grover's Algorithm. They are used when:

- The number of solutions is unknown.
- Multiple solutions exist in a search space.

PART A: Quantum Counting Algorithm

---

1. Introduction to Quantum Counting

In many search problems, we know:

- Total number of possible states (N)

But we do not know:

- Number of correct solutions (M)

Example:

A database contains:

$$N = 1,000,000$$

records.

But we do not know how many records satisfy a condition.

Quantum counting solves this problem.

---

2. Definition

Exam Definition:

> **Quantum counting is a quantum algorithm that estimates the number of solutions in an unstructured search space by combining Grover's algorithm with Quantum Phase Estimation.**

---

3. Need for Quantum Counting

Classical approach:

Check every item:

$$O(N)$$

operations.

For huge search spaces:

- Very slow.
- Computationally expensive.

Quantum counting estimates the number of solutions faster.

---

4. Relationship with Grover's Algorithm

Quantum counting is based on the Grover operator:

$$G = D \cdot O$$

where:

- O = Oracle
- D = Diffusion operator

Grover's algorithm:

Finds the solution.

Quantum counting:

Finds:

$$\text{Number of solutions}$$

---

5. Main Idea of Quantum Counting

Quantum counting uses:

1. Grover operator
2. Eigenvalue estimation
3. Quantum Fourier Transform (QFT)

---

6. Components of Quantum Counting

The main components are:

1. Oracle

Marks the valid solutions.

---

2. Grover Operator

Performs amplitude amplification.

$$G = D \cdot O$$

3. Quantum Phase Estimation (QPE)

Measures the phase of Grover operator.

4. Quantum Fourier Transform

Converts phase information into a number.

7. Working of Quantum Counting

The process occurs in steps.

Step 1: Initialize Quantum Registers

Two registers are prepared:

Counting Register

Stores phase information.

Search Register

Stores database states.

Step 2: Apply Hadamard Gates

The search space is converted into superposition.

$$|s\rangle = \frac{1}{\sqrt{N}} \sum |x\rangle$$

Step 3: Apply Grover Operator

The Grover operator is repeatedly applied.

$$G^k$$

Step 4: Perform Phase Estimation

The phase of Grover operator is measured.

The eigenvalue is:

$$e^{i\theta}$$

Step 5: Apply Quantum Fourier Transform

QFT converts phase information into:

$$M$$

(number of solutions)

---

8. Mathematical Relation

The Grover operator rotation angle:

$$\theta$$

is related to the number of solutions:

$$M$$

by:

$$\sin^2(\theta) = \frac{M}{N}$$

Therefore:

$$M = N \sin^2(\theta)$$

---

9. Quantum Counting Circuit

Basic representation:

```
Counting Register
|0) —H—QPE————QFT†—Measurement

Search Register
|0) —H—Grover Operator————
```

10. Applications of Quantum Counting

---

1. Database Analysis

Counting records satisfying conditions.

2. Search Problems

Finding number of valid solutions.

3. Cryptography

Estimating possible keys.

4. Probability Estimation

Counting successful outcomes.

PART B: Multiple Target Search

11. Introduction

The original Grover algorithm assumes:

Only one target exists.

But many practical problems have multiple solutions.

Example:

A database contains many records matching a query.

12. Definition

Exam Definition:

> **Multiple target search is an extension of Grover's algorithm where more than one marked state exists in the search space, allowing faster identification of any valid solution.**

13. Mathematical Representation

Suppose:

Total states:

$$N$$

Number of target states:

$$M$$

where:

$$M < N$$

Example:

Database:

$$N = 1000$$

Matching records:

$$M = 20$$

---

14. Modified Grover Complexity

For multiple targets:

$$\boxed{O(\sqrt{\frac{N}{M}})}$$

---

15. Why is Multiple Target Search Faster?

When more solutions exist:

The probability of finding a correct answer is naturally higher.

Therefore fewer Grover iterations are needed.

---

16. Example Calculation

Given:

$$N = 1,000,000$$

Number of solutions:

$$M = 100$$

Complexity:

$$\sqrt{\frac{N}{M}}$$

Substitute:

$$= \sqrt{\frac{1000000}{100}}$$

$$= \sqrt{10000}$$

$$= 100$$

Only about:

$$100$$

iterations are required.

---

17. Working of Multiple Target Search

Steps:

Step 1:

Prepare equal superposition.

$$|s\rangle$$

Step 2:

Oracle marks all target states.

Step 3:

Diffusion operator amplifies all marked states.

Step 4:

Measurement returns one of the target states.

---

18. Example

Database:

$$N = 8$$

States:

$$|000\rangle$$

to

$$|111\rangle$$

Suppose valid states:

$$|010\rangle$$

$$|101\rangle$$

Oracle marks both.

Grover amplification increases probability of these states.

Measurement gives one valid answer.

---

19. Quantum Counting vs Multiple Target Search

Feature	Quantum Counting	Multiple Target Search
Goal	Count number of solutions	Find a solution
Uses	Grover + QFT	Modified Grover
Output	Number of solutions	Target state
Complexity	Depends on phase estimation	$O(\sqrt{N/M})$

---

20. Importance of These Algorithms

They improve:

- Large-scale database searching.
- Optimization problems.
- Scientific computing.
- Machine learning search.
- Cryptanalysis.

---

21. Limitations

1. Quantum Hardware Requirement

Needs fault-tolerant quantum systems.

2. Oracle Construction

Creating an oracle is difficult.

3. Noise Sensitivity

Errors affect phase estimation.

22. Frequently Asked University Questions

2-Mark Questions

1. What is quantum counting?
2. What is the role of QFT in quantum counting?
3. Define multiple target search.
4. Write complexity of multiple target search.

5-Mark Questions

1. Explain quantum counting algorithm.
2. Describe multiple target search using Grover's algorithm.
3. Explain the role of phase estimation.

10-Mark Questions

1. Explain quantum counting with circuit and mathematical formulation.
2. Explain Grover's algorithm for multiple target search.
3. Compare quantum counting and multiple target search.

Formula Box (Very Important)

Number of Solutions:

$$\boxed{M = N \sin^2(\theta)}$$

Multiple Target Search Complexity:

$$\boxed{O(\sqrt{\frac{N}{M}})}$$

Grover Operator:

$$\boxed{G = D \cdot O}$$

UNIT–III Completed ✅

Unit III Summary:

Topic	Main Concept
Unstructured Search	Searching without order
Grover Algorithm	Quantum search using amplitude amplification
Oracle	Marks correct state
Diffusion Operator	Amplifies probability
Quantum Counting	Counts number of solutions
Multiple Target Search	Search with multiple answers

Next: UNIT–IV – Quantum Algorithms for Image Processing

First Topic:

Quantum Image Representation

We will study:

1. Need for quantum image processing
2. Classical image representation
3. Quantum image encoding
4. Pixel representation using qubits
5. Basic quantum image models

This unit connects **quantum computing with computer vision and image processing**.

> ok next

UNIT–IV: Quantum Algorithms for Image Processing

Topic 1 – Introduction to Quantum Image Processing (QIP) and Quantum Image Representation

> **Weightage:** ⭐⭐⭐⭐⭐ (Important Unit–IV Foundation Topic)
> Quantum Image Processing combines **quantum computing principles** with **digital image processing** to process and represent images using quantum states.

---

1. Introduction to Quantum Image Processing (QIP)

Digital images contain huge amounts of data.

A classical image stores information using:

- Pixels
- Intensity values
- Color information

For large images:

- Storage requirement increases.
- Processing time increases.

Quantum computing provides new methods to:

- Store images efficiently.
- Process images using quantum algorithms.
- Perform operations faster using quantum parallelism.

---

2. Definition of Quantum Image Processing

Exam Definition:

> **Quantum Image Processing is an emerging field that applies quantum computing principles to represent, process, and analyze images using quantum states and quantum algorithms.**

---

3. Need for Quantum Image Processing

Classical image processing faces challenges:

- Large memory requirement.
- High computational cost.

- Slow processing for large datasets.

Examples:

- Medical imaging
- Satellite images
- Computer vision
- Artificial intelligence

Quantum image processing aims to provide:

- Efficient image representation.
- Faster image operations.
- Quantum parallel processing.

---

4. Classical Image Representation

Before understanding quantum images, we need to understand classical representation.

A digital image is represented as a matrix of pixels.

Example:

A grayscale image:

$$I = \begin{bmatrix} 20 & 50 & 80 \\ 30 & 60 & 90 \\ 40 & 70 & 100 \end{bmatrix}$$

Each value represents pixel intensity.

---

5. Pixel Representation

A pixel contains:

1. Position information

$$(x, y)$$

2. Intensity or color information

Example:

Grayscale:

$$0 - 255$$

where:

- $0 \rightarrow$ black

- 255 → white

---

6. Classical Image Storage

For an image of size:

$$M \times N$$

pixels.

Memory required:

$$M \times N$$

values.

Example:

A:

$$1024 \times 1024$$

grayscale image:

Number of pixels:

$$1024^2$$

$$= 1,048,576$$

Each pixel needs storage.

---

7. Quantum Representation of Images

In quantum computing, image information is encoded into qubits.

A quantum image consists of:

1. Position qubits
2. Color/intensity qubits

---

8. Basic Concept of Quantum Image Encoding

A quantum image state is represented as:

$$|Image\rangle$$

It contains:

- Pixel locations

- Pixel values

General form:

$$|I\rangle = \sum_x \sum_y |xy\rangle |C_{xy}\rangle$$

where:

- $|xy\rangle$ represents pixel position.
- $|C_{xy}\rangle$ represents pixel intensity/color.

---

9. Quantum Image Representation Components

A quantum image contains:

1. Coordinate Information

The location of pixels.

Example:

$$(x, y)$$

---

2. Pixel Information

The intensity/color value.

Example:

$$C$$

---

10. Number of Qubits Required

For an image of size:

$$2^n \times 2^n$$

Position requires:

$$2n$$

qubits.

Example:

Image size:

$$256 \times 256$$

Since:

$$256 = 2^8$$

Position qubits:

$$2 \times 8 = 16$$

qubits.

---

11. Quantum Image Encoding

Quantum image encoding means converting classical image data into quantum states.

The main steps are:

Step 1:

Read classical image.

Example:

Pixel matrix.

↓

Step 2:

Convert pixel values into binary form.

Example:

Intensity:

$$255$$

Binary:

$$11111111$$

↓

Step 3:

Store binary information using qubits.

↓

Step 4:

Create quantum image state.

---

12. Types of Quantum Image Representation Models

Important models are:

1. FRQI
2. NEQR

These are included in your syllabus.

---

13. FRQI Model

Flexible Representation of Quantum Images

Proposed by:

$$\text{Le et al., } 2011$$

Definition:

> **FRQI is a quantum image representation model that stores pixel colors using quantum probability amplitudes.**

---

14. Basic Idea of FRQI

FRQI represents an image using:

- Position qubits
- One color qubit

The quantum image state:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos\theta_i|0\rangle + \sin\theta_i|1\rangle)|i\rangle$$

where:

- $|i\rangle$ = pixel position
- θ_i = color angle

---

15. FRQI Working

Each pixel intensity is converted into an angle.

Example:

Dark pixel:

$$\theta \approx 0$$

Bright pixel:

$$\theta \approx \frac{\pi}{2}$$

The intensity is stored in the probability amplitude.

---

16. Advantages of FRQI

- Simple representation.
- Uses fewer qubits.
- Supports quantum image operations.

---

17. Limitations of FRQI

- Color information is stored indirectly.
- Measuring pixels is probabilistic.
- Requires repeated measurements.

---

18. NEQR Model

Novel Enhanced Quantum Representation

Definition:

> **NEQR is a quantum image representation model that stores pixel intensity values directly using computational basis states.**

---

19. Basic Idea of NEQR

Unlike FRQI:

FRQI:

Stores intensity as amplitudes.

NEQR:

Stores intensity as binary values.

Example:

Pixel value:

$$200$$

Binary:

$$11001000$$

Stored directly in qubits.

---

20. NEQR Representation

General form:

$$|I\rangle = \frac{1}{2^n}\sum_{y,x}|f(y,x)\rangle|yx\rangle$$

where:

- $f(y,x)$ = pixel intensity
- yx = position

---

21. FRQI vs NEQR Comparison

Feature	FRQI	NEQR
Full form	Flexible Representation of Quantum Images	Novel Enhanced Quantum Representation
Color storage	Amplitude	Binary basis states
Pixel accuracy	Probability based	Exact values
Qubits	Less	More
Representation	Continuous	Discrete

---

22. Applications of Quantum Image Processing

---

1. Medical Imaging

Examples:

- MRI analysis
- CT scans

---

2. Satellite Image Processing

Used for:

- Earth observation
- Remote sensing

---

3. Computer Vision

Applications:

- Object recognition
- Image classification

---

4. Artificial Intelligence

Quantum machine learning with images.

---

23. Challenges in Quantum Image Processing

1. Hardware Limitations

Current quantum computers have limited qubits.

---

2. Noise

Quantum states are sensitive.

---

3. Encoding Cost

Converting classical images to quantum form is expensive.

---

4. Measurement Problem

Reading quantum images destroys the state.

24. Frequently Asked University Questions

2-Mark Questions

1. Define Quantum Image Processing.
2. What is FRQI?
3. What is NEQR?
4. How are pixels represented in quantum images?

5-Mark Questions

1. Explain quantum image representation.
2. Describe FRQI model.
3. Explain NEQR model.

10-Mark Questions

1. Explain quantum image representation models FRQI and NEQR.
2. Compare classical and quantum image representation.
3. Discuss applications and challenges of quantum image processing.

Formula Box (Important)

Quantum Image State:

$$|I\rangle = \sum_{x,y} |xy\rangle |C_{xy}\rangle$$

FRQI:

$$|I\rangle = \frac{1}{2^n} \sum_i (\cos\theta_i |0\rangle + \sin\theta_i |1\rangle)|i\rangle$$

NEQR:

$$|I\rangle = \frac{1}{2^n} \sum_{x,y} |f(x,y)\rangle |xy\rangle$$

Quick Revision

- Quantum Image Processing uses quantum computing for image tasks.
- Images are represented using qubits.
- Two important models:
  - **FRQI** → stores pixel information using amplitudes.
  - **NEQR** → stores pixel values directly in binary form.
- Quantum image contains:
  - Position information
  - Pixel information
- Major challenges:
  - Noise
  - Hardware limitations
  - Measurement issues

Next Topic

Quantum Image Representation Models – FRQI (Detailed Explanation)

We will cover:

- FRQI architecture
- Mathematical formulation
- Encoding process
- Circuit representation
- Advantages and limitations
- Semester exam answer format (10 marks)

> ok next

UNIT–IV: Quantum Algorithms for Image Processing

Topic 2 – FRQI Model (Flexible Representation of Quantum Images)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> FRQI is one of the earliest and most widely studied quantum image representation models. It

provides a method to encode grayscale and color images into quantum states.

---

1. Introduction to FRQI

FRQI stands for:

$$\boxed{\text{Flexible Representation of Quantum Images}}$$

It was proposed by:

$$\boxed{\text{Le et al. (2011)}}$$

The main objective of FRQI is:

> To represent an image using a small number of qubits by combining pixel position information and pixel intensity information.

---

2. Definition

Exam Definition:

> **FRQI is a quantum image representation model that encodes pixel intensity information into the probability amplitudes of a single qubit and uses additional qubits to represent pixel positions.**

---

3. Basic Idea of FRQI

A classical image contains:

- Pixel position
- Pixel intensity

Example:

```
Image

Pixel Position → (x,y)
Pixel Value → Brightness
```

FRQI stores:

1. Position information using computational basis states.
2. Pixel intensity using quantum amplitudes.

---

4. Structure of FRQI State

A quantum image in FRQI is represented as:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos\theta_i|0\rangle + \sin\theta_i|1\rangle)|i\rangle$$

Where:

Symbol	Meaning
(	I\rangle)
(	i\rangle)
θ_i	Color/intensity angle
n	Number of position qubits

5. Qubit Requirement in FRQI

For an image:

$$2^n \times 2^n$$

pixels.

Position requires:

$$2n$$

qubits.

Additionally:

One qubit stores intensity.

Therefore:

$$\boxed{2n + 1 \text{ qubits}}$$

are required.

6. Example

Consider a:

$$2 \times 2$$

image.

Number of pixels:

$$4$$

Position states:

$$|00\rangle$$

$$|01\rangle$$

$$|10\rangle$$

$$|11\rangle$$

Position qubits:

$$2$$

Color qubit:

$$1$$

Total:

$$3$$

qubits.

---

7. Pixel Encoding in FRQI

In FRQI, pixel intensity is converted into an angle.

The intensity range:

$$0 - 255$$

is mapped to:

$$0 \leq \theta \leq \frac{\pi}{2}$$

---

8. Intensity Mapping

The color qubit is:

$$|\psi_i\rangle = \cos\theta_i|0\rangle + \sin\theta_i|1\rangle$$

Dark Pixel

For black:

$$\theta_i = 0$$

Therefore:

$$|\psi_i\rangle = |0\rangle$$

Bright Pixel

For white:

$$\theta_i = \frac{\pi}{2}$$

Therefore:

$$|\psi_i\rangle = |1\rangle$$

---

9. FRQI Encoding Process

The encoding process consists of steps:

Step 1: Classical Image Input

Example:

Pixel matrix:

$$\begin{bmatrix} 0 & 255 \\ 128 & 200 \end{bmatrix}$$

Step 2: Convert Pixel Values to Angles

Each intensity is converted:

$$Intensity \rightarrow \theta$$

Example:

Black:

$$0 \rightarrow 0$$

White:

$$255 \rightarrow \frac{\pi}{2}$$

Step 3: Assign Pixel Positions

Each pixel receives a binary position.

Example:

Pixel	Position
P0	00
P1	01

Pixel	Position
P2	10
P3	11

Step 4: Create Quantum State

The final FRQI state contains:

- Position qubits
- Color qubit

10. FRQI Circuit Representation

A simplified FRQI circuit:

```
Position Qubits
|00) ──────────────
|01) ──────────────
|10) ──────────────
Color Qubit
|0) ──Rotation(θ)──
```

The rotation operation:

$$R_y(2\theta)$$

is used to encode intensity.

11. Working Principle of FRQI

The intensity information is stored through quantum rotation.

A rotation gate changes:

$$|0\rangle$$

into:

$$\cos\theta|0\rangle + \sin\theta|1\rangle$$

Thus:

Different pixel values produce different quantum states.

12. Image Operations Using FRQI

FRQI supports quantum image processing operations such as:

1. Image Compression

Quantum representation can store images efficiently.

2. Image Retrieval

Finding image information using quantum search.

3. Image Transformation

Operations:

- Rotation
- Scaling
- Translation

4. Image Enhancement

Improving image quality.

13. Advantages of FRQI

1. Efficient Representation

Requires fewer qubits compared with direct encoding.

2. Unified Representation

Can represent:

- Grayscale images
- Color images

3. Supports Quantum Operations

Compatible with quantum algorithms.

4. Uses Quantum Parallelism

All pixels are represented together.

14. Limitations of FRQI

1. Probabilistic Measurement

Pixel values are stored as amplitudes.

Measurement gives probabilities.

2. Difficult Image Retrieval

Extracting exact pixel values requires repeated measurements.

3. Complex Processing

Quantum circuits become complicated for large images.

4. Limited Precision

Intensity representation depends on angle precision.

15. FRQI vs Classical Image Representation

Feature	Classical Image	FRQI
Storage	Pixel values	Quantum states
Basic unit	Bit	Qubit
Processing	Sequential	Quantum parallelism
Pixel storage	Direct values	Probability amplitudes
Memory	Large	Potentially reduced

16. FRQI vs NEQR

Feature	FRQI	NEQR
Encoding	Amplitude based	Binary basis based
Intensity	Angle θ	Binary values
Accuracy	Probabilistic	Exact
Qubits	Less	More
Measurement	Difficult	Easier

17. Applications of FRQI

FRQI is used in:

1. Quantum Computer Vision

Image analysis using quantum algorithms.

2. Pattern Recognition

Finding patterns in images.

3. Medical Imaging

Processing:

- MRI
- CT images

4. Quantum Machine Learning

Image classification tasks.

18. Frequently Asked University Questions

2-Mark Questions

1. Expand FRQI.
2. Who proposed FRQI?
3. How is pixel intensity represented in FRQI?

4. How many qubits are required for a $2^n \times 2^n$ image?

5-Mark Questions

1. Explain the FRQI model.
2. Describe FRQI encoding process.
3. Write advantages and limitations of FRQI.

10-Mark Questions

1. Explain FRQI quantum image representation model with mathematical formulation.
2. Describe the encoding process of FRQI.
3. Compare FRQI and NEQR models.

Formula Box (Very Important)

FRQI State:

$$\boxed{|I\rangle = \frac{1}{2^n}\sum_i (\cos\theta_i|0\rangle + \sin\theta_i|1\rangle)|i\rangle}$$

Qubit Requirement:

$$\boxed{2n+1}$$

Intensity Encoding:

$$\boxed{|\psi_i\rangle = \cos\theta_i|0\rangle + \sin\theta_i|1\rangle}$$

Quick Revision

- FRQI = Flexible Representation of Quantum Images.
- Proposed in 2011.
- Uses:
  - Position qubits → pixel location
  - One color qubit → intensity
- Pixel intensity is represented as an angle.

- Total qubits:

$$2n + 1$$

- Advantage:
  - Efficient representation.
- Limitation:
  - Difficult measurement and retrieval.

Next Topic

NEQR Model (Novel Enhanced Quantum Representation)

We will study:

- NEQR architecture
- Mathematical representation
- Pixel encoding using binary states
- Circuit implementation
- FRQI vs NEQR comparison
- Advantages and limitations

This is the second major model in **UNIT–IV** and frequently asked in semester exams.

> ok next

UNIT–IV: Quantum Algorithms for Image Processing

Topic 3 – NEQR Model (Novel Enhanced Quantum Representation)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> NEQR is an advanced quantum image representation model that improves the limitations of FRQI by storing pixel intensity values directly using quantum basis states.

---

1. Introduction to NEQR

NEQR stands for:

$$\boxed{\text{Novel Enhanced Quantum Representation}}$$

It was proposed to provide:

- More accurate image representation.
- Direct storage of pixel values.
- Better image retrieval compared to FRQI.

---

2. Definition

Exam Definition:

> **NEQR is a quantum image representation model that encodes image pixel values using computational basis states of qubits, while pixel positions are represented by separate coordinate qubits.**

---

3. Motivation Behind NEQR

FRQI has some limitations:

- Pixel values are stored using amplitudes.
- Measurement gives probabilistic results.
- Exact pixel retrieval is difficult.

NEQR solves this by:

- Storing pixel intensity as binary information.
- Using basis states instead of probability amplitudes.

---

4. Basic Concept of NEQR

A digital image contains:

1. Pixel position

$$(x, y)$$

2. Pixel intensity

$$f(x, y)$$

NEQR stores both using qubits:

```
Quantum Image

+----------------+
| Intensity      |
| Qubits         |
+----------------+
| Position       |
| Qubits         |
+----------------+
```

5. NEQR Mathematical Representation

The quantum image state is represented as:

$$\boxed{|I\rangle = \frac{1}{2^n}\sum_{y=0}^{2^n-1}\sum_{x=0}^{2^n-1}|f(y,x)\rangle|yx\rangle}$$

Where:

Symbol	Meaning
(	I\rangle)
$f(y,x)$	Pixel intensity value
(	yx\rangle)
n	Number of position qubits

6. Pixel Representation in NEQR

Unlike FRQI:

FRQI:

$$Intensity \rightarrow Angle$$

NEQR:

$$Intensity \rightarrow Binaryvalue$$

Example:

Pixel intensity:

$$200$$

Binary representation:

$$11001000$$

Stored directly as:

$$|11001000\rangle$$

7. Qubit Requirement in NEQR

For an image:

$$2^n \times 2^n$$

pixels.

Position Qubits:

For x-coordinate:

$$n$$

qubits

For y-coordinate:

$$n$$

qubits

Total position qubits:

$$2n$$

Intensity Qubits:

For grayscale image:

Usually:

$$q$$

qubits.

Example:

8-bit grayscale:

$$q = 8$$

Therefore total qubits:

$$\boxed{2n + q}$$

---

8. Example

Consider:

$$4 \times 4$$

image.

Since:

$$4 = 2^2$$

Position qubits:

$$2n = 4$$

For 8-bit grayscale:

$$q = 8$$

Total qubits:

$$4 + 8$$

$$\boxed{12 \text{ qubits}}$$

---

9. NEQR Encoding Process

The encoding process involves several steps.

Step 1: Read Classical Image

Example:

$$\begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix}$$

Step 2: Convert Pixel Values into Binary

Example:

Pixel value:

$$10$$

Binary:

$$00001010$$

Step 3: Assign Position Coordinates

Example:

Pixel	Position
10	00
20	01
30	10
40	11

Step 4: Generate Quantum State

The final state contains:

- Intensity qubits.
- Coordinate qubits.

10. NEQR Circuit Representation

Simplified circuit:

```
Position Register
|00⟩ ──────────────

Intensity Register
|00000000⟩ ──Encoding──
```

Quantum gates used:

- NOT gates
- Controlled-NOT gates
- Multi-controlled gates

11. Working Principle of NEQR

NEQR uses computational basis states.

Each pixel value corresponds to a unique quantum state.

Example:

Pixel:

$$5$$

Binary:

$$00000101$$

Quantum state:

$$|00000101\rangle$$

The measurement gives the exact binary value.

12. Advantages of NEQR

1. Exact Pixel Representation

Pixel values are stored directly.

2. Better Image Retrieval

Measurement produces accurate intensity values.

3. Suitable for Image Processing

Supports:

- Edge detection
- Image transformation
- Filtering

4. Easy Conversion

Binary representation is similar to classical images.

13. Limitations of NEQR

1. More Qubits Required

Compared with FRQI.

Example:

8-bit image requires:

8

intensity qubits.

2. Complex Circuit Design

Encoding requires many quantum gates.

3. Hardware Requirements

Large images need many qubits.

14. NEQR vs FRQI

Feature	FRQI	NEQR
Full form	Flexible Representation of Quantum Images	Novel Enhanced Quantum Representation
Pixel storage	Amplitude	Basis states
Intensity representation	Angle θ	Binary values
Accuracy	Probabilistic	Exact
Measurement	Difficult	Simple
Qubit requirement	Lower	Higher
Encoding	Rotation gates	Logic gates

15. NEQR vs Classical Image Representation

Feature	Classical Image	NEQR
Storage unit	Bits	Qubits
Pixel representation	Binary numbers	Quantum basis states
Processing	Sequential	Quantum operations
Parallelism	Limited	High
Memory	Large	Potentially reduced

16. Applications of NEQR

1. Quantum Image Processing

Basic image manipulation.

2. Edge Detection

Detecting boundaries in images.

3. Pattern Recognition

Finding objects and patterns.

4. Medical Imaging

Processing:

- MRI
- CT scans

5. Quantum Machine Learning

Image classification models.

17. NEQR-Based Image Processing Operations

NEQR supports:

1. Image Retrieval

Extracting pixel information.

2. Image Transformation

Operations:

- Rotation
- Scaling
- Translation

3. Image Enhancement

Improving image quality.

4. Edge Detection

Finding intensity changes.

18. FRQI and NEQR Summary

FRQI:

- Efficient storage.
- Uses amplitude encoding.
- Requires fewer qubits.
- Retrieval is difficult.

NEQR:

- Accurate storage.
- Uses binary encoding.
- Requires more qubits.
- Retrieval is easier.

---

19. Frequently Asked University Questions

2-Mark Questions

1. Expand NEQR.
2. How does NEQR represent pixel values?
3. What is the qubit requirement of NEQR?
4. Differentiate FRQI and NEQR.

5-Mark Questions

1. Explain NEQR model.
2. Describe NEQR encoding process.
3. Explain advantages of NEQR.

10-Mark Questions

1. Explain NEQR quantum image representation with mathematical formulation.
2. Compare FRQI and NEQR models.

3. Discuss applications and limitations of NEQR.

Formula Box (Very Important)

NEQR State:

$$\boxed{|I\rangle = \frac{1}{2^n}\sum_{x,y}|f(x,y)\rangle|xy\rangle}$$

Total Qubits:

$$\boxed{2n+q}$$

where:

- $2n \rightarrow$ position qubits
- $q \rightarrow$ intensity qubits

Quick Revision

- NEQR = Novel Enhanced Quantum Representation.
- Stores pixel values directly as binary states.
- Uses:
  - Position qubits
  - Intensity qubits
- Pixel retrieval is more accurate than FRQI.
- Requires more qubits.
- Important for:
  - Edge detection
  - Image processing
  - Computer vision

Next Topic

Quantum Image Processing Techniques – Edge Detection

We will cover:

1. Importance of edge detection
2. Classical edge detection methods

3. Quantum edge detection concept
4. Quantum operators used
5. Quantum edge detection algorithms
6. Applications and limitations

This completes the major **Quantum Image Representation Models** of UNIT–IV.

> ok next

UNIT–IV: Quantum Algorithms for Image Processing

Topic 4 – Quantum Image Processing Techniques: Edge Detection

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> Edge detection is one of the important applications of quantum image processing. It identifies boundaries and significant intensity changes in images using quantum operations.

---

1. Introduction to Edge Detection

An image consists of different regions with different intensity values.

An **edge** represents a sudden change in pixel intensity.

Examples:

- Object boundaries
- Shape outlines
- Texture changes

Example

Original Image:

```
White background + Black object
```

Edge:

```
Boundary between white and black
```

2. Definition of Edge Detection

Exam Definition:

Edge detection is an image processing technique used to identify points where the intensity of an image changes significantly, representing object boundaries and important features.

---

3. Importance of Edge Detection

Edge detection is used for:

- Object recognition
- Image segmentation
- Pattern detection
- Computer vision
- Medical image analysis

---

4. Classical Edge Detection Methods

Before quantum edge detection, classical methods are used.

Important classical techniques:

---

1. Sobel Operator

Uses gradient calculation.

Detects:

- Horizontal edges
- Vertical edges

---

2. Prewitt Operator

Similar to Sobel.

Uses convolution masks.

---

3. Canny Edge Detection

A multi-stage algorithm:

Steps:

1. Noise reduction

2. Gradient calculation
3. Non-maximum suppression
4. Edge tracking

5. Limitations of Classical Edge Detection

Classical approaches face problems:

1. Large Computation

High-resolution images require more processing.

2. Memory Requirement

Large images need large storage.

3. Sequential Processing

Pixels are processed individually.

Quantum computing provides new approaches using:

- Superposition
- Quantum parallelism
- Quantum interference

6. Quantum Edge Detection Concept

Quantum edge detection works by:

1. Encoding image into quantum states.
2. Applying quantum operators.
3. Comparing neighboring pixels.
4. Detecting intensity differences.

7. Basic Idea

An edge occurs when:

$$|I(x,y) - I(x+1,y)|$$

is large.

Quantum algorithms calculate these differences using quantum operations.

---

8. Quantum Image Edge Detection Process

The general process:

```
Classical Image
       ↓
Quantum Image Encoding
       ↓
Quantum Operations
       ↓
Pixel Comparison
       ↓
Edge Extraction
       ↓
Measurement
```

9. Quantum Image Representation for Edge Detection

Edge detection can use:

- FRQI
- NEQR

models.

FRQI Approach

Pixel intensity is stored as:

$$\cos\theta|0\rangle + \sin\theta|1\rangle$$

Edges are detected by comparing amplitude differences.

NEQR Approach

Pixel intensity is stored as binary values.

Example:

Pixel A:

$$00101010$$

Pixel B:

$$11101010$$

Difference indicates possible edge.

---

10. Quantum Operators Used in Edge Detection

Important quantum operations:

10.1 Hadamard Gate

Used for:

- Creating superposition.
- Parallel processing.

Symbol:

$$H$$

10.2 Controlled-NOT Gate (CNOT)

Used for:

- Comparing pixel values.
- Creating correlations.

10.3 Rotation Gates

Used for:

- Modifying pixel amplitudes.

Examples:

$$R_x, R_y, R_z$$

10.4 Quantum Measurement

Used for:

- Extracting final edge information.

---

11. Quantum Edge Detection Algorithm (General Steps)

Step 1: Image Encoding

Convert classical image into quantum representation.

Using:

- FRQI
- NEQR

Step 2: Create Superposition

Apply Hadamard gates.

All pixels can be processed simultaneously.

Step 3: Pixel Comparison

Quantum circuits compare:

$$I(x, y)$$

with neighboring pixels:

$$I(x + 1, y)$$

Step 4: Edge Identification

If intensity difference is high:

Edge = detected

Step 5: Measurement

Extract edge image.

---

12. Quantum Edge Detection Using NEQR

NEQR is suitable because it stores exact pixel values.

Example:

Two neighboring pixels:

Pixel 1:

$$01001100$$

Pixel 2:

$$01001101$$

Difference:

$$00000001$$

Small difference:

No strong edge.

Large difference:

Strong edge.

---

13. Quantum Edge Detection Using FRQI

FRQI stores:

$$\theta$$

values.

Pixel state:

$$|\psi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$$

Two pixels:

$$\theta_1, \theta_2$$

Difference:

$$|\theta_1 - \theta_2|$$

Large difference indicates edge.

---

14. Advantages of Quantum Edge Detection

---

1. Quantum Parallelism

Multiple pixels can be processed simultaneously.

2. Reduced Computation

Potentially faster for large images.

3. Efficient Representation

Images stored as quantum states.

4. Better Processing Capability

Useful for complex image datasets.

15. Limitations of Quantum Edge Detection

1. Current Hardware Limitations

Quantum computers have limited qubits.

2. Noise

Quantum states are sensitive to errors.

3. Image Conversion Cost

Encoding classical images requires time.

4. Measurement Difficulty

Quantum measurement destroys the state.

16. Applications

1. Medical Imaging

Examples:

- Tumor boundary detection
- MRI analysis

2. Satellite Imaging

Used for:

- Land boundary detection
- Object identification

3. Autonomous Vehicles

Used for:

- Road detection
- Object boundaries

4. Security Systems

Used for:

- Face recognition
- Object tracking

17. Quantum vs Classical Edge Detection

Feature	Classical Edge Detection	Quantum Edge Detection
Processing	Sequential	Quantum parallel
Representation	Pixels	Quantum states
Storage	Bits	Qubits
Speed	Depends on image size	Potential quantum speedup
Hardware	Available	Developing

18. Frequently Asked University Questions

2-Mark Questions

1. Define edge detection.
2. Why is edge detection important?
3. Name quantum image representation models used for edge detection.
4. What is the role of CNOT gate?

5-Mark Questions

1. Explain quantum edge detection technique.
2. Describe the steps involved in quantum image processing.
3. Explain edge detection using NEQR.

10-Mark Questions

1. Explain quantum edge detection algorithm with architecture.
2. Discuss quantum techniques used in image processing.
3. Compare classical and quantum edge detection.

Formula Box (Important)

Edge Difference:

$$\boxed{D = |I(x,y) - I(x+1,y)|}$$

Large:

$$D \rightarrow Edge$$

FRQI Pixel State:

$$\boxed{|\psi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle}$$

Quick Revision

- Edge detection identifies intensity changes.
- Quantum edge detection uses:
  - Superposition
  - Quantum gates
  - Quantum image models
- FRQI:
  - Uses amplitude comparison.
- NEQR:
  - Uses binary pixel comparison.
- Applications:
  - Medical imaging
  - Satellite images
  - Computer vision

UNIT–IV Completed ✅

UNIT–IV Summary:

Topic	Main Concept
Quantum Image Representation	Encoding images using qubits
FRQI	Amplitude-based image representation
NEQR	Binary-based image representation
Edge Detection	Finding boundaries using quantum methods

Next: UNIT–V – Application-Oriented Quantum Algorithms

First Topic:

Quantum Optimization

We will study:

1. Introduction to quantum optimization
2. Classical vs quantum optimization
3. Variational Quantum Eigensolver (VQE)
4. Quantum Approximate Optimization Algorithm (QAOA)

5. Real-world applications (logistics, scheduling)

This begins the final unit of your syllabus.

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 1 – Introduction to Quantum Optimization

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – UNIT–V Foundation Topic)
> Quantum optimization is one of the most practical applications of quantum computing. It focuses on solving complex optimization problems that are difficult for classical computers.

---

1. Introduction to Optimization

Optimization means finding the **best possible solution** from a set of possible solutions.

The best solution may mean:

- Minimum cost
- Maximum profit
- Shortest distance
- Maximum efficiency
- Minimum time

Examples of Optimization Problems

1. Route Planning

Finding the shortest delivery route.

Example:

A logistics company wants the shortest path for delivery trucks.

---

2. Scheduling

Creating the best timetable.

Example:

Assigning:

- Employees
- Machines

- Resources

efficiently.

---

3. Finance

Finding the best investment portfolio.

---

4. Machine Learning

Finding optimal model parameters.

---

2. Mathematical Formulation of Optimization

An optimization problem generally consists of:

Objective Function

The function that needs to be optimized.

Represented as:

$$f(x)$$

Constraints

Conditions that solutions must satisfy.

Example:

$$x \leq 10$$

Solution Space

The collection of all possible solutions.

General optimization form:

$$\text{Minimize/Maximize } f(x)$$

Subject to:

$$g(x) \leq 0$$

---

3. Classical Optimization

Classical computers solve optimization problems using algorithms like:

- Gradient descent
- Linear programming
- Dynamic programming
- Branch and bound

---

4. Challenges in Classical Optimization

Many optimization problems become difficult when the search space increases.

Example:

Suppose:

$$N = 2^{100}$$

possible solutions.

Checking all possibilities becomes impossible.

Problems with huge search spaces are called:

$$\boxed{\text{Combinatorial Optimization Problems}}$$

---

5. What is Quantum Optimization?

Definition:

> **Quantum optimization uses quantum computing principles such as superposition, entanglement, and quantum interference to solve optimization problems more efficiently than classical approaches.**

---

6. Principles Behind Quantum Optimization

Quantum optimization uses:

6.1 Superposition

A quantum computer can represent multiple states simultaneously.

Example:

A qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

6.2 Entanglement

Quantum states can become correlated.

This allows complex relationships between variables.

6.3 Quantum Interference

Correct solutions can be amplified.

Incorrect solutions can be reduced.

7. Quantum Optimization Workflow

General process:

```
Optimization Problem
        ↓
Mathematical Model
        ↓
Quantum Encoding
        ↓
Quantum Algorithm
        ↓
Measurement
        ↓
Optimal Solution
```

8. Types of Quantum Optimization Algorithms

Important algorithms:

1. Variational Quantum Eigensolver (VQE)

Used for:

- Finding minimum energy states.
- Quantum chemistry problems.

2. Quantum Approximate Optimization Algorithm (QAOA)

Used for:

- Combinatorial optimization.
- Scheduling.
- Routing problems.

---

9. Variational Quantum Eigensolver (VQE)

9.1 Introduction

VQE is a hybrid quantum-classical algorithm.

It combines:

- Quantum computer
- Classical optimizer

9.2 Definition

Exam Definition:

> **Variational Quantum Eigensolver (VQE) is a hybrid quantum algorithm that finds the lowest eigenvalue of a Hamiltonian by optimizing parameters of a quantum circuit.**

---

10. Need for VQE

Many problems in physics and chemistry require finding:

$$\text{Minimum energy state}$$

Example:

Molecules:

- Hydrogen
- Lithium hydride

Need:

Ground state energy calculation.

11. Working of VQE

VQE works in steps:

Step 1: Prepare Quantum State

A parameterized quantum circuit creates:

$$|\psi(\theta)\rangle$$

where:

$$\theta$$

are adjustable parameters.

Step 2: Measure Energy

The quantum computer calculates:

$$E(\theta)$$

Step 3: Classical Optimization

A classical optimizer updates:

$$\theta$$

to minimize:

$$E(\theta)$$

Step 4: Repeat

The process continues until minimum energy is obtained.

---

12. VQE Mathematical Representation

The energy is calculated using:

$$\boxed{E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle}$$

where:

- H = Hamiltonian operator
- $|\psi(\theta)\rangle$ = quantum state

---

13. VQE Architecture

```
Quantum Circuit
      ↓
Measurement
      ↓
Energy Calculation
      ↓
Classical Optimizer
      ↓
Update Parameters
      ↓
Repeat
```

14. Advantages of VQE

---

1. Suitable for NISQ Devices

Works on noisy intermediate-scale quantum computers.

---

2. Hybrid Approach

Uses classical computers for optimization.

---

3. Efficient Energy Calculation

Useful for chemistry and physics.

---

15. Limitations of VQE

---

1. Optimization Difficulty

Finding correct parameters is challenging.

2. Noise Effects

Quantum errors affect accuracy.

3. Many Iterations Required

Requires repeated measurements.

16. Applications of VQE

1. Quantum Chemistry

Finding molecular energies.

2. Material Science

Designing new materials.

3. Drug Discovery

Molecular simulation.

4. Physics Simulation

Studying quantum systems.

17. VQE Summary

Feature	Description
Full form	Variational Quantum Eigensolver
Type	Hybrid quantum-classical
Purpose	Find minimum eigenvalue
Optimization	Classical optimizer

Feature	Description
Quantum part	Parameterized circuit
Applications	Chemistry, physics

18. Frequently Asked University Questions

2-Mark Questions

1. Define quantum optimization.
2. What is VQE?
3. What is an objective function?
4. Mention two applications of quantum optimization.

5-Mark Questions

1. Explain quantum optimization.
2. Describe the working of VQE.
3. Explain the importance of hybrid quantum algorithms.

10-Mark Questions

1. Explain VQE algorithm with architecture.
2. Discuss quantum optimization techniques.
3. Compare classical and quantum optimization.

Formula Box (Important)

Optimization Function:

$$\boxed{\text{Optimize } f(x)}$$

VQE Energy:

$$\boxed{E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle}$$

Quick Revision

- Optimization finds the best solution.
- Quantum optimization uses:
  - Superposition
  - Entanglement
  - Interference
- Main algorithms:
  - VQE
  - QAOA
- VQE:
  - Hybrid quantum-classical algorithm.
  - Finds minimum energy states.
  - Uses parameterized circuits.

Next Topic

Variational Quantum Eigensolver (VQE) – Detailed Working

We will cover:

1. Quantum circuit ansatz
2. Hamiltonian representation
3. Variational principle
4. Classical optimizer role
5. Step-by-step VQE algorithm
6. Applications and limitations

This topic is a **frequently asked 10-mark question** in Quantum Computing exams.

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 2 – Variational Quantum Eigensolver (VQE) – Detailed Explanation

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> VQE is one of the most important quantum algorithms designed for **NISQ (Noisy Intermediate-Scale Quantum)** computers. It combines quantum computation with classical optimization techniques.

1. Introduction to VQE

VQE stands for:

$$\boxed{\text{Variational Quantum Eigensolver}}$$

It is a **hybrid quantum-classical algorithm**.

It uses:

- Quantum computer → Performs quantum state preparation and measurement.
- Classical computer → Optimizes parameters.

---

2. Definition

Exam Definition:

> **Variational Quantum Eigensolver (VQE) is a hybrid quantum algorithm that finds the minimum eigenvalue of a Hamiltonian by optimizing a parameterized quantum circuit using the variational principle.**

---

3. Motivation Behind VQE

Many scientific problems require finding the lowest energy state of a system.

Example:

A molecule has many possible energy states.

The most stable state is:

$$\boxed{\text{Ground State Energy}}$$

Finding this energy is difficult for classical computers.

---

4. Basic Concept of VQE

VQE uses the idea:

> The lowest possible energy of a quantum system can be found by minimizing the expectation value of its Hamiltonian.

The energy is:

$$E = \langle \psi | H | \psi \rangle$$

---

where:

- H = Hamiltonian operator
- $|\psi\rangle$ = quantum state

---

5. Variational Principle

The foundation of VQE is the variational principle.

It states:

> For any trial quantum state, the calculated energy is always greater than or equal to the true ground state energy.

Mathematically:

$$\boxed{E(\psi) \geq E_0}$$

where:

- E_0 = actual minimum energy.

Therefore:

Finding minimum:

$$E(\theta)$$

gives the ground state.

---

6. Components of VQE

VQE contains four major components:

---

1. Parameterized Quantum Circuit (Ansatz)

A quantum circuit with adjustable parameters.

Represented as:

$$|\psi(\theta)\rangle$$

---

2. Hamiltonian Operator

Represents the energy of the system.

Example:

$$H$$

---

3. Quantum Measurement

Measures the expectation value:

$$\langle H \rangle$$

---

4. Classical Optimizer

Updates circuit parameters.

Examples:

- Gradient descent
- COBYLA
- SPSA

---

7. VQE Architecture

The complete architecture:

```
        Parameters θ
             ↓
Classical Optimizer
             |
             ↓
Parameterized Quantum Circuit
             ↓
Quantum Measurement
             ↓
Energy Calculation
             ↓
Feedback to Optimizer
             ↓
Minimum Energy State
```

8. Working of VQE Algorithm

VQE works in an iterative process.

Step 1: Initialize Parameters

Choose initial values:

$$\theta_1, \theta_2, ..., \theta_n$$

Step 2: Prepare Quantum State

The quantum circuit creates:

$$|\psi(\theta)\rangle$$

Example:

$$|\psi(\theta)\rangle = U(\theta)|0\rangle$$

where:

$$U(\theta)$$

is a parameterized unitary circuit.

Step 3: Apply Hamiltonian Measurement

The quantum computer measures:

$$E(\theta)$$

using:

$$\boxed{E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle}$$

Step 4: Classical Optimization

The classical optimizer checks:

Is energy minimum?

If:

$$E(\theta)$$

is high:

Update parameters.

Step 5: Repeat

The process continues until:

$$E(\theta)$$

converges.

Step 6: Obtain Ground State

The final optimized state:

$$|\psi(\theta^*)\rangle$$

represents the minimum energy state.

---

9. Quantum Circuit Ansatz

The ansatz is the heart of VQE.

It is a parameterized circuit.

General form:

$$|\psi(\theta)\rangle = U(\theta)|0\rangle$$

It contains:

Single Qubit Gates

Examples:

- Rotation gates
- Phase gates

Entangling Gates

Example:

- CNOT gate

Used to create correlations.

---

10. Hamiltonian Representation

A Hamiltonian describes the energy of a quantum system.

For quantum computing:

$$H = \sum_i c_i P_i$$

where:

- c_i = coefficients

- P_i = Pauli operators

Example:

$$H = c_1 I + c_2 X + c_3 Y + c_4 Z$$

11. Example: Molecular Energy Calculation

Consider a molecule:

$$H_2$$

The goal:

Find ground state energy.

Steps:

1. Encode molecule into Hamiltonian.
2. Prepare quantum state.
3. Measure energy.
4. Optimize parameters.

Final result:

Lowest energy configuration.

12. Role of Classical Optimizer

The optimizer controls:

$$\theta$$

parameters.

It minimizes:

$$E(\theta)$$

Common optimizers:

Optimizer	Type
Gradient Descent	Derivative based
COBYLA	Gradient free
SPSA	Noise tolerant

13. Advantages of VQE

1. Works on NISQ Devices

Does not require fault-tolerant quantum computers.

2. Hybrid Approach

Uses classical computers for difficult optimization.

3. Flexible

Can solve different Hamiltonians.

4. Lower Circuit Depth

Compared with some quantum algorithms.

14. Limitations of VQE

1. Optimization Challenges

Finding global minimum is difficult.

2. Quantum Noise

Measurement errors affect results.

3. Circuit Design

Choosing good ansatz is challenging.

4. Slow Convergence

Many iterations may be required.

15. Applications of VQE

1. Quantum Chemistry

Most important application.

Used for:

- Molecular simulation
- Chemical reactions

2. Drug Discovery

Simulating molecular structures.

3. Material Science

Finding new materials.

4. Physics

Solving quantum systems.

5. Machine Learning

Optimization of quantum models.

16. VQE vs Classical Methods

Feature	Classical Method	VQE
Processor	Classical computer	Quantum + Classical
State representation	Memory bits	Quantum states
Optimization	Classical algorithms	Hybrid optimization
Problem size	Limited by memory	Potentially larger
Energy calculation	Direct computation	Quantum measurement

17. VQE Algorithm Summary

Step	Operation
1	Initialize parameters
2	Prepare quantum state
3	Measure Hamiltonian
4	Calculate energy
5	Optimize parameters
6	Repeat until minimum

---

18. Frequently Asked University Questions

2-Mark Questions

1. Expand VQE.
2. What is the purpose of VQE?
3. Define variational principle.
4. What is an ansatz?

5-Mark Questions

1. Explain the working of VQE.
2. Describe components of VQE.
3. Explain the role of classical optimizer.

10-Mark Questions

1. Explain VQE architecture and working with diagram.
2. Discuss the applications and limitations of VQE.
3. Explain the variational principle used in VQE.

Formula Box (Very Important)

Quantum State:

$$\boxed{|\psi(\theta)\rangle = U(\theta)|0\rangle}$$

Energy Expectation:

$$\boxed{E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle}$$

Variational Principle:

$$\boxed{E(\psi) \geq E_0}$$

Quick Revision

- VQE = Variational Quantum Eigensolver.
- It is a hybrid quantum-classical algorithm.
- Main goal:
  - Find minimum eigenvalue of Hamiltonian.
- Uses:
  - Parameterized quantum circuit.
  - Measurement.
  - Classical optimization.
- Applications:
  - Chemistry
  - Drug discovery
  - Material science

Next Topic

Quantum Approximate Optimization Algorithm (QAOA)

We will study:

1. Introduction and motivation
2. QAOA architecture
3. Cost Hamiltonian
4. Mixing Hamiltonian
5. Working steps
6. Applications in logistics and scheduling
7. VQE vs QAOA comparison

This is another **high-priority 10-mark topic in UNIT–V**.

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 3 – Quantum Approximate Optimization Algorithm (QAOA)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> QAOA is one of the most important quantum optimization algorithms used for solving **combinatorial optimization problems** such as routing, scheduling, and resource allocation.

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1. Introduction to QAOA

QAOA stands for:

$$\boxed{\text{Quantum Approximate Optimization Algorithm}}$$

It was introduced by:

$$\boxed{\text{Farhi, Goldstone, and Gutmann (2014)}}$$

QAOA is a:

$$\boxed{\text{Hybrid Quantum-Classical Algorithm}}$$

similar to VQE.

It combines:

- Quantum circuits
- Classical optimization

to find approximate solutions to optimization problems.

---

2. Definition

Exam Definition:

> **QAOA is a hybrid quantum algorithm that solves combinatorial optimization problems by preparing parameterized quantum states and optimizing them to maximize or minimize a cost function.**

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3. Need for QAOA

Many real-world problems involve selecting the best solution among many possibilities.

Examples:

- Traveling salesman problem

- Job scheduling
- Traffic optimization
- Network design

For N variables:

Number of possible solutions:

$$2^N$$

Checking all solutions classically becomes expensive.

---

4. QAOA Basic Idea

QAOA converts an optimization problem into a quantum problem.

The main idea:

1. Represent solutions as quantum states.
2. Assign a cost value to each solution.
3. Increase probability of better solutions.
4. Measure the best solution.

---

5. QAOA Components

QAOA mainly consists of:

1. Cost Hamiltonian
2. Mixing Hamiltonian
3. Parameterized quantum circuit
4. Classical optimizer

---

6. Optimization Problem Formulation

An optimization problem is represented as:

$$\text{Minimize/Maximize } C(x)$$

where:

$$C(x)$$

is called the:

$$\boxed{\text{Cost Function}}$$

Example:

Finding minimum travel distance:

$$C(x) = \text{Total Distance}$$

7. Cost Hamiltonian

7.1 Definition

Exam Definition:

> **The cost Hamiltonian is a quantum operator that represents the objective function of an optimization problem and assigns energy values to possible solutions.**

It is represented as:

$$\boxed{H_C}$$

The eigenvalues of:

$$H_C$$

represent solution costs.

Example:

For a solution:

$$|x\rangle$$

the cost Hamiltonian gives:

$$H_C|x\rangle = C(x)|x\rangle$$

8. Mixing Hamiltonian

8.1 Definition

> **The mixing Hamiltonian is used to create transitions between possible solutions and explore the search space.**

It is represented as:

$$\boxed{H_M}$$

The common mixing Hamiltonian:

$$H_M = \sum_i X_i$$

where:

$$X_i$$

is Pauli-X operator.

9. QAOA Quantum State

The QAOA state is:

$$|\psi(\gamma, \beta)\rangle$$

It is created by applying two operators:

Cost Operator

$$U(C, \gamma)$$

Mixing Operator

$$U(B, \beta)$$

Mathematically:

$$|\psi(\gamma, \beta)\rangle = U(B, \beta)U(C, \gamma)|+\rangle$$

10. QAOA Circuit Structure

Basic QAOA circuit:

```
Initial State
     ↓
Hadamard Gates
     ↓
Cost Hamiltonian
     ↓
```

```
Mixing Hamiltonian
      ↓
Measurement
      ↓
Classical Optimization
```

11. Working of QAOA

QAOA works through iterations.

Step 1: Initialize Qubits

All qubits are placed in:

$$|+\rangle$$

state.

Using Hadamard gates:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Step 2: Apply Cost Operator

The cost operator:

$$U(C,\gamma) = e^{-i\gamma H_C}$$

adds phase based on solution quality.

Step 3: Apply Mixing Operator

The mixing operator:

$$U(B,\beta) = e^{-i\beta H_M}$$

allows movement between solutions.

Step 4: Measurement

The quantum state is measured.

The result represents a candidate solution.

Step 5: Classical Optimization

A classical optimizer updates:

$$\gamma, \beta$$

parameters.

The process repeats until the best solution is obtained.

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12. QAOA Parameter Layers

QAOA has layers represented by:

$$p$$

For:

$$p = 1$$

One cost and one mixing operation are applied.

For:

$$p > 1$$

Multiple layers are applied.

Increasing:

$$p$$

generally improves solution quality.

---

13. Mathematical Representation

The final state:

$$|\psi_p(\gamma, \beta)\rangle$$

is:

$$\boxed{|\psi_p\rangle = \prod_{k=1}^{p} U(B, \beta_k) U(C, \gamma_k) |+\rangle^{\otimes n}}$$

The objective is:

$$\boxed{\max\langle\psi_p|H_C|\psi_p\rangle}$$

14. Applications of QAOA

1. Logistics Optimization

Problems:

- Vehicle routing
- Delivery planning
- Supply chain optimization

Example:

Find shortest route for delivery trucks.

2. Scheduling Problems

Used for:

- Employee scheduling
- Machine scheduling
- University timetables

Example:

Assign classes without conflicts.

3. Traffic Optimization

Finding:

- Best traffic flow
- Shortest travel time

4. Finance

Applications:

- Portfolio optimization
- Risk management

5. Machine Learning

Used in:

- Feature selection
- Model optimization

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15. Advantages of QAOA

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1. Suitable for NISQ Devices

Works on current quantum processors.

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2. General Optimization Framework

Can solve many combinatorial problems.

---

3. Hybrid Approach

Uses classical optimization effectively.

---

4. Scalable

Performance improves with higher layers.

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16. Limitations of QAOA

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1. Approximate Solutions

Does not always guarantee the perfect solution.

---

2. Parameter Optimization Difficulty

Finding:

$$\gamma, \beta$$

is challenging.

3. Noise Sensitivity

Quantum errors affect results.

4. Circuit Depth

Higher layers require deeper circuits.

17. VQE vs QAOA Comparison

Feature	VQE	QAOA
Purpose	Find minimum energy	Solve optimization problems
Main operator	Hamiltonian	Cost + Mixing Hamiltonian
Application	Chemistry	Logistics, scheduling
Output	Ground state energy	Optimal solution
Type	Hybrid	Hybrid

18. QAOA vs Classical Optimization

Feature	Classical Optimization	QAOA
Processing	Classical bits	Qubits
Search	Sequential	Quantum state exploration
Solution	Exact sometimes	Approximate
Speed	Problem dependent	Potential quantum advantage

19. Frequently Asked University Questions

2-Mark Questions

1. Expand QAOA.
2. Who introduced QAOA?
3. What is cost Hamiltonian?
4. What is mixing Hamiltonian?

5-Mark Questions

1. Explain the working of QAOA.
2. Describe the role of cost and mixing Hamiltonians.
3. Explain QAOA applications.

10-Mark Questions

1. Explain QAOA architecture and working with equations.
2. Discuss applications of QAOA in real-world optimization.
3. Compare VQE and QAOA.

Formula Box (Very Important)

QAOA State:

$$\boxed{|\psi_p\rangle = \prod_{k=1}^{p} U(B, \beta_k) U(C, \gamma_k) |+\rangle^{\otimes n}}$$

Cost Operator:

$$\boxed{U(C, \gamma) = e^{-i\gamma H_C}}$$

Mixing Operator:

$$\boxed{U(B, \beta) = e^{-i\beta H_M}}$$

Quick Revision

- QAOA = Quantum Approximate Optimization Algorithm.
- Introduced in 2014.
- Hybrid quantum-classical algorithm.
- Uses:
  - Cost Hamiltonian
  - Mixing Hamiltonian
- Main applications:
  - Logistics
  - Scheduling

- Routing
- Finance

- Finds approximate optimal solutions.

Next Topic

Applications of Quantum Optimization

Logistics and Scheduling Problems

We will study:

1. Quantum optimization in logistics
2. Vehicle routing problem
3. Supply chain optimization
4. Scheduling optimization
5. Real-world examples
6. Advantages and limitations

This will complete the **Quantum Optimization section of UNIT–V**.

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 4 – Applications of Quantum Optimization: Logistics and Scheduling

> **Weightage:** ⭐⭐⭐⭐☆ (Important – 5 to 10 Marks Question)
> Quantum optimization has major applications in solving real-world problems where the number of possible solutions grows exponentially. Two important applications mentioned in the syllabus are **logistics** and **scheduling**.

---

1. Introduction to Quantum Optimization Applications

Many industries face optimization problems such as:

- Finding shortest routes.
- Allocating limited resources.
- Creating efficient schedules.

- Reducing operational costs.

Classical algorithms may struggle when:

- Number of variables increases.
- Search space becomes extremely large.

Quantum optimization algorithms like:

- QAOA
- VQE-based optimization methods

provide new approaches for solving these problems.

---

2. Logistics Optimization

Definition:

> **Logistics optimization is the process of finding the most efficient way to transport goods, manage resources, and minimize cost and time.**

---

3. Challenges in Logistics

A logistics company must decide:

- Which route should vehicles take?
- How should goods be distributed?
- Which vehicle should deliver which package?
- How can fuel consumption be reduced?

---

4. Important Logistics Optimization Problems

4.1 Vehicle Routing Problem (VRP)

Definition:

> **Vehicle Routing Problem is an optimization problem that determines the best routes for a fleet of vehicles to deliver goods to multiple locations while minimizing distance and cost.**

Example:

A delivery company has:

- 10 trucks
- 100 customers

The goal:

Find the shortest routes for all trucks.

Classical Challenge

If there are:

$$n$$

locations.

Possible routes:

$$(n-1)!$$

For large:

$$n$$

the number of possibilities becomes extremely large.

---

5. Quantum Approach for Vehicle Routing

QAOA can solve VRP by converting it into a mathematical optimization problem.

Step 1: Define Cost Function

The cost represents:

- Distance
- Fuel cost
- Travel time

Example:

$$C(x) = Total\ Distance$$

Step 2: Convert into Hamiltonian

The cost function is converted into:

$$H_C$$

Step 3: Apply QAOA

QAOA searches for a route with minimum cost.

Step 4: Measurement

The final quantum state gives the optimized route.

6. Applications of Quantum Optimization in Logistics

1. Route Planning

Finding:

- Shortest path
- Minimum fuel route

2. Supply Chain Optimization

Optimizing:

- Inventory
- Warehouses
- Transportation

3. Fleet Management

Improving:

- Vehicle utilization
- Delivery schedules

4. Traffic Management

Optimizing:

- Traffic signals
- Vehicle movement

---

7. Example: Delivery Route Optimization

Problem:

A company has to deliver packages to:

A, B, C, D

Possible routes:

```
Depot → A → B → C → D
```

or

```
Depot → C → A → D → B
```

Classical approach:

Checks many routes.

Quantum approach:

QAOA creates a quantum state containing possible routes and increases the probability of better routes.

---

8. Scheduling Optimization

Definition:

> **Scheduling optimization is the process of assigning tasks, resources, or activities to available time slots in the most efficient way.**

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9. Scheduling Problems

Examples:

- University timetable generation.
- Employee scheduling.

- Machine scheduling.
- Cloud resource scheduling.

---

10. Types of Scheduling Problems

10.1 Job Scheduling Problem

A company has:

- Multiple jobs.
- Multiple machines.

Need:

Assign jobs to machines efficiently.

Goal:

Minimize:

- Processing time.
- Machine idle time.

10.2 Employee Scheduling

Problem:

Assign employees to shifts.

Constraints:

- Working hours.
- Availability.
- Required staff.

10.3 University Timetable Scheduling

Need to assign:

- Subjects.
- Teachers.
- Classrooms.
- Time slots.

Without conflicts.

11. Quantum Approach for Scheduling

QAOA can represent scheduling as:

$$Optimization\ Problem$$

The objective function may minimize:

$$Cost = Conflicts + Time + Resource\ Usage$$

Then:

1. Convert cost function into Hamiltonian.
2. Apply QAOA.
3. Optimize parameters.
4. Obtain best schedule.

12. Example: University Timetable Optimization

Problem:

Create timetable for:

- 50 subjects
- 20 classrooms
- 30 teachers

Constraints:

- Teacher cannot handle two classes simultaneously.
- Classroom cannot have multiple classes.
- Students should not have overlapping subjects.

Classical approach:

Searches many combinations.

Quantum approach:

QAOA searches possible schedules efficiently.

13. Quantum Optimization Workflow for Real Problems

```
Real-World Problem
        ↓
Mathematical Model
        ↓
Cost Function Creation
        ↓
Hamiltonian Conversion
        ↓
Quantum Algorithm
(QAOA/VQE)
        ↓
Measurement
        ↓
Optimized Solution
```

14. Advantages of Quantum Optimization Applications

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1. Handles Large Search Spaces

Quantum systems can explore multiple possibilities.

---

2. Better Resource Utilization

Reduces:

- Cost
- Time
- Energy consumption

---

3. Faster Decision Making

Useful for complex problems.

---

4. Real-Time Optimization

Potentially useful in dynamic environments.

---

15. Limitations

1. Current Quantum Hardware

Limited number of qubits.

2. Noise Problems

Errors affect optimization accuracy.

3. Hybrid Requirement

Most algorithms still require classical computers.

4. No Guaranteed Speedup

Quantum advantage depends on the problem.

16. Classical vs Quantum Optimization in Logistics

Feature	Classical Optimization	Quantum Optimization
Search method	Traditional algorithms	Quantum algorithms
Representation	Bits	Qubits
Large search space	Difficult	Potential advantage
Algorithm examples	Linear programming, heuristics	QAOA, VQE
Current availability	High	Developing

17. Classical vs Quantum Scheduling

Feature	Classical Scheduling	Quantum Scheduling
Complexity handling	Limited for large problems	Better exploration
Optimization	Sequential search	Quantum parallelism
Approach	Rules and heuristics	Hamiltonian-based optimization

18. Frequently Asked University Questions

2-Mark Questions

1. Define logistics optimization.
2. What is Vehicle Routing Problem?
3. Define scheduling optimization.
4. Name the quantum algorithm used for scheduling problems.

5-Mark Questions

1. Explain quantum optimization applications in logistics.
2. Explain scheduling optimization using QAOA.
3. Describe Vehicle Routing Problem.

10-Mark Questions

1. Explain applications of quantum optimization in logistics and scheduling.
2. Describe how QAOA solves real-world optimization problems.
3. Discuss advantages and limitations of quantum optimization.

Quick Revision

- Quantum optimization solves complex decision problems.
- Important applications:
    - Logistics
    - Scheduling
    - Routing
    - Supply chain
- QAOA is mainly used.
- Steps:
    1. Create cost function.
    2. Convert to Hamiltonian.
    3. Apply quantum algorithm.
    4. Measure optimized solution.

UNIT–V Remaining Topic

Next Topic:

Quantum Machine Learning (QML)

We will cover:

1. Introduction to quantum machine learning
2. Classical ML vs Quantum ML
3. Quantum classifiers
4. Quantum neural networks
5. Applications and challenges

This is the final major concept of your syllabus.

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 5 – Quantum Machine Learning (QML): Introduction and Fundamentals

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> Quantum Machine Learning (QML) is an emerging field that combines **quantum computing** with **machine learning** to develop algorithms capable of solving complex data-processing problems.

---

1. Introduction to Quantum Machine Learning (QML)

Machine Learning (ML) allows computers to learn patterns from data and make predictions.

Examples:

- Image classification
- Speech recognition
- Recommendation systems
- Medical diagnosis

However, modern ML faces challenges:

- Huge datasets.

- High computational requirements.
- Complex optimization problems.

Quantum computing provides new methods using:

- Superposition
- Entanglement
- Quantum parallelism

to improve machine learning algorithms.

---

2. Definition of Quantum Machine Learning

Exam Definition:

> **Quantum Machine Learning is a field that combines quantum computing techniques with machine learning algorithms to process data, train models, and perform learning tasks using quantum systems.**

---

3. Relationship Between Quantum Computing and Machine Learning

Quantum Computing provides:

- Quantum data representation.
- Quantum algorithms.
- Faster optimization methods.

Machine Learning provides:

- Learning algorithms.
- Pattern recognition.
- Prediction techniques.

Together:

$$\boxed{QuantumComputing + MachineLearning = QuantumMachineLearning}$$

---

4. Why Quantum Machine Learning?

Classical machine learning problems:

- Require large memory.
- Need high computational power.

- Optimization can be expensive.

Quantum ML aims to improve:

- Data processing speed.
- Feature extraction.
- Optimization.
- Model training.

---

5. Basic Concept of QML

A typical QML system contains:

```
Classical Data
      ↓
Quantum Encoding
      ↓
Quantum Circuit
      ↓
Measurement
      ↓
Classical Processing
      ↓
Prediction / Classification
```

6. Components of Quantum Machine Learning

QML consists of four major components:

6.1 Data Encoding

Classical data must be converted into quantum states.

Methods:

- Basis encoding
- Amplitude encoding
- Angle encoding

Example:

Classical vector:

$$x = (x_1, x_2)$$

Converted into:

$$|\psi\rangle$$

6.2 Quantum Circuit

A quantum circuit processes encoded data.

It uses:

- Quantum gates
- Parameterized operations

Example:

$$U(\theta)$$

6.3 Measurement

Quantum states are measured to obtain results.

Measurement converts:

Quantum information

↓

Classical information

6.4 Classical Optimization

A classical optimizer adjusts parameters.

Examples:

- Gradient descent
- Adam optimizer

---

7. Types of Quantum Machine Learning

Important QML approaches:

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1. Quantum Classification

Used for:

- Data classification.
- Pattern recognition.

---

2. Quantum Regression

Used for:

- Predicting numerical values.

---

3. Quantum Clustering

Used for:

- Grouping similar data.

---

4. Quantum Neural Networks

Quantum version of neural networks.

---

8. Quantum Classification

Definition:

> **Quantum classification is a machine learning technique where quantum algorithms classify data into different categories using quantum circuits.**

Example:

Classifying:

- Images
- Medical data
- Text data

---

9. Quantum Classifier Working

Steps:

Step 1: Data Encoding

Input data is converted into quantum states.

Example:

$$x \rightarrow |\psi\rangle$$

Step 2: Apply Quantum Circuit

Parameterized circuit processes data.

$$U(\theta)$$

Step 3: Measurement

Output is measured.

Step 4: Classification

The result determines the class.

---

10. Quantum Support Vector Machine (QSVM)

One important quantum classifier is:

$$\boxed{\text{Quantum Support Vector Machine}}$$

It is the quantum version of classical SVM.

Classical SVM:

Finds a hyperplane separating classes.

QSVM:

Uses quantum feature mapping to represent data in high-dimensional spaces.

---

11. Quantum Feature Mapping

Feature mapping converts data into quantum states.

Mathematically:

$$x \rightarrow |\phi(x)\rangle$$

The quantum system creates complex feature spaces.

12. Advantages of Quantum Classifiers

1. High-Dimensional Feature Space

Quantum states can represent complex patterns.

2. Faster Similarity Calculation

Quantum kernels may improve classification.

3. Handles Complex Data

Useful for:

- Images
- Scientific data

13. Limitations of Quantum Classifiers

1. Data Encoding Problem

Converting large datasets into quantum states is difficult.

2. Hardware Limitations

Current quantum computers have limited qubits.

3. Noise

Quantum errors reduce accuracy.

14. Quantum Neural Networks (QNN)

Definition:

> **Quantum Neural Network is a quantum computing model inspired by classical neural networks where quantum circuits replace traditional neurons and learn parameters through optimization.**

15. Classical Neural Network vs QNN

Classical Neural Network:

```
Input
 ↓
Neurons
 ↓
Weights
 ↓
Output
```

Quantum Neural Network:

```
Input Data

 ↓

Quantum Encoding

 ↓

Parameterized Quantum Circuit

 ↓

Measurement

 ↓

Output
```

16. Structure of Quantum Neural Network

A QNN contains:

1. Input Layer

Receives encoded quantum data.

2. Quantum Circuit Layer

Contains:

- Rotation gates
- Entanglement gates

---

3. Measurement Layer

Produces output.

---

17. Working of QNN

Step 1:

Encode input data.

$$x \rightarrow |\psi(x)\rangle$$

Step 2:

Apply parameterized circuit:

$$U(\theta)$$

Step 3:

Measure output.

Step 4:

Update parameters using optimizer.

Step 5:

Repeat until accuracy improves.

---

18. Applications of Quantum Machine Learning

1. Image Processing

Applications:

- Image classification
- Object recognition

2. Healthcare

Applications:

- Disease prediction
- Drug discovery

3. Finance

Applications:

- Fraud detection
- Risk analysis

4. Cybersecurity

Applications:

- Threat detection
- Pattern analysis

5. Natural Language Processing

Applications:

- Text classification
- Language understanding

19. Advantages of QML

1. Quantum Parallelism

Processes multiple states simultaneously.

2. Improved Optimization

Quantum algorithms help solve complex optimization problems.

3. Better Feature Representation

Quantum states provide richer representations.

4. Future Scalability

Potential advantage for large datasets.

20. Challenges of QML

1. Limited Quantum Hardware

Current devices have:

- Few qubits
- High error rates

2. Data Loading Problem

Classical data conversion is expensive.

3. Noise

Quantum errors affect learning.

4. Lack of Quantum Advantage Proof

Not all ML tasks benefit from quantum computing.

21. Classical ML vs Quantum ML

Feature	Classical ML	Quantum ML
Processing unit	CPU/GPU	Quantum processor
Data unit	Bits	Qubits
Learning	Classical algorithms	Quantum circuits
Feature space	Limited	High-dimensional quantum space
Optimization	Classical	Quantum-enhanced

22. Frequently Asked University Questions

2-Mark Questions

1. Define Quantum Machine Learning.
2. What is a quantum classifier?
3. What is QNN?
4. Mention two applications of QML.

5-Mark Questions

1. Explain the concept of QML.
2. Describe quantum classifiers.
3. Explain quantum neural networks.

10-Mark Questions

1. Explain Quantum Machine Learning architecture and applications.
2. Discuss quantum classifiers and QNN.
3. Compare classical ML and quantum ML.

Formula Box (Important)

Quantum Data Encoding:

$$\boxed{x \rightarrow |\psi(x)\rangle}$$

Parameterized Quantum Circuit:

$$\boxed{|\psi(\theta)\rangle = U(\theta)|0\rangle}$$

Quick Revision

- QML combines quantum computing and machine learning.
- Main components:
  - Data encoding
  - Quantum circuit
  - Measurement
  - Optimization
- Important algorithms:
  - Quantum classifiers
  - Quantum neural networks
  - QSVM
- Applications:
  - Healthcare
  - Finance
  - Image processing
  - Cybersecurity

UNIT–V Completed ✅

Complete UNIT–V Summary:

Topic	Main Concept
Quantum Optimization	Finding best solutions using quantum algorithms
VQE	Minimum energy calculation
QAOA	Optimization problems
Logistics & Scheduling	Real-world optimization applications
QML	Machine learning using quantum systems

Full Subject Completion Status:

✅ UNIT–I Fundamentals of Quantum Computing
✅ UNIT–II Quantum Gates and Circuits
✅ UNIT–III Quantum Search Algorithms
✅ UNIT–IV Quantum Algorithms for Image Processing
✅ UNIT–V Application-Oriented Quantum Algorithms

Your complete **Generative Quantum Computing for Engineers** syllabus is now covered unit-wise with semester-oriented explanations.

> ok next

We have completed the entire **Generative Quantum Computing for Engineers syllabus (UNIT–I to UNIT–V)**.

Since you said **"ok next"**, the next step in semester preparation will be:

Quantum Computing – Semester Revision Phase

We will start with:

Topic 1 – Important 2-Mark Questions and Answers (UNIT–I)

This will help for quick exam preparation.

UNIT–I: Fundamentals of Quantum Computing

Important 2-Mark Questions with Answers

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1. What is Quantum Computing?

Answer:

Quantum computing is a computing approach that uses the principles of **quantum mechanics** to process information using quantum bits (qubits) instead of classical bits.

It uses phenomena like:

- Superposition
- Entanglement
- Quantum interference

to perform computations.

---

2. Differentiate Classical Computing and Quantum Computing.

Classical Computing	Quantum Computing
Uses bits	Uses qubits
Bit represents 0 or 1	Qubit can represent 0, 1, or both
Uses classical logic gates	Uses quantum gates
Sequential processing	Quantum parallelism
Based on Boolean logic	Based on quantum mechanics

---

3. What is a Qubit?

Answer:

A **qubit (quantum bit)** is the basic unit of quantum information.

Unlike a classical bit, which can exist only as:

$$0 \text{ or } 1$$

a qubit can exist as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where:

- α and β are probability amplitudes.

---

4. What is Superposition?

Answer:

Superposition is a quantum property where a qubit can exist in a combination of multiple states simultaneously.

Mathematically:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Before measurement, the qubit contains both possibilities.

---

5. What is Quantum Measurement?

Answer:

Quantum measurement is the process of observing a quantum state, which converts the superposition state into a definite classical state.

For:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Measurement gives:

- 0 with probability:

$$|\alpha|^2$$

- 1 with probability:

$$|\beta|^2$$

---

6. What is the Bloch Sphere?

Answer:

The Bloch sphere is a geometric representation of a single qubit state.

A qubit can be represented as:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

where:

- θ represents polar angle.
- φ represents phase angle.

---

7. What is Dirac Notation?

Answer:

Dirac notation is a mathematical notation used to represent quantum states.

It consists of:

Ket notation:

$$|\psi\rangle$$

represents a quantum state.

Bra notation:

$$\langle\psi|$$

represents the conjugate state.

8. What is Entanglement?

Answer:

Entanglement is a quantum phenomenon where two or more qubits become strongly correlated, such that the state of one qubit depends on another even when separated.

Example:

Bell state:

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

9. What are Quantum Gates?

Answer:

Quantum gates are operations that manipulate qubits while preserving quantum properties.

Examples:

- Pauli gates
- Hadamard gate
- CNOT gate
- Phase gates

10. What is Quantum Hardware?

Answer:

Quantum hardware refers to physical systems used to implement quantum computers.

Major technologies:

- Superconducting qubits
- Trapped ions
- Photonic quantum computers
- Quantum dots

11. List Quantum Computing Hardware Platforms.

Answer:

Major quantum hardware platforms:

1. Superconducting quantum computers
2. Trapped ion systems
3. Photonic quantum computers
4. Neutral atom systems
5. Quantum dots

---

12. What is Quantum Programming?

Answer:

Quantum programming is the process of creating algorithms and circuits that run on quantum computers.

It involves:

- Quantum gates
- Quantum circuits
- Qubit manipulation

---

13. Name Quantum Programming Platforms.

Answer:

Popular quantum programming platforms:

1. IBM Quantum Experience
2. Qiskit
3. Google Cirq
4. Microsoft Q#
5. Amazon Braket

---

14. What is Quantum Parallelism?

Answer:

Quantum parallelism is the ability of quantum computers to process multiple quantum states simultaneously using superposition.

15. What is Quantum Interference?

Answer:

Quantum interference is the phenomenon where quantum states combine to increase the probability of correct solutions and reduce incorrect solutions.

UNIT–I Important Long Questions (10 Marks)

1. Explain classical and quantum computing differences.
2. Explain qubits and superposition with mathematical representation.
3. Describe Bloch sphere representation of qubits.
4. Explain Dirac notation and quantum mechanics basics.
5. Explain quantum hardware platforms and programming environments.
6. Discuss quantum entanglement with examples.

Next Topic:

UNIT–I Important 5-Mark and 10-Mark Answers (Exam Writing Format)

We will prepare:

- Introduction
- Diagram
- Mathematical equations
- Advantages
- Applications
- Conclusion

for scoring maximum marks.

> ok next

UNIT–I: Fundamentals of Quantum Computing

Important 5-Mark & 10-Mark Exam Answers (Writing Format)

These answers are prepared in **semester exam style** with:

- Definition
- Explanation
- Mathematical representation
- Diagrams (where needed)
- Advantages / Applications
- Conclusion

Q1. Explain Classical Computing and Quantum Computing Differences (10 Marks)

Introduction

Computing technology has evolved from classical computers based on bits to quantum computers based on quantum mechanical principles. Classical computers process information using binary bits, whereas quantum computers use quantum bits (qubits).

Classical Computing

Classical computers use:

$$\boxed{\text{Bit}}$$

as the fundamental unit of information.

A bit can exist in only two states:

$$0 \quad \text{or} \quad 1$$

Examples:

- Computers
- Smartphones
- Servers

Quantum Computing

Quantum computers use:

$$\boxed{\text{Qubit}}$$

as the basic information unit.

A qubit can exist as a combination of states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where:

- α and β are probability amplitudes.

Comparison Between Classical and Quantum Computing

Classical Computing	Quantum Computing
Uses bits	Uses qubits
Bit represents 0 or 1	Qubit represents superposition of 0 and 1
Uses Boolean logic	Uses quantum logic
Uses logic gates	Uses quantum gates
Information processing is sequential	Uses quantum parallelism
Based on classical physics	Based on quantum mechanics
Less suitable for some complex problems	Suitable for certain complex problems

---

Key Quantum Advantages

1. Superposition

A qubit can represent multiple possibilities simultaneously.

Example:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

---

2. Entanglement

Multiple qubits can become correlated.

Example:

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

---

3. Quantum Interference

Enhances correct solutions and reduces incorrect possibilities.

Applications Comparison

Classical Computers	Quantum Computers
Web browsing	Cryptography
Office applications	Drug discovery
Gaming	Optimization problems
Database systems	Quantum simulation

Conclusion

Classical computers are powerful for everyday applications, while quantum computers provide new computational methods for solving complex scientific and optimization problems using quantum principles.

Q2. Explain Qubits and Their Representation (10 Marks)

Introduction

A qubit is the fundamental unit of quantum information. It is the quantum equivalent of a classical bit.

Definition of Qubit

A qubit is a two-level quantum system that can exist in a combination of basis states.

The general qubit state is:

$$\boxed{|\psi\rangle = \alpha|0\rangle + \beta|1\rangle}$$

Where:

- $|0\rangle$ and $|1\rangle$ are computational basis states.
- α and β are complex probability amplitudes.

Probability Conditions

The total probability must be:

$$|\alpha|^2 + |\beta|^2 = 1$$

Classical Bit vs Qubit

Classical Bit	Qubit
Stores 0 or 1	Stores superposition
Uses electrical signals	Uses quantum states
Two possible states	Infinite possible states on Bloch sphere

Properties of Qubit

1. Superposition

A qubit can exist in multiple states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

2. Measurement

Measurement collapses the qubit into:

$$|0\rangle$$

or

$$|1\rangle$$

Probability:

State 0:

$$P(0) = |\alpha|^2$$

State 1:

$$P(1) = |\beta|^2$$

3. Entanglement

Multiple qubits can share a quantum relationship.

Example:

$$|\phi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Qubit Representation Using Bloch Sphere

A single qubit can be represented as:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

where:

- θ → rotation angle
- φ → phase angle

Applications of Qubits

- Quantum cryptography
- Quantum simulation
- Quantum algorithms
- Optimization problems

Conclusion

Qubits are the foundation of quantum computing. Their ability to exist in superposition and interact through entanglement provides quantum computers with powerful computational capabilities.

Q3. Explain Bloch Sphere Representation of Qubit (5 Marks)

Introduction

The Bloch sphere is a geometric method to visualize a single qubit state.

Mathematical Representation

A qubit can be represented as:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

Components of Bloch Sphere

North Pole

Represents:

$$|0\rangle$$

South Pole

Represents:

$$|1\rangle$$

Equator

Represents equal superposition states.

Example:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Importance of Bloch Sphere

- Visualizes qubit states.
- Represents quantum operations.
- Helps understand rotations.

Quantum Gates on Bloch Sphere

Examples:

X Gate

Rotation around X-axis.

Y Gate

Rotation around Y-axis.

Z Gate

Rotation around Z-axis.

Conclusion

Bloch sphere provides an intuitive representation of qubit states and helps analyze quantum operations.

UNIT–I Remaining Topics:

Next we will cover:

Dirac Notation, Quantum Mechanics Basics, Entanglement and Quantum Hardware Overview (10 Marks Answers)

These are high-weightage theory questions in semester exams.

> ok next

UNIT–I: Fundamentals of Quantum Computing

Topic 4 – Dirac Notation, Quantum Mechanics Basics, and Quantum States

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 5 & 10 Marks Question)
> Dirac notation is the mathematical language used to represent quantum states, operations, and measurements in quantum computing.

---

1. Introduction to Quantum Mechanics in Computing

Quantum computing is based on the principles of **quantum mechanics**, which describes the behavior of very small particles such as:

- Electrons
- Photons
- Atoms

Unlike classical physics, quantum mechanics allows particles to exist in multiple states and exhibit unique properties such as:

- Superposition
- Entanglement
- Interference

---

2. Basic Principles of Quantum Mechanics

The important quantum mechanical concepts used in quantum computing are:

1. Quantum states
2. Superposition

3. Measurement
4. Probability amplitudes
5. Entanglement

---

3. Quantum State

A quantum state describes the condition of a quantum system.

It is represented using:

$$|\psi\rangle$$

called a:

$$\boxed{\text{Ket}}$$

Example:

A qubit state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where:

- $|0\rangle$ and $|1\rangle$ are basis states.
- α and β are probability amplitudes.

---

4. Dirac Notation (Bra-Ket Notation)

Definition:

> **Dirac notation is a mathematical notation used to represent quantum states and perform calculations in quantum mechanics.**

It was introduced by:

$$\boxed{\text{Paul Dirac}}$$

It consists of two parts:

1. Ket
2. Bra

---

5. Ket Notation

A ket represents a quantum state.

Symbol:

$$|\psi\rangle$$

Examples:

Zero state:

$$|0\rangle$$

One state:

$$|1\rangle$$

A qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

---

6. Bra Notation

A bra represents the conjugate transpose of a ket.

Symbol:

$$\langle\psi|$$

If:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Then:

$$\langle\psi| = \alpha^*\langle 0| + \beta^*\langle 1|$$

---

where:

$$*$$

represents complex conjugate.

---

7. Inner Product

The inner product combines a bra and ket:

$$\langle\phi|\psi\rangle$$

It gives a scalar value representing the overlap between two quantum states.

Example:

$$\langle 0|0\rangle = 1$$

$$\langle 0|1\rangle = 0$$

Properties:

Same states:

$$\langle \psi|\psi\rangle = 1$$

(normalization)

Different orthogonal states:

$$\langle 0|1\rangle = 0$$

8. Outer Product

The outer product is:

$$|\psi\rangle\langle\phi|$$

It produces a matrix/operator.

Used in:

- Quantum gates
- Quantum measurements

Example:

$$|0\rangle\langle 0|$$

represents projection onto state 0.

9. Quantum State Normalization

A quantum state must satisfy:

$$\boxed{\langle \psi|\psi\rangle = 1}$$

For:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

the condition is:

$$|\alpha|^2 + |\beta|^2 = 1$$

---

10. Quantum Measurement

Definition:

> **Quantum measurement is the process of observing a quantum state, causing it to collapse into one of its basis states.**

Before measurement:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After measurement:

Either:

$$|0\rangle$$

or:

$$|1\rangle$$

---

11. Measurement Probability

For:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Probability of obtaining:

State 0:

$$P(0) = |\alpha|^2$$

State 1:

$$P(1) = |\beta|^2$$

---

12. Quantum Operators

Quantum operations are represented by operators.

A quantum gate performs:

$$|\psi'\rangle = U|\psi\rangle$$

---

where:

- U = unitary operator
- $|\psi\rangle$ = input state

---

13. Unitary Operators

A quantum operation must satisfy:

$$U^\dagger U = I$$

Properties:

- Reversible
- Preserve probability

Examples:

- Hadamard gate
- Pauli gates
- Rotation gates

---

14. Superposition Using Dirac Notation

A qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

contains both states before measurement.

Example:

Equal superposition:

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Probability:

$$P(0) = \frac{1}{2}$$

$$P(1) = \frac{1}{2}$$

15. Importance of Dirac Notation in Quantum Computing

Dirac notation is used for:

- Representing qubits.
- Describing quantum gates.
- Expressing quantum algorithms.
- Representing entangled states.
- Describing measurements.

16. Summary Table

Concept	Representation
Quantum state	(
Ket	(
Bra	(\langle\psi
Inner product	(\langle\phi
Outer product	(
Normalization	(\langle\psi
Measurement probability	(

17. Frequently Asked University Questions

2-Mark Questions

1. What is Dirac notation?
2. Define ket notation.
3. Define bra notation.
4. What is normalization condition?
5. What is quantum measurement?

5-Mark Questions

1. Explain Dirac notation with examples.
2. Explain quantum measurement.
3. Describe quantum states and probability amplitudes.

10-Mark Questions

1. Explain Dirac notation and its importance in quantum computing.
2. Explain basic principles of quantum mechanics used in quantum computing.
3. Describe quantum states, measurement, and operators.

Quick Revision

- Dirac notation is the language of quantum mechanics.
- Ket:

$$|\psi\rangle$$

represents quantum state.

- Bra:

$$\langle\psi|$$

represents conjugate state.

- Inner product:

$$\langle\phi|\psi\rangle$$

gives overlap.

- Measurement collapses quantum states.
- Quantum states must satisfy:

$$\langle\psi|\psi\rangle = 1$$

Next Topic:

Quantum Entanglement and Quantum Hardware Overview

We will cover:

1. Entanglement concept
2. Bell states

3. Applications of entanglement
4. Quantum hardware technologies
5. Superconducting qubits
6. Trapped ion computers
7. Photonic quantum computers
8. Quantum programming platforms

(Important for UNIT–I completion)

> ok next

UNIT–I: Fundamentals of Quantum Computing

Topic 5 – Quantum Entanglement and Quantum Hardware Overview

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> This topic covers two important concepts:
>
> 1. **Quantum Entanglement** – a fundamental quantum phenomenon
> 2. **Quantum Hardware** – physical systems used to build quantum computers

---

PART–A: Quantum Entanglement

---

1. Introduction to Entanglement

In classical computing, information between two systems is independent.

Example:

Two classical bits:

$$A = 0, \quad B = 1$$

The state of one bit does not affect the other.

In quantum computing, two or more qubits can become connected through a special relationship called **entanglement**.

---

2. Definition of Quantum Entanglement

Exam Definition:

Quantum entanglement is a quantum mechanical phenomenon in which two or more qubits become correlated such that the state of one qubit is directly related to the state of another qubit, even when they are separated by large distances.

---

3. Basic Idea of Entanglement

Two qubits can exist in a combined quantum state:

$$|\psi\rangle_{AB}$$

where:

- A → first qubit
- B → second qubit

The state cannot be described independently.

---

4. Mathematical Representation of Entanglement

A general two-qubit state:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

If the state cannot be separated into:

$$|\psi_A\rangle \otimes |\psi_B\rangle$$

then the qubits are entangled.

---

5. Bell States (Important)

Bell states are the simplest examples of maximally entangled states.

There are four Bell states:

---

1. First Bell State

$$\boxed{|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

Meaning:

If first qubit is measured as 0:

Second qubit will always be 0.

If first qubit is 1:

Second qubit will always be 1.

---

2. Second Bell State

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

---

3. Third Bell State

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

---

4. Fourth Bell State

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

---

6. Creating Entanglement Using Quantum Gates

Entanglement is usually created using:

1. Hadamard Gate
2. CNOT Gate

Circuit:

```
Qubit 1 ──H─────●──
                │
Qubit 2 ────────X──
```

Steps:

Step 1:

Initial state:

$$|00\rangle$$

Step 2:

Apply Hadamard gate:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Step 3:

Apply CNOT gate:

Final state:

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

---

7. Properties of Entanglement

---

1. Non-local Correlation

The states of qubits are connected regardless of distance.

---

2. Cannot be Explained Classically

Classical probability cannot reproduce entanglement behavior.

---

3. Measurement Correlation

Measuring one qubit determines information about another.

---

8. Applications of Quantum Entanglement

---

1. Quantum Cryptography

Used in:

- Secure communication
- Quantum key distribution

Example:

$$\text{BB84 Protocol}$$

---

2. Quantum Teleportation

Transfers quantum states between locations.

3. Quantum Computing

Used in:

- Quantum algorithms
- Quantum circuits

4. Quantum Networking

Enables future quantum internet.

9. Entanglement vs Superposition

Superposition	Entanglement
Single qubit phenomenon	Multiple qubit phenomenon
Combination of states	Correlation between states
Example: (\alpha	0\rangle+\beta
Creates quantum parallelism	Creates quantum connections

PART–B: Quantum Hardware Overview

10. Introduction to Quantum Hardware

Quantum hardware refers to the physical systems that implement quantum computation.

A quantum computer requires:

- Qubits
- Control systems
- Measurement devices
- Error correction techniques

11. Requirements of Quantum Hardware

An ideal quantum computer should satisfy:

1. Qubit Stability

Qubits should maintain their quantum state.

---

2. Low Error Rate

Quantum operations should be accurate.

---

3. Scalability

Should support millions of qubits.

---

4. Fast Measurement

Results should be measured accurately.

---

12. Major Quantum Hardware Technologies

Important platforms:

1. Superconducting Qubits
2. Trapped Ion Quantum Computers
3. Photonic Quantum Computers
4. Quantum Dots
5. Neutral Atom Systems

---

13. Superconducting Qubits

Definition:

> **Superconducting quantum computers use electrical circuits cooled to extremely low temperatures to create artificial atoms that behave as qubits.**

Examples:

- IBM Quantum computers

- Google Quantum processors

Working Principle

Uses:

$$\text{Josephson Junction}$$

A superconducting circuit behaves like a quantum two-level system.

Advantages

- Fast gate operations
- Easy integration with electronics

Limitations

- Requires extremely low temperature
- Sensitive to noise

14. Trapped Ion Quantum Computers

Definition:

> **Trapped ion quantum computers use charged atoms trapped using electromagnetic fields as qubits.**

Examples:

- IonQ systems

Working Principle

Information is stored in:

- Electronic energy levels of ions

Advantages

- Very high accuracy
- Long coherence time

Limitations

- Slow operation speed
- Difficult scaling

---

15. Photonic Quantum Computers

Definition:

> **Photonic quantum computers use photons (particles of light) as qubits.**

Information is encoded using:

- Photon polarization
- Phase

Advantages

- Operate at room temperature
- Fast communication

Limitations

- Difficult photon interaction
- Complex hardware

---

16. Quantum Dots

Definition:

Quantum dots use electrons confined in semiconductor structures as qubits.

Advantages:

- Similar to classical semiconductor technology

Limitations:

- Difficult control

17. Comparison of Quantum Hardware Platforms

Technology	Qubit Type	Advantage	Limitation
Superconducting	Electrical circuits	Fast operations	Requires cooling
Trapped Ion	Atoms	High accuracy	Slow
Photonic	Photons	Long distance communication	Hard control
Quantum Dots	Electrons	Semiconductor compatible	Complex fabrication

18. Quantum Programming Platforms

Quantum programming platforms allow developers to create quantum circuits and run them on simulators or real quantum computers.

19. Major Quantum Programming Platforms

1. IBM Qiskit

Developed by:

IBM

Features:

- Python-based
- Circuit design
- Quantum simulation

2. Google Cirq

Developed by:

Google

Features:

- Quantum circuit development
- Research applications

---

3. Microsoft Q#

Features:

- Quantum algorithm development
- Integration with .NET

---

4. Amazon Braket

Provides:

- Cloud access to multiple quantum hardware systems

---

20. Applications of Quantum Programming

- Designing quantum algorithms
- Testing quantum circuits
- Running simulations
- Developing quantum applications

Frequently Asked University Questions

2-Mark Questions

1. Define quantum entanglement.
2. What are Bell states?
3. Mention two quantum hardware technologies.
4. What is a superconducting qubit?
5. Name quantum programming platforms.

5-Mark Questions

1. Explain quantum entanglement with examples.
2. Describe different quantum hardware technologies.

3. Explain quantum programming platforms.

10-Mark Questions

1. Explain quantum entanglement and its applications.
2. Discuss different quantum hardware architectures.
3. Explain quantum programming platforms and their features.

UNIT–I Complete Summary ✅

Concept	Key Idea
Classical vs Quantum	Bits vs Qubits
Qubit	Quantum information unit
Superposition	Multiple states
Measurement	State collapse
Bloch Sphere	Qubit visualization
Dirac Notation	Quantum mathematics
Entanglement	Quantum correlation
Hardware	Physical quantum systems
Platforms	Qiskit, Cirq, Q#

Next Topic:

UNIT–II: Quantum Gates and Circuits

We will start with:

1. Introduction to Quantum Gates
2. Pauli Gates (X, Y, Z)
3. Hadamard Gate
4. Phase Gates
5. Rotation Gates

(UNIT–II is highly important for circuit-based questions.)

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 1 – Introduction to Quantum Gates and Quantum Operations

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 5 & 10 Marks Question)
> Quantum gates are the fundamental building blocks of quantum circuits. They perform operations on qubits similar to how classical logic gates operate on bits.

---

1. Introduction to Quantum Gates

In classical computing:

- Information is processed using logic gates.
- Examples:
  - AND gate
  - OR gate
  - NOT gate

In quantum computing:

- Information is processed using **quantum gates**.
- Quantum gates manipulate the state of qubits.

---

2. Definition of Quantum Gate

Exam Definition:

> **A quantum gate is a reversible mathematical operation that transforms one quantum state into another while preserving the probability of the quantum system.**

A quantum gate is represented as:

$$\boxed{U}$$

and operates as:

$$|\psi'\rangle = U|\psi\rangle$$

where:

- $|\psi\rangle$ = input quantum state
- $|\psi'\rangle$ = output quantum state

- U = quantum gate operator

---

3. Properties of Quantum Gates

Quantum gates have the following properties:

---

1. Unitary Operation

A quantum gate must satisfy:

$$\boxed{U^\dagger U = I}$$

where:

- $U^\dagger$ = conjugate transpose of U
- I = identity matrix

---

2. Reversible

Every quantum operation can be reversed.

If:

$$|\psi'\rangle = U|\psi\rangle$$

then:

$$|\psi\rangle = U^{-1}|\psi'\rangle$$

---

3. Preserve Probability

Quantum gates maintain:

$$|\alpha|^2 + |\beta|^2 = 1$$

---

4. Quantum State Representation

A single qubit is represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Matrix representation:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

A quantum gate is a matrix:

$$U = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

After applying gate:

$$|\psi'\rangle = U|\psi\rangle$$

---

5. Single-Qubit Quantum Gates

Single-qubit gates operate on one qubit.

Important single-qubit gates:

1. Pauli Gates
2. Hadamard Gate
3. Phase Gates
4. Rotation Gates

---

6. Multi-Qubit Quantum Gates

Multi-qubit gates operate on two or more qubits.

Examples:

- CNOT Gate
- Toffoli Gate
- SWAP Gate

---

7. Quantum Circuit

Definition:

> **A quantum circuit is a sequence of quantum gates applied to qubits to perform quantum computation.**

A quantum circuit contains:

1. Input qubits

2. Quantum gates
3. Measurement operations

---

8. Representation of Quantum Circuit

Example:

Single qubit circuit:

```
|ψ⟩ ──H──X──M
```

where:

- H → Hadamard gate
- X → Pauli-X gate
- M → Measurement

---

9. Quantum Circuit Components

---

1. Qubit Lines

Represent quantum information flow.

Example:

```
|0⟩ ─────────
```

2. Quantum Gates

Operations performed on qubits.

Example:

```
──H──
```

3. Measurement

Converts quantum information into classical output.

Symbol:

```
──M──
```

10. Difference Between Classical and Quantum Gates

Classical Gates	Quantum Gates
Operate on bits	Operate on qubits
May be irreversible	Always reversible
Boolean operations	Unitary operations
AND, OR, NOT	Pauli, Hadamard, CNOT

11. Quantum Gate Classification

Quantum gates are classified into:

A. Single-Qubit Gates

Operate on one qubit.

Examples:

- X gate
- Y gate
- Z gate
- H gate
- S gate
- T gate

B. Multi-Qubit Gates

Operate on multiple qubits.

Examples:

- CNOT
- Toffoli
- SWAP

12. Universal Quantum Gates

Definition:

> **A universal quantum gate set is a collection of quantum gates that can perform any quantum computation when combined together.**

Examples:

- Hadamard + CNOT + Rotation gates

13. Importance of Quantum Gates

Quantum gates are important because they:

- Create superposition.
- Generate entanglement.
- Perform quantum algorithms.
- Transform quantum states.

14. Applications of Quantum Gates

Used in:

- Quantum algorithms
- Quantum simulation
- Quantum cryptography
- Quantum machine learning

15. Important Quantum Gate Symbols

Gate	Symbol	Purpose
Pauli-X	X	Bit flip
Pauli-Y	Y	Bit and phase flip
Pauli-Z	Z	Phase flip
Hadamard	H	Creates superposition
Phase	S,T	Changes phase
CNOT	⊕	Creates entanglement

16. Frequently Asked University Questions

2-Mark Questions

1. Define quantum gate.
2. What is a unitary operation?

3. What is a quantum circuit?
4. Define universal quantum gate.
5. List single-qubit gates.

5-Mark Questions

1. Explain quantum gates and their properties.
2. Describe quantum circuits.
3. Explain classification of quantum gates.

10-Mark Questions

1. Explain quantum gates with mathematical representation.
2. Describe quantum circuits and reversible computation.
3. Explain the importance of quantum gates in quantum computing.

Quick Revision

- Quantum gates manipulate qubits.
- Represented by unitary matrices.
- Must be reversible.
- Main categories:
  - Single-qubit gates
  - Multi-qubit gates
- Quantum circuits are combinations of quantum gates.

Next Topic:

Pauli Quantum Gates (X, Y, Z Gates)

We will cover:

1. Pauli-X gate (Bit flip)
2. Pauli-Y gate (Bit + phase flip)
3. Pauli-Z gate (Phase flip)
4. Matrix representation

5. Effect on qubit states
6. Circuit examples

(Important for UNIT–II numerical and theory questions.)

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 2 – Pauli Quantum Gates (X, Y, Z Gates)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 5 & 10 Marks Question)
> Pauli gates are the basic single-qubit quantum gates. They are named after physicist **Wolfgang Pauli** and are represented using Pauli matrices.

---

1. Introduction to Pauli Gates

Pauli gates are single-qubit gates that perform transformations on quantum states.

There are three Pauli gates:

1. Pauli-X Gate
2. Pauli-Y Gate
3. Pauli-Z Gate

They are represented by:

$$\boxed{X, Y, Z}$$

---

2. Pauli Gate Mathematical Representation

A qubit state is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Matrix form:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Each Pauli gate is represented by a unitary matrix.

---

3. Pauli-X Gate (Bit Flip Gate)

Definition:

Pauli-X gate flips the state of a qubit from $|0\rangle$ to $|1\rangle$ and $|1\rangle$ to $|0\rangle$.

It is equivalent to the classical NOT gate.

Matrix Representation

$$\boxed{X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}$$

Operation on Basis States

For state:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Applying X:

$$X|0\rangle$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Result:

$$\boxed{|1\rangle}$$

Similarly:

$$\boxed{X|1\rangle = |0\rangle}$$

Effect on General Qubit

Given:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After X gate:

$$X|\psi\rangle = \alpha|1\rangle + \beta|0\rangle$$

Bloch Sphere Interpretation

Pauli-X rotates the qubit:

$$180^\circ$$

around the X-axis.

Circuit Symbol

```
|ψ⟩ ──X──
```

Applications

- Bit flipping
- Quantum error correction
- Quantum algorithms

---

4. Pauli-Y Gate

Definition:

Pauli-Y gate performs both bit flip and phase flip operations on a qubit.

Matrix Representation

$$\boxed{Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}}$$

Operation on Basis States

For:

$$|0\rangle$$

Applying Y:

$$Y|0\rangle = i|1\rangle$$

For:

$$|1\rangle$$

$$Y|1\rangle = -i|0\rangle$$

The factor:

$$i$$

represents a phase change.

Effect on General Qubit

For:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After Y:

$$Y|\psi\rangle = -i\beta|0\rangle + i\alpha|1\rangle$$

Bloch Sphere Interpretation

Pauli-Y rotates the state:

$$180^\circ$$

around the Y-axis.

Circuit Symbol

```
|ψ⟩ ──Y──
```

Applications

- Quantum state manipulation
- Quantum error correction

---

5. Pauli-Z Gate (Phase Flip Gate)

Definition:

Pauli-Z gate changes the phase of the |1⟩ state while keeping the |0⟩ state unchanged.

Matrix Representation

$$\boxed{Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}$$

Operation on Basis States

For:

$$|0\rangle$$

$$Z|0\rangle = |0\rangle$$

For:

$$|1\rangle$$

$$Z|1\rangle = -|1\rangle$$

Effect on General Qubit

For:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After Z:

$$\boxed{Z|\psi\rangle = \alpha|0\rangle - \beta|1\rangle}$$

Bloch Sphere Interpretation

Pauli-Z rotates the qubit:

$$180^\circ$$

around the Z-axis.

Circuit Symbol

```
|ψ⟩ ──Z──
```

Applications

- Phase correction
- Quantum algorithms
- Error correction

6. Comparison of Pauli Gates

Gate	Matrix	Operation	Rotation
X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Bit flip	X-axis
Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	Bit + phase flip	Y-axis
Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Phase flip	Z-axis

7. Pauli Gates and Classical Logic

Classical Operation	Quantum Equivalent
NOT Gate	Pauli-X
Phase operation	Pauli-Z
Combined operation	Pauli-Y

8. Important Properties of Pauli Gates

All Pauli gates are:

1. Unitary

$$U^\dagger U = I$$

2. Hermitian

$$U^\dagger = U$$

3. Self-Inverse

Applying twice returns original state:

$$X^2 = Y^2 = Z^2 = I$$

Example:

$$X(X|0\rangle) = |0\rangle$$

9. Pauli Gates in Quantum Error Correction

Quantum systems are sensitive to errors.

Common errors:

1. Bit flip error
2. Phase flip error

Correction:

Error	Correction Gate
Bit flip	X gate
Phase flip	Z gate
Both errors	Y gate

---

10. Frequently Asked University Questions

2-Mark Questions

1. What is Pauli-X gate?
2. Write the matrix of Pauli-Z gate.
3. Which gate performs bit flip operation?
4. What is the effect of Pauli-Y gate?

5-Mark Questions

1. Explain Pauli gates with matrices.
2. Describe Pauli-X, Y, and Z operations.
3. Explain applications of Pauli gates.

10-Mark Questions

1. Explain Pauli quantum gates with mathematical representation.
2. Describe the effect of Pauli gates on qubit states.
3. Compare different Pauli gates.

Quick Revision

Pauli-X:

$$|0\rangle \leftrightarrow |1\rangle$$

(Bit flip)

Pauli-Y:

$$\text{Bit flip} + \text{Phase flip}$$

Pauli-Z:

$$|1\rangle \rightarrow -|1\rangle$$

(Phase flip)

Next Topic:

Hadamard Gate (H Gate)

We will cover:

1. Definition
2. Matrix representation
3. Creating superposition
4. Effect on |0⟩ and |1⟩ states
5. Quantum circuit examples
6. Applications

(Hadamard gate is one of the **most important gates in quantum computing**.)

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 3 – Hadamard Gate (H Gate)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 5 & 10 Marks Question)
> The Hadamard gate is one of the most important single-qubit quantum gates because it creates **superposition states**, which is the foundation of quantum parallelism.

---

1. Introduction to Hadamard Gate

The Hadamard gate is a single-qubit quantum gate represented by:

$$\boxed{H}$$

It transforms computational basis states into superposition states.

Definition:

> **The Hadamard gate is a quantum gate that creates an equal superposition of the basis states |0⟩ and |1⟩ by applying a 90° rotation followed by a reflection on the Bloch sphere.**

---

2. Matrix Representation of Hadamard Gate

The Hadamard gate is represented by the matrix:

$$\boxed{H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}$$

For a qubit:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

After applying H:

$$|\psi'\rangle = H|\psi\rangle$$

---

3. Operation of Hadamard Gate on Basis States

The two computational basis states are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

---

4. Applying Hadamard to |0⟩

$$H|0\rangle$$

Substitute:

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Result:

$$\boxed{H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}}$$

This is called:

$$\boxed{\text{Equal Superposition State}}$$

Probability:

$$P(0) = \frac{1}{2}$$

$$P(1) = \frac{1}{2}$$

---

5. Applying Hadamard to $|1\rangle$

$$H|1\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Result:

$$\boxed{H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}}$$

---

6. Hadamard Creates Superposition

Before applying H:

$$|0\rangle$$

is a definite state.

After applying H:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

the qubit exists in both states simultaneously.

This property enables:

- Quantum parallelism
- Quantum algorithms

---

7. Hadamard Gate on General Qubit

Given:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Applying H:

$$H|\psi\rangle$$

$$= \frac{1}{\sqrt{2}}\left[(\alpha+\beta)|0\rangle + (\alpha-\beta)|1\rangle\right]$$

---

8. Circuit Representation

The symbol of Hadamard gate:

```
|0⟩ ──H──
```

Creating superposition:

```
|0⟩ ──H──
```

Output:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

---

9. Hadamard Gate and Quantum Measurement

Suppose:

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

After measurement:

Probability of:

$$|0\rangle = \frac{1}{2}$$

Probability of:

$$|1\rangle = \frac{1}{2}$$

Measurement destroys the superposition.

10. Important Properties of Hadamard Gate

1. Unitary Operation

$$H^\dagger H = I$$

2. Self-Inverse

Applying H twice returns the original state:

$$\boxed{H^2 = I}$$

Example:

$$H(H|0\rangle) = |0\rangle$$

3. Creates Superposition

Main function of H gate.

11. Bloch Sphere Interpretation

The Hadamard gate performs a rotation on the Bloch sphere.

It transforms:

$$|0\rangle$$

into:

$$|+\rangle$$

where:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

It transforms:

$$|1\rangle$$

into:

$$|-\rangle$$

where:

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

---

12. Hadamard Basis States

The states created by Hadamard are called:

$$\boxed{\text{Hadamard Basis}}$$

They are:

Positive State:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Negative State:

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

---

13. Applications of Hadamard Gate

---

1. Quantum Algorithms

Used in:

- Grover's algorithm
- Shor's algorithm

---

2. Quantum Parallelism

Creates multiple possibilities simultaneously.

---

3. Entanglement Creation

Combined with CNOT gate:

$$H + CNOT$$

creates Bell states.

---

4. Quantum Cryptography

Used in quantum key distribution protocols.

---

14. Creating Entanglement Using H Gate

Circuit:

```
Qubit 1 —H———●———
           |
Qubit 2 ————X———
```

Initial state:

$$|00\rangle$$

After H:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

After CNOT:

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

A Bell state is created.

---

15. Hadamard Gate vs Pauli Gates

Feature	Hadamard Gate	Pauli Gates
Symbol	H	X,Y,Z
Main function	Creates superposition	State transformation
Operation	Basis change	Flip/phase operation
Single qubit	Yes	Yes

---

16. Frequently Asked University Questions

2-Mark Questions

1. Define Hadamard gate.
2. Write the matrix representation of H gate.
3. What is the purpose of Hadamard gate?
4. What state is produced by H|0⟩?

5-Mark Questions

1. Explain Hadamard gate with matrix representation.
2. Explain how H gate creates superposition.
3. Describe applications of Hadamard gate.

10-Mark Questions

1. Explain Hadamard gate and its role in quantum computing.
2. Derive the output states of H|0⟩ and H|1⟩.
3. Explain creation of superposition and entanglement using Hadamard gate.

Quick Revision

Hadamard Matrix:

$$\boxed{H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}$$

Important Transformations:

$$\boxed{H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}}$$

$$\boxed{H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}}$$

Key Point:

Hadamard gate converts definite quantum states into superposition states and is the starting point of many quantum algorithms.

Next Topic:

Phase Gates (S Gate and T Gate)

We will cover:

1. Phase shift concept
2. S gate
3. T gate
4. Matrix representation
5. Effect on qubit states
6. Applications in quantum circuits

(Important for understanding rotation and advanced quantum gates.)

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 4 – Phase Gates (S Gate and T Gate)

> **Weightage:** ⭐⭐⭐⭐ (Important – 5 Marks Question)
> Phase gates are single-qubit quantum gates that modify the **phase of quantum states** without changing their probability amplitudes.

---

1. Introduction to Phase Gates

In quantum mechanics, every qubit has two important properties:

1. **Amplitude** → determines probability of measurement.
2. **Phase** → determines interference behavior.

Phase gates modify the phase component of a qubit.

Definition:

> **A phase gate is a quantum gate that changes the phase of the $|1\rangle$ state while keeping the $|0\rangle$ state unchanged.**

---

2. General Phase Shift Gate

A general phase gate is represented as:

$$\boxed{P(\theta)}$$

Matrix representation:

$$\boxed{P(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}}$$

Where:

- θ = phase angle
- $e^{i\theta}$ = phase factor

---

3. Operation on Qubit

For a general qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After applying phase gate:

$$P(\theta)|\psi\rangle$$

Result:

$$\boxed{\alpha|0\rangle + e^{i\theta}\beta|1\rangle}$$

Only the phase of the $|1\rangle$ component changes.

---

4. S Gate (Phase Gate)

Definition:

> **The S gate is a phase gate that rotates the phase of a qubit by 90° (π/2 radians).**

It is also called:

$$\boxed{\sqrt{Z}\text{ gate}}$$

because:

$$S^2 = Z$$

---

5. Matrix Representation of S Gate

$$\boxed{S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}}$$

---

where:

$$i = e^{i\pi/2}$$

---

6. Effect of S Gate on Basis States

For $|0\rangle$:

$$S|0\rangle = |0\rangle$$

No change.

For $|1\rangle$:

$$S|1\rangle = i|1\rangle$$

Phase changes by:

$$90^\circ$$

---

7. Effect on General Qubit

Given:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After S gate:

$$\boxed{S|\psi\rangle = \alpha|0\rangle + i\beta|1\rangle}$$

---

8. Bloch Sphere Interpretation of S Gate

The S gate rotates the qubit state around the Z-axis by:

$$\frac{\pi}{2}$$

or:

90°

9. S Gate Circuit Symbol

```
|ψ⟩ ──S──
```

10. Applications of S Gate

Used in:

- Quantum phase manipulation
- Quantum Fourier Transform
- Quantum algorithms
- Error correction

---

11. T Gate (π/8 Gate)

Definition:

The T gate is a phase gate that rotates the phase of the |1⟩ state by 45° (π/4 radians).

It is also called:

$$\boxed{\pi/8 \text{ gate}}$$

---

12. Matrix Representation of T Gate

$$\boxed{T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}}$$

or:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix}$$

---

13. Effect of T Gate on Basis States

For |0⟩:

$$T|0\rangle = |0\rangle$$

For $|1\rangle$:

$$T|1\rangle = e^{i\pi/4}|1\rangle$$

14. Effect on General Qubit

Given:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After T gate:

$$\boxed{T|\psi\rangle = \alpha|0\rangle + e^{i\pi/4}\beta|1\rangle}$$

15. Relationship Between T, S and Z Gates

Important relationships:

$$\boxed{T^2 = S}$$

$$\boxed{S^2 = Z}$$

Therefore:

$$\boxed{T^4 = Z}$$

16. Comparison Between Phase Gates

Feature	S Gate	T Gate
Other name	√Z Gate	π/8 Gate
Phase rotation	90°	45°
Angle	π/2	π/4
Matrix	$\begin{bmatrix}1 & 0\\0 & i\end{bmatrix}$	$\begin{bmatrix}1 & 0\\0 & e^{i\pi/4}\end{bmatrix}$
Operation	Phase shift	Smaller phase shift

17. Phase Gate vs Pauli-Z Gate

Pauli-Z	Phase Gate
Phase change = 180°	Phase change can be any angle
Matrix: Z	Matrix: P(θ)
Special case of phase gate	General phase operation

18. Importance of Phase Gates

Phase gates are important because:

1. Control Quantum Interference

They modify phase relationships between states.

2. Quantum Algorithms

Used in:

- Quantum Fourier Transform
- Shor's Algorithm

3. Universal Quantum Computation

Combined with:

- Hadamard gate
- CNOT gate

they form a universal gate set.

19. Frequently Asked University Questions

2-Mark Questions

1. What is a phase gate?
2. Write the matrix of S gate.
3. What is the phase angle of T gate?
4. Why is T gate called π/8 gate?

5-Mark Questions

1. Explain S and T gates with matrices.
2. Describe phase shift operation.
3. Compare S and T gates.

10-Mark Questions

1. Explain phase gates and their applications.
2. Derive the effect of S and T gates on qubit states.
3. Explain the importance of phase gates in quantum computation.

Quick Revision

S Gate:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

Phase rotation:

$$90^\circ$$

T Gate:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Phase rotation:

$$45^\circ$$

Key Concept:

Phase gates do not change measurement probabilities; they only modify the phase of quantum states, which affects interference.

Next Topic:

Rotation Gates (Rx, Ry, Rz Gates)

We will cover:

1. Concept of quantum rotations
2. Rx gate
3. Ry gate
4. Rz gate
5. Matrix representations
6. Bloch sphere rotations

7. Applications in quantum circuits

(Important for advanced quantum circuit design.)

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 5 – Rotation Gates (Rx, Ry, Rz Gates)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 5 & 10 Marks Question)
> Rotation gates are fundamental single-qubit gates used to rotate the state of a qubit around the **X, Y, and Z axes** of the Bloch sphere.

---

1. Introduction to Rotation Gates

A qubit state can be visualized as a point on the **Bloch sphere**.

Quantum operations can rotate this point around different axes.

The three basic rotation gates are:

1. **Rx Gate** → Rotation around X-axis
2. **Ry Gate** → Rotation around Y-axis
3. **Rz Gate** → Rotation around Z-axis

---

2. Definition of Rotation Gate

Exam Definition:

> **A rotation gate is a quantum gate that rotates the state vector of a qubit around a specific axis of the Bloch sphere by a given angle.**

General form:

$$\boxed{R(\theta) = e^{-i\theta\sigma/2}}$$

where:

- θ = rotation angle
- σ = Pauli matrix (X, Y, or Z)

---

3. General Rotation Operator

A rotation operation is represented as:

$$|\psi'\rangle = R(\theta)|\psi\rangle$$

The rotation depends on the axis:

$$R_x(\theta), R_y(\theta), R_z(\theta)$$

4. Rx Gate (Rotation Around X-axis)

Definition:

Rx gate rotates a qubit state around the X-axis of the Bloch sphere by an angle θ.

Matrix Representation

$$\boxed{R_x(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -i\sin(\frac{\theta}{2}) \\ -i\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix}}$$

Operation on Qubit

For:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After Rx:

$$R_x(\theta)|\psi\rangle$$

produces rotation around X-axis.

Special Cases

When:

$$\theta = \pi$$

Rx becomes:

$$R_x(\pi) = -iX$$

which is equivalent to Pauli-X up to a global phase.

Bloch Sphere Interpretation

Rotation:

$$\boxed{\theta \text{ around X-axis}}$$

Circuit Symbol

```
|ψ⟩ ──Rx(θ)──
```

Applications

- Quantum state preparation
- Quantum algorithms
- Quantum machine learning circuits

---

5. Ry Gate (Rotation Around Y-axis)

Definition:

Ry gate rotates a qubit state around the Y-axis of the Bloch sphere.

Matrix Representation

$$\boxed{R_y(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix}}$$

Effect on Basis States

For:

$$|0\rangle$$

after rotation:

$$R_y(\theta)|0\rangle$$

becomes a combination of:

$$|0\rangle$$

and:

$$|1\rangle$$

Special Case

For:

$$\theta = \pi$$

$$R_y(\pi)$$

performs a rotation similar to Pauli-Y gate.

Bloch Sphere Interpretation

Rotation:

$$\boxed{\theta \text{ around Y-axis}}$$

Circuit Symbol

```
|ψ⟩ ──Ry(θ)──
```

Applications

- State preparation
- Variational quantum algorithms
- Quantum neural networks

6. Rz Gate (Rotation Around Z-axis)

Definition:

> **Rz gate rotates a qubit state around the Z-axis of the Bloch sphere and changes the phase of the quantum state.**

Matrix Representation

$$\boxed{R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}}$$

Operation on Qubit

For:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After Rz:

$$\boxed{R_z(\theta)|\psi\rangle = \alpha e^{-i\theta/2}|0\rangle + \beta e^{i\theta/2}|1\rangle}$$

Bloch Sphere Interpretation

Rotation:

$$\boxed{\theta \text{ around Z-axis}}$$

Circuit Symbol

```
|ψ⟩ ──Rz(θ)──
```

Applications

- Phase control
- Quantum Fourier Transform
- Quantum algorithms

---

7. Comparison of Rotation Gates

Gate	Axis of Rotation	Matrix
Rx(θ)	X-axis	Uses Pauli-X
Ry(θ)	Y-axis	Uses Pauli-Y
Rz(θ)	Z-axis	Uses Pauli-Z

---

8. Relation Between Rotation and Pauli Gates

Rotation gates are generated using Pauli matrices.

X Rotation:

$$R_x(\theta) = e^{-i\theta X/2}$$

Y Rotation:

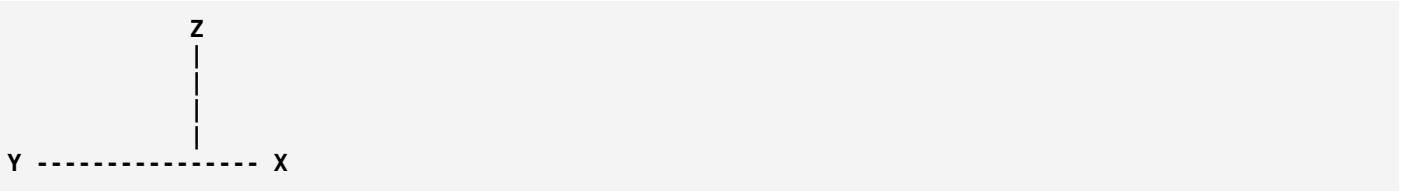
$$R_y(\theta) = e^{-i\theta Y/2}$$

Z Rotation:

$$R_z(\theta) = e^{-i\theta Z/2}$$

9. Rotation Gates and Bloch Sphere

The Bloch sphere has three axes:



Rotation gates move the qubit state along these axes.

10. Importance of Rotation Gates

Rotation gates are important because they:

1. Create Arbitrary Quantum States

Any qubit state can be prepared using rotations.

2. Used in Variational Algorithms

Examples:

- VQE
- QAOA
- Quantum Neural Networks

3. Control Quantum Phase

Used for:

- Interference
- Quantum algorithms

11. Applications of Rotation Gates

1. Quantum Machine Learning

Used as trainable parameters in quantum circuits.

2. Quantum Optimization

Used in:

- VQE
- QAOA

3. Quantum Simulation

Used to represent physical systems.

12. Frequently Asked University Questions

2-Mark Questions

1. What are rotation gates?
2. Name three rotation gates.
3. Around which axis does Rx rotate?
4. Write the matrix of Rz gate.

5-Mark Questions

1. Explain Rx, Ry, and Rz gates.
2. Describe rotation operations on Bloch sphere.
3. Explain applications of rotation gates.

10-Mark Questions

1. Explain different rotation gates with mathematical representation.

2. Describe the role of rotation gates in quantum circuits.
3. Explain Bloch sphere rotations using Rx, Ry, and Rz.

Quick Revision

Rx:

$$\text{Rotation around X-axis}$$

Ry:

$$\text{Rotation around Y-axis}$$

Rz:

$$\text{Rotation around Z-axis}$$

Important Point:

Rotation gates allow precise control of qubit states and are essential for designing advanced quantum algorithms and variational quantum circuits.

Next Topic:

Multi-Qubit Gates

We will cover:

1. CNOT Gate
2. Toffoli Gate
3. SWAP Gate
4. Truth tables
5. Matrix representation
6. Entanglement creation using multi-qubit gates

(Highly important for UNIT–II exam questions.)

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 6 – Multi-Qubit Gates (CNOT, Toffoli, SWAP Gates)

Weightage: ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
Multi-qubit gates operate on two or more qubits and are essential for creating **entanglement**, performing reversible computation, and implementing quantum algorithms.

---

1. Introduction to Multi-Qubit Gates

A single-qubit gate can manipulate only one qubit.

However, quantum computing requires interaction between qubits.

For this purpose, **multi-qubit gates** are used.

Definition:

> **Multi-qubit gates are quantum gates that operate on two or more qubits simultaneously and create correlations between quantum states.**

---

2. Need for Multi-Qubit Gates

Multi-qubit gates are required for:

- Creating entanglement
- Quantum communication
- Quantum algorithms
- Reversible computation
- Quantum error correction

---

3. Two-Qubit System

Two qubits can exist in four basis states:

$$|00\rangle$$

$$|01\rangle$$

$$|10\rangle$$

$$|11\rangle$$

A general two-qubit state:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

---

4. Controlled-NOT Gate (CNOT Gate)

Definition:

> **The Controlled-NOT (CNOT) gate is a two-qubit gate that flips the target qubit only when the control qubit is in the state $|1\rangle$.**

CNOT has:

- One control qubit
- One target qubit

---

5. CNOT Gate Symbol

Circuit representation:

```
Control ──●──
          |
Target  ──⊕──
```

Where:

- ● → Control qubit
- ⊕ → Target qubit

---

6. Working of CNOT Gate

The control qubit remains unchanged.

The target qubit changes:

$$|0\rangle \leftrightarrow |1\rangle$$

when control = 1.

---

7. CNOT Truth Table

Input	Output
00	00
01	01
10	11
11	10

Explanation:

Input:

$$|00\rangle$$

Control = 0

No change:

$$|00\rangle$$

Input:

$$|10\rangle$$

Control = 1

Target flips:

$$|11\rangle$$

---

8. CNOT Matrix Representation

The CNOT matrix is:

$$\boxed{CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}}$$

---

9. CNOT and Entanglement

CNOT is one of the most important gates for creating entanglement.

Circuit:

```
Q1 ──H──●──
        │
Q2 ─────⊕──
```

Initial state:

$$|00\rangle$$

After Hadamard:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

After CNOT:

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

A Bell state is created.

---

10. Applications of CNOT Gate

Used in:

- Quantum teleportation
- Quantum error correction
- Quantum algorithms
- Entanglement generation

---

11. Toffoli Gate (CCNOT Gate)

Definition:

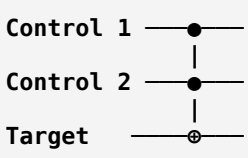
> **The Toffoli gate is a three-qubit controlled-controlled-NOT gate that flips the target qubit only when both control qubits are in state |1⟩.**

It contains:

- Two control qubits
- One target qubit

---

12. Toffoli Gate Symbol



13. Working of Toffoli Gate

Target changes only when:

$$Control_1 = 1$$

and

$$Control_2 = 1$$

14. Toffoli Truth Table

Input	Output
000	000
001	001
010	010
011	011
100	100
101	101
110	110
111	110?

Correction:

For input:

$$|111\rangle$$

Target qubit flips:

$$|110\rangle$$

Complete:

Input	Output
000	000
001	001
010	010
011	011
100	100
101	101
110	110
111	110

15. Toffoli Matrix

It is an 8×8 matrix.

It acts as identity except:

$$|110\rangle$$

and

$$|111\rangle$$

are swapped.

16. Applications of Toffoli Gate

Used in:

- Reversible computation
- Quantum arithmetic circuits
- Quantum algorithms

17. SWAP Gate

Definition:

> **The SWAP gate exchanges the states of two qubits.**

Example:

Input:

$$|01\rangle$$

Output:

$$|10\rangle$$

18. SWAP Gate Symbol

```
Q1 ──×──
     |
Q2 ──×──
```

19. SWAP Truth Table

Input	Output
00	00
01	10
10	01
11	11

20. SWAP Matrix Representation

$$SWAP = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

21. Applications of SWAP Gate

Used for:

- Moving quantum information
- Quantum communication
- Connecting distant qubits

22. Comparison of Multi-Qubit Gates

Gate	Number of Qubits	Function
CNOT	2	Controlled bit flip
Toffoli	3	Two-control bit flip
SWAP	2	Exchange qubit states

23. Multi-Qubit Gates in Quantum Circuits

A complete quantum circuit often combines:

Example:

```
Input Qubits
      ↓
Hadamard Gate
      ↓
CNOT Gate
      ↓
Measurement
      ↓
Classical Output
```

24. Importance of Multi-Qubit Gates

Multi-qubit gates enable:

1. Entanglement

Creates quantum correlations.

---

2. Quantum Parallelism

Allows multiple computations.

---

3. Complex Algorithms

Used in:

- Grover's algorithm
- Shor's algorithm
- Quantum simulation

Frequently Asked University Questions

2-Mark Questions

1. Define CNOT gate.
2. What is the function of SWAP gate?
3. What is a Toffoli gate?
4. Which gate creates entanglement?

5-Mark Questions

1. Explain CNOT gate with truth table.
2. Describe SWAP gate.
3. Explain Toffoli gate.

10-Mark Questions

1. Explain different multi-qubit gates with circuits and truth tables.
2. Describe CNOT, Toffoli, and SWAP gates.
3. Explain the role of multi-qubit gates in quantum computing.

Quick Revision

CNOT:

$$\text{Control} = 1 \rightarrow \text{Flip Target}$$

Toffoli:

$$\text{Both Controls} = 1 \rightarrow \text{Flip Target}$$

SWAP:

$$|ab\rangle \rightarrow |ba\rangle$$

Next Topic:

Quantum Circuits and Reversible Computation

We will cover:

1. Quantum circuit model
2. Circuit representation
3. Reversible computation
4. Classical vs reversible computation
5. Importance of reversibility in quantum computing

(Important for UNIT–II long-answer questions.)

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 7 – Quantum Circuits and Reversible Computation

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> Quantum circuits provide the framework for implementing quantum algorithms. Reversible computation is a fundamental requirement because all quantum operations must preserve information.

PART–A: Quantum Circuits

---

1. Introduction to Quantum Circuits

A quantum circuit is a sequence of quantum gates applied to qubits to perform a computation.

Similar to classical circuits that use logic gates, quantum circuits use:

- Quantum gates
- Qubits
- Measurement operations

---

2. Definition of Quantum Circuit

Exam Definition:

> **A quantum circuit is a computational model that represents a sequence of quantum operations performed on qubits to transform input quantum states into output states.**

---

3. Components of a Quantum Circuit

A quantum circuit consists of:

---

1. Qubits

Qubits are the input information carriers.

Represented as:

$$|0\rangle, |1\rangle$$

---

2. Quantum Gates

Operations that modify qubit states.

Examples:

- X gate
- H gate
- CNOT gate

---

3. Measurement

Converts quantum states into classical information.

Symbol:

$$M$$

4. Representation of Quantum Circuit

Example of a single qubit circuit:

```
|0⟩ ──H──X──M
```

Operations:

1. Start with $|0\rangle$
2. Apply Hadamard gate
3. Apply Pauli-X gate
4. Measure result

5. Multi-Qubit Circuit Example

Creating entanglement:

```
Q1 |0⟩ ──H──●──M
           |
Q2 |0⟩ ─────⊕──M
```

Steps:

Step 1:

Initial state:

$$|00\rangle$$

Step 2:

Apply H gate:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

Step 3:

Apply CNOT:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

A Bell state is generated.

6. Quantum Circuit Operations

Quantum circuits perform:

1. State Preparation

Creating desired quantum states.

Example:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

using H gate.

2. Quantum Transformation

Applying gates to modify states.

Example:

$$|\psi'\rangle = U|\psi\rangle$$

3. Measurement

Obtaining classical output.

7. Quantum Circuit Model

The quantum circuit model consists of:

```
Input Qubits
      ↓
Quantum Gates
      ↓
Measurement
      ↓
Classical Result
```

8. Circuit Depth

Definition:

Circuit depth represents the number of sequential layers of quantum operations.

Example:

```
Layer 1     Layer 2     Layer 3
 H            X           M
```

Depth:

$$3$$

Lower circuit depth is preferred because:

- Less noise
- Fewer errors

---

9. Quantum Circuit Example: Quantum NOT Operation

Circuit:

```
|0⟩ ──X──
```

Input:

$$|0\rangle$$

Output:

$$|1\rangle$$

---

10. Importance of Quantum Circuits

Quantum circuits are used for:

- Implementing algorithms
- Quantum simulations
- Cryptography
- Optimization problems
- Machine learning

PART–B: Reversible Computation

11. Introduction to Reversible Computation

In classical computing, many operations lose information.

Example:

AND gate:

Inputs:

$$00, 01, 10, 11$$

Outputs:

$$0, 0, 0, 1$$

Different inputs produce the same output.

Information is lost.

Quantum computing requires reversible operations.

---

12. Definition of Reversible Computation

Exam Definition:

> **Reversible computation is a computational process in which every output state uniquely determines the input state, allowing the computation to be reversed.**

Mathematically:

If:

$$f(x) = y$$

then there exists an inverse function:

$$f^{-1}(y) = x$$

---

13. Properties of Reversible Computation

---

1. One-to-One Mapping

Each input has a unique output.

Example:

$$A \to B$$

and:

$$B \to A$$

---

2. No Information Loss

All input information is preserved.

---

3. Invertible Operations

Every operation has an inverse.

---

14. Reversible Logic Gates

Examples:

1. NOT Gate
2. CNOT Gate
3. Toffoli Gate
4. Fredkin Gate

---

15. NOT Gate as Reversible Gate

Classical NOT operation:

$$0 \to 1$$

$$1 \to 0$$

Applying twice:

$$NOT(NOT(x)) = x$$

Therefore:

$$NOT^{-1} = NOT$$

---

16. CNOT Gate as Reversible Gate

CNOT transformation:

$$|a, b\rangle \rightarrow |a, a \oplus b\rangle$$

The original input can be recovered.

17. Toffoli Gate as Universal Reversible Gate

The Toffoli gate is important because it can implement any classical reversible computation.

It is used for:

- Quantum arithmetic
- Quantum algorithms

18. Classical vs Reversible Computation

Classical Computation	Reversible Computation
Information may be lost	No information loss
Many-to-one mapping	One-to-one mapping
Not always invertible	Always invertible
Uses AND, OR gates	Uses reversible gates

19. Why Quantum Computing Requires Reversibility?

Quantum evolution follows:

$$|\psi'\rangle = U|\psi\rangle$$

where:

$$U^\dagger U = I$$

This means quantum operations must be:

- Unitary
- Reversible

20. Applications of Reversible Computation

Used in:

1. Quantum Computing

Foundation of quantum circuits.

2. Low Power Computing

Reduces energy loss.

3. Quantum Algorithms

Used in:

- Shor's algorithm
- Grover's algorithm

21. Advantages of Reversible Computation

1. Preserves information.
2. Enables quantum operations.
3. Reduces theoretical energy loss.
4. Supports quantum algorithms.

Frequently Asked University Questions

2-Mark Questions

1. Define quantum circuit.
2. What is reversible computation?
3. Why are quantum gates reversible?
4. Give examples of reversible gates.

5-Mark Questions

1. Explain quantum circuit model.
2. Describe reversible computation.
3. Explain reversible quantum gates.

10-Mark Questions

1. Explain quantum circuits with suitable examples.

2. Discuss reversible computation and its importance in quantum computing.
3. Explain the relationship between quantum circuits and reversible operations.

Quick Revision

Quantum Circuit:

$$\boxed{Qubits + QuantumGates + Measurement}$$

Reversible Computation:

$$\boxed{Input \leftrightarrow Output}$$

Important Points:

- Quantum circuits use unitary operations.
- All quantum gates are reversible.
- CNOT and Toffoli are important reversible gates.
- Reversibility prevents information loss.

Next Topic:

Quantum Phenomena in Computation

We will cover:

1. Quantum Interference
2. Quantum Parallelism
3. Quantum Noise
4. Quantum Error Basics

(Last major topic of UNIT–II before moving to UNIT–III Grover's Algorithm.)

> ok next

UNIT–II: Quantum Gates and Circuits

Topic 8 – Quantum Phenomena: Interference, Parallelism, Noise and Error Basics

Weightage: ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
Quantum phenomena are the fundamental principles that give quantum computers their computational power. The major phenomena are:

1. Quantum Interference
2. Quantum Parallelism
3. Quantum Noise
4. Quantum Errors

PART–A: Quantum Interference

---

1. Introduction to Quantum Interference

In quantum mechanics, quantum states behave like waves.

When two quantum states combine, their amplitudes can either:

- Increase each other
- Cancel each other

This phenomenon is called **quantum interference**.

---

2. Definition of Quantum Interference

Exam Definition:

> **Quantum interference is a quantum phenomenon in which probability amplitudes of quantum states combine to increase the probability of desired outcomes and decrease the probability of unwanted outcomes.**

---

3. Types of Interference

There are two types:

---

1. Constructive Interference

When amplitudes combine positively.

Effect:

- Probability increases.

Example:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

---

2. Destructive Interference

When amplitudes cancel each other.

Effect:

- Probability decreases or becomes zero.

---

4. Interference in Quantum Computing

Quantum algorithms use interference to:

- Amplify correct answers
- Suppress incorrect answers

Example:

Grover's Algorithm

Uses amplitude amplification through interference.

---

5. Example of Quantum Interference

Suppose:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After quantum operations, amplitudes can change:

$$\alpha \to \alpha'$$

$$\beta \to \beta'$$

The probability:

$$|\alpha'|^2$$

and:

$$|\beta'|^2$$

changes due to interference.

---

6. Importance of Quantum Interference

Used in:

- Quantum algorithms
- Quantum search
- Quantum simulation
- Quantum communication

PART–B: Quantum Parallelism

---

7. Introduction to Quantum Parallelism

Quantum parallelism is the ability of quantum computers to process multiple possibilities simultaneously using superposition.

---

8. Definition of Quantum Parallelism

Exam Definition:

> **Quantum parallelism is the ability of a quantum computer to perform computations on multiple quantum states simultaneously due to the superposition property of qubits.**

---

9. Classical vs Quantum Parallelism

Classical Computer:

A bit can be:

$$0 \quad \text{or} \quad 1$$

Quantum Computer:

A qubit can be:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

For two qubits:

Classical:

possible states are stored separately.

Quantum:

$$\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

all exist simultaneously.

---

10. Example of Quantum Parallelism

For n qubits:

Number of possible states:

$$\boxed{2^n}$$

Example:

3 qubits:

$$2^3 = 8$$

possible states.

---

11. Role in Quantum Algorithms

Quantum parallelism is used in:

- Grover's search algorithm
- Shor's factoring algorithm
- Quantum simulation

---

12. Limitation of Quantum Parallelism

Important:

Quantum parallelism does not mean we can directly read all results.

Measurement gives only one output.

Therefore, algorithms require interference.

PART–C: Quantum Noise

---

13. Introduction to Quantum Noise

Quantum systems are extremely sensitive to environmental effects.

These unwanted disturbances are called quantum noise.

---

14. Definition of Quantum Noise

Exam Definition:

> **Quantum noise is any unwanted disturbance that affects the state of qubits and reduces the accuracy of quantum computation.**

---

15. Sources of Quantum Noise

---

1. Environment Interaction

Qubits interact with surroundings.

Example:

- Temperature
- Electromagnetic fields

---

2. Hardware Imperfections

Errors occur due to:

- Gate operation problems
- Measurement errors

---

3. Decoherence

Loss of quantum information due to interaction with environment.

---

16. Quantum Decoherence

Definition:

> **Decoherence is the process by which a quantum system loses its quantum properties due to interaction with the environment.**

Effects:

- Loss of superposition
- Loss of entanglement

---

17. Types of Quantum Noise

---

1. Bit Flip Noise

Changes:

$$|0\rangle \leftrightarrow |1\rangle$$

Corrected by:

$$X$$

gate.

---

2. Phase Flip Noise

Changes:

$$|1\rangle \rightarrow -|1\rangle$$

Corrected by:

$$Z$$

gate.

---

3. Bit-Phase Flip Noise

Combination of:

- Bit flip
- Phase flip

Corrected by:

$$Y$$

gate.

PART–D: Quantum Error Basics

18. Introduction to Quantum Errors

Because qubits are sensitive, errors can occur during:

- Storage
- Transmission
- Computation
- Measurement

19. Definition of Quantum Error

Exam Definition:

> **Quantum error is an unwanted change in the quantum state of a qubit caused by noise or imperfections during quantum computation.**

20. Common Quantum Errors

1. Bit Flip Error

Original:

$$|0\rangle$$

Error:

$$|1\rangle$$

Correction:

$$X$$

gate.

2. Phase Flip Error

Original:

$$|1\rangle$$

Error:

$$-|1\rangle$$

Correction:

$$Z$$

gate.

---

3. Bit-Phase Flip Error

Combination of:

$$X + Z$$

Correction:

$$Y$$

gate.

---

21. Quantum Error Correction

Definition:

> **Quantum error correction is a technique used to protect quantum information from errors caused by noise and decoherence.**

Methods:

- Quantum error correcting codes
- Repetition codes
- Surface codes

---

22. Difference Between Classical and Quantum Errors

Classical Error	Quantum Error
Bit changes 0 to 1	Quantum state changes
Easy duplication	Cannot copy unknown quantum states
Simple correction	Requires quantum error correction

---

23. No-Cloning Theorem

Definition:

> **The no-cloning theorem states that an unknown quantum state cannot be copied perfectly.**

This makes quantum error correction challenging.

24. Importance of Quantum Error Correction

It helps:

- Increase reliability
- Reduce computation errors
- Build fault-tolerant quantum computers

25. Comparison Table

Phenomenon	Meaning	Importance
Interference	Combining amplitudes	Improves correct solutions
Parallelism	Multiple states processing	Computational advantage
Noise	External disturbance	Causes errors
Error correction	Protects qubits	Improves reliability

Frequently Asked University Questions

2-Mark Questions

1. Define quantum interference.
2. What is quantum parallelism?
3. What is quantum noise?
4. Define decoherence.
5. What is quantum error correction?

5-Mark Questions

1. Explain quantum interference.
2. Describe quantum parallelism.
3. Explain quantum noise and errors.

10-Mark Questions

1. Explain quantum phenomena used in quantum computing.
2. Discuss interference, parallelism, noise, and error correction.
3. Explain quantum errors and their correction methods.

UNIT–II Complete Summary ✅

Topic	Key Concept
Quantum Gates	Operations on qubits
Pauli Gates	Bit and phase changes
Hadamard Gate	Creates superposition
Phase Gates	Control phase
Rotation Gates	Rotate qubit states
CNOT	Creates entanglement
Toffoli	Controlled-controlled operation
SWAP	Exchange qubits
Quantum Circuits	Sequence of gates
Reversible Computation	No information loss
Interference	Amplitude manipulation
Parallelism	Multiple quantum states
Noise	Quantum disturbance

Next Topic:

UNIT–III: Quantum Search Algorithms

Starting with:

1. Unstructured Search Problem

2. Classical Search vs Quantum Search

3. Introduction to Grover's Algorithm

(UNIT–III focuses mainly on **Grover's Algorithm**, a very important 10-mark topic.)

ok next

UNIT–III: Quantum Search Algorithms

Topic 1 – Introduction to Quantum Search Algorithms and Unstructured Search Problem

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> UNIT–III mainly focuses on **Grover's Algorithm**, which provides a quadratic speedup for searching an unsorted database.

---

1. Introduction to Quantum Search Algorithms

Search problems are among the most common computational problems.

Examples:

- Finding a person's record in a database
- Searching an item in a list
- Finding a solution to an optimization problem

In classical computing, searching usually requires checking elements one by one.

Quantum computing uses quantum properties such as:

- Superposition
- Interference
- Amplitude amplification

to improve search efficiency.

---

2. Definition of Quantum Search Algorithm

Exam Definition:

> **A quantum search algorithm is a quantum algorithm that uses quantum mechanical principles to find desired information from a search space faster than classical algorithms.**

---

3. Search Problem

A search problem consists of:

1. A collection of possible solutions.

2. A condition to identify the correct solution.

Mathematically:

Given a function:

$$f(x)$$

where:

$$f(x) = \begin{cases} 1, & \text{if x is the desired item} \\ 0, & \text{otherwise} \end{cases}$$

The goal is to find:

$$x$$

such that:

$$f(x) = 1$$

---

4. Types of Search Problems

Search problems are mainly classified into:

---

1. Structured Search

The data has an organized pattern.

Examples:

- Sorted arrays
- Binary search trees

Classical algorithms can efficiently solve these.

---

2. Unstructured Search

The data has no order or pattern.

Examples:

- Searching an unsorted database
- Finding an unknown element in a list

This is the problem solved by Grover's algorithm.

---

5. Unstructured Search Problem

Definition:

> **An unstructured search problem is a problem where the search space has no known organization, and the only way to find the solution is by checking possible candidates.**

Example:

Suppose we have:

$$N$$

unsorted database entries.

One entry is correct.

We need to find it.

---

6. Classical Solution to Unstructured Search

The classical approach is:

Linear Search

Steps:

1. Check first element.
2. Compare with required item.
3. Continue until found.

Worst case:

The required item may be the last element.

Number of searches:

$$\boxed{N}$$

Time complexity:

$$\boxed{O(N)}$$

---

7. Example of Classical Search

Database:

$$[5, 8, 2, 9, 1]$$

Find:

$$9$$

Checking:

Step 1:

$$5 \neq 9$$

Step 2:

$$8 \neq 9$$

Step 3:

$$2 \neq 9$$

Step 4:

$$9 = 9$$

Found.

Maximum:

$$5$$

operations.

---

8. Quantum Approach to Search

Quantum computers represent all possible states using superposition.

For N items:

$$|0\rangle, |1\rangle, |2\rangle, ..., |N-1\rangle$$

are represented simultaneously.

A quantum state:

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

This creates an equal superposition of all possible search states.

---

9. Quantum Search Advantage

Classical search:

$$O(N)$$

operations.

Quantum search:

$$O(\sqrt{N})$$

operations.

This is a quadratic speedup.

---

10. Example Comparison

Suppose database size:

$$N = 1,000,000$$

Classical Search:

Maximum operations:

$$1,000,000$$

Quantum Search:

Operations:

$$\sqrt{1,000,000}$$

$$= 1000$$

Huge improvement.

---

11. Role of Oracle in Search

Quantum search uses an important component called an **oracle**.

Definition:

An oracle is a quantum operation that identifies the correct solution by changing its phase without revealing the answer directly.

The oracle performs:

$$|x\rangle \rightarrow (-1)^{f(x)}|x\rangle$$

If:

$$f(x) = 1$$

phase changes.

If:

$$f(x) = 0$$

state remains unchanged.

---

12. Limitations of Classical Search

Classical unstructured search:

- Requires checking each element.
- Time increases linearly with database size.
- Not efficient for large datasets.

---

13. Advantages of Quantum Search

Quantum search provides:

1. Faster Searching

$$O(\sqrt{N})$$

complexity.

---

2. Reduced Computational Cost

Requires fewer operations.

---

3. Applications in Many Fields

Used in:

- Database search

- Cryptography
- Optimization
- Artificial intelligence

14. Applications of Quantum Search Algorithms

1. Database Search

Finding information from large databases.

2. Cryptography

Searching encryption keys.

3. Optimization Problems

Finding best solutions.

Examples:

- Scheduling
- Route planning

4. Artificial Intelligence

Improving search-based AI methods.

15. Classical Search vs Quantum Search

Feature	Classical Search	Quantum Search
Search method	Sequential checking	Quantum operations
Algorithm	Linear Search	Grover's Algorithm
Complexity	O(N)	O(√N)
Principle	Classical logic	Superposition + interference
Speed	Slower	Faster

16. Important Terms for UNIT–III

Term	Meaning
Search Space	Collection of possible solutions
Oracle	Marks correct solution
Amplitude	Probability strength of state
Amplitude Amplification	Increasing correct state's probability
Grover Iteration	One search improvement step

Frequently Asked University Questions

2-Mark Questions

1. Define quantum search algorithm.
2. What is unstructured search?
3. What is the complexity of classical search?
4. What is the advantage of quantum search?

5-Mark Questions

1. Explain unstructured search problem.
2. Compare classical and quantum search.
3. Explain the role of oracle in quantum search.

10-Mark Questions

1. Explain unstructured search problem and quantum search approach.
2. Compare classical search algorithms with quantum search algorithms.
3. Explain the need for Grover's algorithm.

Quick Revision

Classical Search:

$$O(N)$$

Quantum Search:

$$O(\sqrt{N})$$

Main Idea:

Quantum search uses superposition to explore multiple possibilities and interference to amplify the correct answer.

Next Topic:

Grover's Algorithm

We will cover:

1. Introduction
2. Working principle
3. Oracle function
4. Initialization
5. Amplitude amplification
6. Mathematical derivation
7. Circuit representation
8. Applications

(Grover's Algorithm is the most important topic in UNIT–III.)

> ok next

UNIT–III: Quantum Search Algorithms

Topic 2 – Grover's Algorithm: Introduction and Basic Principle

> **Weightage:** ⭐⭐⭐⭐⭐ (Most Important – 10 Marks Question)
> Grover's Algorithm is one of the most famous quantum algorithms that provides a **quadratic speedup** for searching an unstructured database.

---

1. Introduction to Grover's Algorithm

Grover's algorithm was proposed by:

$$\boxed{\text{Lov Grover in 1996}}$$

It is a quantum search algorithm used to find a specific item from an unstructured collection.

Definition:

> **Grover's algorithm is a quantum algorithm that searches an unsorted database of N elements in O(√N) time using quantum superposition and amplitude amplification.**

---

2. Problem Statement of Grover's Algorithm

Given:

- A database containing N possible elements.
- Only one element satisfies the search condition.

Find the correct element.

Mathematically:

We have a function:

$$f(x)$$

such that:

$$f(x) = \begin{cases} 1, & \text{if x is the required solution} \\ 0, & \text{otherwise} \end{cases}$$

The goal is to find:

$$x$$

where:

$$f(x) = 1$$

---

3. Classical vs Grover Search

Classical Linear Search

For N items:

Worst case:

$$N$$

queries.

Complexity:

$$\boxed{O(N)}$$

Grover’s Algorithm

Number of queries:

$$\sqrt{N}$$

Complexity:

$$\boxed{O(\sqrt{N})}$$

4. Basic Idea of Grover’s Algorithm

Grover's algorithm works using three quantum principles:

1. Superposition

All possible states are created simultaneously.

Example:

For 2 qubits:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

exist together.

2. Oracle

The correct answer is marked.

3. Amplitude Amplification

The probability of the correct answer is increased.

5. Main Steps of Grover’s Algorithm

Grover's algorithm consists of four major steps:

Step 1: Initialization

Create an equal superposition of all possible states.

For N states:

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

Using Hadamard gates:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Example:

For 2 qubits:

Initial:

$$|00\rangle$$

After H gates:

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

Step 2: Oracle Operation

The oracle identifies the target state.

Oracle Definition:

> **A quantum oracle is a black-box quantum operation that marks the correct solution by changing its phase.**

Mathematical operation:

$$O|x\rangle = (-1)^{f(x)}|x\rangle$$

Meaning:

If:

$$f(x) = 1$$

then:

$$|x\rangle \rightarrow -|x\rangle$$

If:

$$f(x) = 0$$

then:

$$|x\rangle \rightarrow |x\rangle$$

Example:

Suppose:

Target state:

$$|11\rangle$$

Oracle changes:

$$|11\rangle \rightarrow -|11\rangle$$

Other states remain unchanged.

Step 3: Amplitude Amplification

After oracle marking:

The probability of the target state is increased.

This is done by the:

$$\boxed{\text{Grover Diffusion Operator}}$$

---

6. Grover Diffusion Operator

The diffusion operator performs:

1. Reflection about average amplitude.
2. Increases target amplitude.

Mathematical form:

$$D = 2|\psi\rangle\langle\psi| - I$$

Where:

- $|\psi\rangle$ = equal superposition state

- I = identity matrix

---

7. Working of Amplitude Amplification

Initially:

All states have equal probability.

Example:

$$\frac{1}{\sqrt{N}}$$

After oracle:

Target state gets negative amplitude.

After diffusion:

Target amplitude becomes larger.

Repeating this process increases the probability of the correct answer.

---

8. Grover Iteration

One complete cycle:

$$\boxed{\text{Oracle} + \text{Diffusion}}$$

is called a:

$$\boxed{\text{Grover Iteration}}$$

Number of iterations:

$$\boxed{\frac{\pi}{4}\sqrt{N}}$$

---

9. Example of Grover Search

Suppose:

$$N = 16$$

items.

Classical search:

$$16$$

queries.

Grover search:

$$\sqrt{16} = 4$$

approximately.

Iterations:

$$\frac{\pi}{4} \times 4$$

≈

$$3$$

iterations.

---

10. Quantum Circuit Representation

Basic Grover circuit:

```
Input Qubits

     |
     H
     |
Superposition
     |
   Oracle
     |
Diffusion Operator
     |
Measurement
     |
Answer
```

Detailed:

```
|0> —H—Oracle—Diffusion—M
|0> —H—Oracle—Diffusion—M
```

11. Mathematical Explanation of Grover Rotation

Grover's algorithm can be viewed as rotation in a two-dimensional space.

The states are:

1. Target state:

$$|w\rangle$$

2. Non-target states:

$$|w^{\perp}\rangle$$

Each Grover iteration rotates the state closer to:

$$|w\rangle$$

12. Advantages of Grover's Algorithm

1. Quadratic speed improvement.

$$O(N) \rightarrow O(\sqrt{N})$$

2. Works for unstructured databases.

3. General search technique.

4. Useful in optimization problems.

13. Limitations of Grover's Algorithm

1. Requires quantum hardware.
2. Cannot provide exponential speedup.
3. Number of iterations must be carefully chosen.
4. Sensitive to noise.

14. Applications of Grover's Algorithm

1. Database Search

Finding records from large databases.

2. Cryptography

Searching symmetric encryption keys.

Example:

Breaking a key of size N requires:

Classical:

$$O(N)$$

Quantum:

$$O(\sqrt{N})$$

---

3. Optimization

Finding optimal solutions.

---

4. Artificial Intelligence

Search-based learning methods.

---

15. Grover’s Algorithm Flow

```
Start
 ↓
Initialize Qubits
 ↓
Create Superposition
 ↓
Apply Oracle
 ↓
Apply Diffusion Operator
 ↓
Repeat √N times
 ↓
Measure
 ↓
Obtain Solution
```

Frequently Asked University Questions

2-Mark Questions

1. Who proposed Grover’s algorithm?

2. What is the complexity of Grover's algorithm?
3. What is an oracle?
4. Define amplitude amplification.

5-Mark Questions

1. Explain the working principle of Grover's algorithm.
2. Describe oracle operation.
3. Explain diffusion operator.

10-Mark Questions

1. Explain Grover's algorithm with steps and circuit.
2. Describe oracle and amplitude amplification in Grover's algorithm.
3. Compare classical search with Grover's quantum search.

Quick Revision

Grover Complexity:

$$\boxed{O(\sqrt{N})}$$

Main Components:

1. Initialization
2. Oracle
3. Diffusion Operator
4. Measurement

Core Idea:

Oracle marks the correct answer, and amplitude amplification increases its probability until measurement reveals the solution.

Next Topic:

Grover's Algorithm – Oracle Function and Amplitude Amplification (Detailed Mathematical Working)

(Important for 10-mark derivation questions.)

> ok next

UNIT–III: Quantum Search Algorithms

Topic 3 – Grover's Algorithm: Oracle Function and Amplitude Amplification (Detailed Working)

> **Weightage:** ⭐⭐⭐⭐⭐ (Most Important – 10 Marks Derivation Question)
> This topic explains the mathematical foundation of Grover's algorithm. The two main components are:
>
> 1. **Oracle Function**
> 2. **Amplitude Amplification (Diffusion Operator)**

PART–A: Oracle Function in Grover's Algorithm

---

1. Introduction to Oracle

In Grover's algorithm, the quantum computer needs a way to identify the correct solution among many possibilities.

This is done using an **oracle**.

Definition:

> **A quantum oracle is a black-box quantum operation that identifies the desired solution by changing its phase while leaving other states unchanged.**

The oracle does not directly give the answer.

It only **marks** the correct state.

---

2. Classical Oracle vs Quantum Oracle

Classical Oracle	Quantum Oracle
Returns 0 or 1	Changes quantum phase

Classical Oracle	Quantum Oracle
Direct output	Does not reveal answer
Function evaluation	Unitary operation

3. Mathematical Representation of Oracle

The oracle operation is represented as:

$$\boxed{O_f}$$

The oracle performs:

$$\boxed{O_f|x\rangle = (-1)^{f(x)}|x\rangle}$$

Where:

- x = search state
- $f(x)$ = search function

4. Oracle Function

The search function is:

$$f(x)$$

defined as:

$$f(x) = \begin{cases} 1, & \text{if x is the solution} \\ 0, & \text{otherwise} \end{cases}$$

5. Oracle Working

There are two cases:

Case 1: Non-target State

If:

$$f(x) = 0$$

then:

$$O_f|x\rangle = (-1)^0|x\rangle$$

Therefore:

$$\boxed{O_f|x\rangle = |x\rangle}$$

No change occurs.

Case 2: Target State

If:

$$f(x) = 1$$

then:

$$O_f|x\rangle = (-1)^1|x\rangle$$

Therefore:

$$\boxed{O_f|x\rangle = -|x\rangle}$$

The phase is flipped.

---

6. Example of Oracle Operation

Consider four states:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Assume:

$$|11\rangle$$

is the target.

Before oracle:

$$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

After oracle:

$$\boxed{|\psi'\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)}$$

Only the target state changes.

7. Oracle Circuit Representation

```
Input States
      |
      H
      |
 Oracle Function
      |
Phase Marking
```

8. Importance of Oracle

The oracle:

- Identifies correct solution.
- Performs phase inversion.
- Enables amplitude amplification.
- Forms the core of Grover's algorithm.

PART–B: Amplitude Amplification

---

9. Introduction

After oracle operation:

The correct state has a negative amplitude.

However, its probability is still not high enough.

Amplitude amplification increases the probability of the correct state.

Definition:

> **Amplitude amplification is a quantum technique that increases the probability amplitude of the desired state using repeated reflections.**

---

10. Grover Diffusion Operator

Amplitude amplification is performed using the:

$$\boxed{\text{Diffusion Operator}}$$

It is represented as:

$$D$$

Mathematical form:

$$\boxed{D = 2|\psi\rangle\langle\psi| - I}$$

Where:

- $|\psi\rangle$ = equal superposition state
- I = identity matrix

11. Working of Diffusion Operator

The diffusion operator performs:

Step 1:

Calculate average amplitude.

Step 2:

Reflect amplitudes around the average.

Step 3:

Increase target amplitude.

12. Amplitude Amplification Example

Suppose:

Four states:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Target:

$$|11\rangle$$

Initial amplitudes:

$$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$$

After oracle:

$$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}$$

Average amplitude:

$$\frac{1}{4}$$

After diffusion:

Target amplitude increases.

Result:

$$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$$

is transformed closer toward:

$$|11\rangle$$

13. Geometrical Interpretation

Grover's algorithm works as a rotation.

The state vector rotates toward the target state.

Each iteration rotates by:

$$\boxed{2\theta}$$

where:

$$\sin\theta = \frac{1}{\sqrt{N}}$$

14. Number of Grover Iterations

Optimal number of iterations:

$$\boxed{R = \frac{\pi}{4}\sqrt{N}}$$

Example:

For:

$$N = 100$$

Iterations:

$$R = \frac{\pi}{4} \times 10$$

≈

$$8$$

iterations.

---

15. Complete Grover Algorithm Flow

```
Initialize Qubits
       ↓
Create Superposition
       ↓
Apply Oracle
       ↓
Apply Diffusion Operator
       ↓
Repeat √N times
       ↓
Measure
       ↓
Find Solution
```

16. Oracle + Diffusion Together

One complete Grover iteration:

$$\boxed{G = DO_f}$$

where:

- O_f = Oracle
- D = Diffusion operator

Repeated application:

$$G^k|\psi\rangle$$

increases target probability.

---

17. Probability Amplification

Initially:

Probability of finding target:

$$\frac{1}{N}$$

After Grover iterations:

Probability approaches:

$$\approx 1$$

Example:

For:

$$N = 1024$$

Initial probability:

$$\frac{1}{1024}$$

After amplification:

Nearly:

$$100\%$$

18. Oracle and Diffusion Comparison

Feature	Oracle	Diffusion
Purpose	Marks solution	Amplifies solution
Operation	Phase inversion	Amplitude reflection
Effect	Changes sign	Increases probability
Position	First step	Second step

19. Advantages of Amplitude Amplification

1. Provides quadratic speedup.
2. Reduces search operations.
3. Improves probability of correct output.
4. Used in many quantum algorithms.

Frequently Asked University Questions

2-Mark Questions

1. What is a quantum oracle?
2. Define amplitude amplification.
3. Write oracle operation formula.
4. What is diffusion operator?

5-Mark Questions

1. Explain oracle function in Grover's algorithm.
2. Describe amplitude amplification.
3. Explain diffusion operator.

10-Mark Questions

1. Explain Grover's algorithm with oracle and diffusion operator.
2. Derive amplitude amplification mathematically.
3. Explain the role of oracle in quantum search.

Quick Revision

Oracle:

$$O_f|x\rangle = (-1)^{f(x)}|x\rangle$$

Marks the correct state.

Diffusion:

$$D = 2|\psi\rangle\langle\psi| - I$$

Amplifies the correct state.

Grover Iteration:

$$\boxed{G = DO_f}$$

Main Concept:

Oracle finds the correct state by phase marking, and diffusion increases its probability through amplitude amplification.

Next Topic:

Applications of Grover's Algorithm

We will cover:

1. Database Search
2. Optimization Problems
3. Cryptography Applications
4. Quantum Counting
5. Multiple Target Search

(Completes UNIT–III.)

> ok next

UNIT–III: Quantum Search Algorithms

Topic 4 – Applications of Grover's Algorithm and Its Variants

> **Weightage:** ⭐⭐⭐⭐ (Important – 5 & 10 Marks Question)
> Grover's algorithm is not limited to database searching. Its ability to amplify the probability of desired solutions makes it useful in many areas such as optimization, cryptography, and counting problems.

PART–A: Applications of Grover's Algorithm

---

1. Database Search

Introduction

The primary application of Grover's algorithm is searching an unstructured database.

In a classical computer:

- Data is searched one by one.

- Requires $O(N)$ operations.

Using Grover's algorithm:

$$\boxed{O(\sqrt{N})}$$

operations are required.

Example:

Suppose a database contains:

$$N = 1,000,000$$

records.

Classical search:

$$1,000,000$$

queries.

Quantum search:

$$\sqrt{1,000,000} = 1000$$

queries.

Working:

1. Prepare superposition of all database entries.
2. Oracle identifies the required record.
3. Diffusion amplifies its probability.
4. Measurement gives the answer.

Applications:

- Searching large databases
- Information retrieval systems
- Data mining

---

2. Cryptography Applications

Introduction

Cryptography relies on the difficulty of searching possible keys.

Grover's algorithm can speed up brute-force attacks.

Classical Key Search

For a key of size:

$$N$$

possible keys exist.

Classical brute force:

$$O(N)$$

Quantum Key Search

Grover reduces:

$$O(N)$$

to:

$$\boxed{O(\sqrt{N})}$$

Example:

For a 128-bit key:

Classical security:

$$2^{128}$$

operations.

Quantum:

$$2^{64}$$

operations.

Impact:

Grover's algorithm reduces the security level of symmetric encryption.

Solution:

Increase key size.

Example:

AES-256 provides stronger protection against quantum attacks.

---

3. Optimization Problems

Introduction

Many optimization problems involve finding the best solution among many possibilities.

Examples:

- Shortest path
- Scheduling
- Resource allocation

Grover's algorithm can accelerate the search for optimal solutions.

Example:

Finding the best schedule among:

$$N$$

possible schedules.

Classical:

$$O(N)$$

search.

Quantum:

$$O(\sqrt{N})$$

search.

Applications:

- Logistics optimization
- Manufacturing planning
- Financial optimization

---

4. Artificial Intelligence Applications

Grover's algorithm can improve search-based AI methods.

Applications:

- Feature selection
- Hyperparameter optimization
- Pattern matching

Example:

Selecting the best feature subset from many possible combinations.

---

5. Machine Learning Applications

Quantum search can help in:

- Searching training parameters
- Finding optimal models
- Data classification

---

6. Quantum Simulation

Quantum simulations require searching through large state spaces.

Grover's algorithm helps reduce computational effort.

Applications:

- Chemistry simulation
- Material science
- Physics research

PART–B: Variants of Grover's Algorithm

---

7. Quantum Counting Algorithm

Introduction

Quantum counting is an extension of Grover's algorithm.

It determines:

- Number of solutions
- Size of marked elements

Definition:

> **Quantum counting is a quantum algorithm that estimates the number of solutions in a search space using Grover's algorithm and quantum phase estimation.**

---

8. Need for Quantum Counting

In normal Grover search:

We know:

- There is one solution.

But sometimes:

- Number of solutions is unknown.

Quantum counting solves this problem.

---

9. Working Principle of Quantum Counting

Quantum counting combines:

1. Grover operator
2. Quantum phase estimation

Steps:

Step 1:

Apply Grover operator.

Step 2:

Measure phase rotation.

Step 3:

Estimate number of solutions.

---

10. Mathematical Concept

If:

$$M$$

solutions exist among:

$$N$$

items,

then:

$$\sin^2(\theta) = \frac{M}{N}$$

Quantum counting estimates:

$$\theta$$

and calculates:

$$M$$

---

11. Applications of Quantum Counting

Used in:

- Database analysis
- Probability estimation
- Search problems with unknown solutions

PART–C: Multiple Target Search

---

12. Introduction

In many cases, there are multiple valid solutions.

Example:

Database contains:

$$M$$

matching records.

Grover's algorithm can be modified to find any one of them.

---

13. Complexity of Multiple Target Search

For:

- N total elements
- M target elements

Number of iterations:

$$\boxed{O(\sqrt{\frac{N}{M}})}$$

---

14. Example

Database size:

$$N = 10000$$

Number of solutions:

$$M = 100$$

Iterations:

$$\sqrt{\frac{10000}{100}}$$

$$= \sqrt{100}$$

$$= 10$$

---

15. Advantages of Multiple Target Search

- Faster search when many solutions exist.
- Reduces number of oracle calls.
- Useful in large databases.

---

16. Comparison of Grover Variants

Algorithm	Purpose	Application
Grover Search	Find one solution	Database search
Quantum Counting	Count solutions	Unknown number of solutions
Multiple Target Search	Find one among many	Multiple matches

---

17. Limitations of Grover's Algorithm

Despite advantages:

1. Only Quadratic Speedup

Not exponential.

---

2. Requires Quantum Hardware

Needs:

- Quantum processors
- Error correction

---

3. Sensitive to Noise

Quantum errors affect results.

---

4. Requires Oracle Construction

Building an efficient oracle can be difficult.

---

18. Grover's Algorithm Summary Diagram

```
            Search Problem
                  ↓
        Create Superposition
                  ↓
            Oracle Function
                  ↓
      Mark Correct Solution
                  ↓
```

```
          Diffusion Operator
                  ↓
        Amplitude Amplification
                  ↓
             Measurement
                  ↓
            Final Answer
```

Frequently Asked University Questions

2-Mark Questions

1. List applications of Grover's algorithm.
2. What is quantum counting?
3. What is multiple target search?
4. What is the complexity of multiple target search?

5-Mark Questions

1. Explain applications of Grover's algorithm.
2. Describe quantum counting algorithm.
3. Explain multiple target search.

10-Mark Questions

1. Discuss applications and variants of Grover's algorithm.
2. Explain quantum counting and multiple target search.
3. Describe how Grover's algorithm improves database search.

UNIT–III Complete Summary ✅

Topic	Key Concept
Unstructured Search	Searching without structure
Grover Algorithm	Quantum search algorithm
Oracle	Marks solution
Amplitude Amplification	Increases probability
Diffusion Operator	Reflection operation

Topic	Key Concept
Database Search	Main application
Quantum Counting	Counts solutions
Multiple Target Search	Search among multiple answers

UNIT–III Completed ✅

Next: UNIT–IV

Quantum Algorithms for Image Processing

First Topic:

Quantum Image Representation

- Introduction
- Classical vs Quantum Image Processing
- Pixel Encoding
- Quantum Image Models

(Important for Unit-IV foundation.)

> ok next

UNIT–IV: Quantum Algorithms for Image Processing

Topic 1 – Introduction to Quantum Image Processing (QIP)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> UNIT–IV focuses on how quantum computing principles are applied to image representation, processing, and analysis. The foundation of this unit is **Quantum Image Representation**.

1. Introduction to Quantum Image Processing (QIP)

Images contain a large amount of data.

A classical image consists of:

- Pixels
- Color values

- Spatial information

As image size increases, processing requires more computational resources.

Quantum computing provides new techniques to represent and process images using qubits.

---

2. Definition of Quantum Image Processing

Exam Definition:

> **Quantum Image Processing (QIP) is a field that combines quantum computing principles with image processing techniques to represent, store, and manipulate images using quantum states.**

---

3. Need for Quantum Image Processing

Classical image processing faces challenges:

- Large storage requirements
- High computational complexity
- Slow processing for huge datasets

Quantum image processing aims to provide:

- Faster computation
- Compact image representation
- Parallel image processing

---

4. Classical Image Representation

In classical computers, images are represented using pixels.

Example:

A grayscale image:

$$M \times N$$

contains:

$$M \times N$$

pixels.

Each pixel stores an intensity value.

Example:

8-bit grayscale:

$$0 - 255$$

values.

A pixel value:

$$P(x, y)$$

represents intensity at position:

$$(x, y)$$

---

5. Quantum Image Representation

Quantum computers represent images using qubits.

A quantum image state contains:

1. Pixel position information
2. Pixel intensity information

General form:

$$\boxed{|Image\rangle}$$

Quantum image representation combines:

- Quantum states
- Superposition
- Entanglement

---

6. Basic Concept of Quantum Image Encoding

Definition:

> **Quantum image encoding is the process of converting classical image information into quantum states that can be processed by quantum computers.**

It involves encoding:

1. Image coordinates

2. Pixel intensity values

into qubits.

---

7. Components of Quantum Image Representation

A quantum image has two main components:

---

1. Position Encoding

Represents the location of pixels.

Example:

Coordinates:

$$(x, y)$$

are stored using qubits.

---

2. Color/Intensity Encoding

Represents pixel values.

Example:

Brightness:

$$0 - 255$$

---

8. Pixel Encoding in Quantum Computing

A pixel can be represented as:

$$|pixel\rangle$$

For grayscale images:

Intensity values are encoded into quantum states.

Example:

Pixel intensity:

$$0$$

represented as:

$$|0\rangle$$

Pixel intensity:

$$255$$

represented using multiple qubits.

---

9. Quantum Representation of Image Coordinates

Suppose an image has:

$$2^n \times 2^n$$

pixels.

The position requires:

$$2n$$

qubits.

Example:

For:

$$4 \times 4$$

image:

Number of position qubits:

$$2 \log_2(4)$$

$$= 4$$

qubits.

---

10. Advantages of Quantum Image Representation

---

1. Compact Storage

Large images can be represented using fewer quantum resources.

---

2. Quantum Parallelism

Multiple pixels can be processed simultaneously.

---

3. Faster Processing

Quantum algorithms can process images efficiently.

4. Improved Image Analysis

Useful for:

- Pattern recognition
- Computer vision
- Medical imaging

11. Classical Image Processing vs Quantum Image Processing

Classical Image Processing	Quantum Image Processing
Uses bits	Uses qubits
Pixel-by-pixel processing	Parallel quantum processing
Large storage requirement	Compact representation
Sequential operations	Quantum operations

12. Quantum Image Processing Pipeline

```
Classical Image
      ↓
Quantum Encoding
      ↓
Quantum Image Representation
      ↓
Quantum Image Algorithm
      ↓
Quantum Measurement
      ↓
Processed Image
```

13. Applications of Quantum Image Processing

1. Medical Imaging

Examples:

- MRI analysis
- Disease detection

---

2. Computer Vision

Applications:

- Object recognition
- Pattern detection

---

3. Satellite Image Processing

Used for:

- Remote sensing
- Earth observation

---

4. Security Systems

Used for:

- Face recognition
- Image authentication

---

14. Challenges in Quantum Image Processing

---

1. Hardware Limitations

Current quantum computers have limited qubits.

---

2. Noise Problem

Quantum states are sensitive to errors.

---

3. Encoding Complexity

Converting classical images into quantum states is challenging.

---

4. Measurement Limitations

Measurement collapses quantum states.

---

15. Important Quantum Image Representation Models

The major models are:

---

1. FRQI

Flexible Representation of Quantum Images

Introduced by:

$$\text{Le et al., 2011}$$

Features:

- Encodes grayscale/color images.
- Uses quantum amplitudes for pixel values.

---

2. NEQR

Novel Enhanced Quantum Representation

Features:

- Stores pixel values directly in computational basis states.
- Provides exact pixel representation.

---

16. FRQI vs NEQR Overview

Feature	FRQI	NEQR
Full form	Flexible Representation of Quantum Images	Novel Enhanced Quantum Representation
Pixel encoding	Amplitude	Basis states
Image type	Color/grayscale	Mostly grayscale
Accuracy	Approximate	Exact

Feature	FRQI	NEQR
Complexity	Lower	Higher

17. Important Terms

Term	Meaning
Qubit	Quantum information unit
Pixel Encoding	Representing pixel data in quantum form
Quantum Image	Image represented using quantum states
FRQI	Amplitude-based representation
NEQR	Basis-state representation

Frequently Asked University Questions

2-Mark Questions

1. Define Quantum Image Processing.
2. What is quantum image representation?
3. What is pixel encoding?
4. Name quantum image representation models.

5-Mark Questions

1. Explain quantum image representation.
2. Compare classical and quantum image processing.
3. Explain pixel encoding.

10-Mark Questions

1. Explain quantum image processing and its applications.
2. Describe quantum image representation techniques.
3. Explain the need for quantum image processing.

Quick Revision

Quantum Image Processing:

$$\boxed{ImageProcessing + QuantumComputing}$$

Main Steps:

1. Encode image
2. Represent using quantum states
3. Apply quantum algorithms
4. Measure output

Key Concept:

Quantum image processing uses qubits to represent image information and exploits superposition and parallelism for efficient image computation.

Next Topic:

Quantum Image Representation Models – FRQI (Flexible Representation of Quantum Images)

We will cover:

1. FRQI concept
2. Mathematical representation
3. Pixel encoding
4. Circuit model
5. Advantages and limitations

(Important Unit-IV 10-mark question.)

> ok next

UNIT–IV: Quantum Algorithms for Image Processing

Topic 2 – FRQI (Flexible Representation of Quantum Images)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> FRQI is one of the first and most important quantum image representation models. It provides a way to represent grayscale and color images using quantum states.

---

1. Introduction to FRQI

FRQI stands for:

$$\boxed{\text{Flexible Representation of Quantum Images}}$$

It was introduced by:

$$\boxed{\text{Le et al. (2011)}}$$

FRQI is a quantum image representation model that encodes:

1. Pixel position information
2. Pixel intensity/color information

using qubits.

---

2. Definition of FRQI

Exam Definition:

> **FRQI is a quantum image representation technique that encodes pixel positions using computational basis states and pixel intensity information using probability amplitudes of a quantum state.**

---

3. Basic Idea of FRQI

A classical image consists of:

- Coordinates
- Pixel values

FRQI converts these into quantum states.

For an image:

$$2^n \times 2^n$$

pixels:

- Position requires:

$$2n$$

qubits.

- Color/intensity requires:

qubit.

Total qubits:

$$\boxed{2n+1}$$

4. Structure of FRQI Representation

The general FRQI state is:

$$\boxed{|I\rangle = \frac{1}{2^n}\sum_{i=0}^{2^{2n}-1}(\cos\theta_i|0\rangle + \sin\theta_i|1\rangle)|i\rangle}$$

Where:

Symbol	Meaning
(	I\rangle)
i	Pixel position
θ_i	Pixel intensity angle
(	i\rangle)
$\cos\theta_i, \sin\theta_i$	Color information

5. Pixel Encoding in FRQI

FRQI does not store pixel values directly.

Instead, pixel intensity is converted into an angle.

The intensity is represented by:

$$\theta_i$$

The color qubit state:

$$|C_i\rangle$$

is:

$$\boxed{|C_i\rangle = \cos\theta_i|0\rangle + \sin\theta_i|1\rangle}$$

6. Mapping Pixel Values to Angles

Pixel intensity:

$$0 \rightarrow \theta = 0$$

Pixel intensity:

$$255 \rightarrow \theta = \frac{\pi}{2}$$

Therefore:

Dark pixel:

$$|0\rangle$$

Bright pixel:

$$|1\rangle$$

7. Example of FRQI Encoding

Consider a simple:

$$2 \times 2$$

image.

Pixels:

$$P_0, P_1, P_2, P_3$$

Position states:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

FRQI state:

$$|I\rangle = \frac{1}{2}[(\cos\theta_0|0\rangle + \sin\theta_0|1\rangle)|00\rangle$$
$$+(\cos\theta_1|0\rangle + \sin\theta_1|1\rangle)|01\rangle$$
$$+(\cos\theta_2|0\rangle + \sin\theta_2|1\rangle)|10\rangle$$
$$+(\cos\theta_3|0\rangle + \sin\theta_3|1\rangle)|11\rangle]$$

8. FRQI Circuit Representation

A general FRQI circuit contains:

```
Position Qubits

|0> ─────H──────────
|0> ─────H──────────

        ↓

Color Encoding Rotation

        ↓

Quantum Image State
```

9. Operations in FRQI

FRQI involves:

---

1. Position Initialization

Hadamard gates create equal superposition of pixel positions.

Example:

$$|00\rangle$$

becomes:

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

---

2. Color Encoding

Rotation gates encode intensity.

Rotation:

$$R_y(\theta)$$

is used.

---

10. Advantages of FRQI

---

1. Compact Representation

An image can be represented using fewer qubits.

Example:

A:

$$2^n \times 2^n$$

image requires:

$$2n + 1$$

qubits.

---

2. Supports Quantum Processing

Can be directly used with quantum algorithms.

---

3. Represents Color Images

FRQI can encode grayscale and color information.

---

4. Uses Quantum Parallelism

Multiple pixels are represented simultaneously.

---

11. Limitations of FRQI

---

1. Pixel Retrieval Problem

Extracting pixel values requires measurement.

---

2. Approximate Encoding

Intensity values are represented by angles.

---

3. Complex Preparation

Creating FRQI states requires complex quantum circuits.

---

4. Measurement Cost

Reading complete images requires many measurements.

12. Applications of FRQI

1. Image Processing

Used for:

- Image transformation
- Filtering

2. Pattern Recognition

Used in:

- Object detection
- Feature extraction

3. Quantum Computer Vision

Used for:

- Quantum image analysis

13. FRQI Advantages and Disadvantages Summary

Advantages	Disadvantages
Requires fewer qubits	Difficult pixel extraction
Supports color images	Intensity is indirect
Uses quantum parallelism	Complex preparation
Suitable for quantum algorithms	Measurement challenges

14. FRQI vs Classical Image Representation

Classical Image	FRQI
Stores pixels directly	Stores amplitudes
Uses bits	Uses qubits
Sequential processing	Quantum parallel processing
Large storage	Compact representation

15. Important Exam Diagram

```
Classical Image
      ↓
Pixel Position Encoding
      ↓
Intensity Encoding
      ↓
Quantum State
      ↓
FRQI Representation
```

Frequently Asked University Questions

2-Mark Questions

1. Expand FRQI.
2. Who introduced FRQI?
3. How does FRQI encode pixel intensity?
4. How many qubits are required for a $2^n \times 2^n$ image?

5-Mark Questions

1. Explain FRQI model.
2. Describe pixel encoding in FRQI.
3. Explain the advantages of FRQI.

10-Mark Questions

1. Explain FRQI quantum image representation with mathematical formulation.

2. Describe FRQI model and its working.
3. Compare FRQI with classical image representation.

Quick Revision

FRQI Formula:

$$\boxed{|I\rangle = \frac{1}{2^n}\sum_i (\cos\theta_i|0\rangle + \sin\theta_i|1\rangle)|i\rangle}$$

Main Idea:

FRQI stores pixel positions in basis states and pixel intensity as amplitudes of a color qubit.

Next Topic:

NEQR (Novel Enhanced Quantum Representation)

We will cover:

1. NEQR concept
2. Mathematical representation
3. Pixel encoding method
4. FRQI vs NEQR comparison
5. Advantages and limitations

(Important 10-mark Unit-IV question.)

> ok next

UNIT–IV: Quantum Algorithms for Image Processing

Topic 3 – NEQR (Novel Enhanced Quantum Representation)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> NEQR is an advanced quantum image representation model that improves the accuracy of pixel representation compared with FRQI.

---

1. Introduction to NEQR

NEQR stands for:

$$\boxed{\text{Novel Enhanced Quantum Representation}}$$

It was introduced to overcome some limitations of FRQI.

FRQI stores pixel intensity using amplitudes, while NEQR stores pixel values directly using computational basis states.

---

2. Definition of NEQR

Exam Definition:

> **NEQR is a quantum image representation model that encodes image pixel values directly into qubit basis states along with their corresponding position information.**

---

3. Need for NEQR

FRQI limitations:

- Pixel values are represented indirectly.
- Image retrieval is difficult.
- Measurement complexity is high.

NEQR solves these problems by:

- Direct pixel encoding
- Exact intensity representation
- Easier image retrieval

---

4. Basic Idea of NEQR

A digital image contains:

1. Pixel position
2. Pixel intensity value

NEQR represents both using qubits.

Example:

A grayscale image:

$$2^n \times 2^n$$

requires:

- Position qubits
- Color qubits

---

5. NEQR Representation Structure

The general NEQR state is:

$$\boxed{|I\rangle = \frac{1}{2^n}\sum_{y=0}^{2^n-1}\sum_{x=0}^{2^n-1}|f(y,x)\rangle|yx\rangle}$$

Where:

Symbol	Meaning
(	I\rangle)
x, y	Pixel coordinates
(	yx\rangle)
(	f(y,x)\rangle)

---

6. Pixel Encoding in NEQR

Unlike FRQI:

FRQI:

$$\text{Intensity} \rightarrow \text{Amplitude}$$

NEQR:

$$\boxed{\text{Intensity} \rightarrow \text{Binary Qubit State}}$$

Example:

For an 8-bit grayscale image:

Pixel intensity range:

$$0 - 255$$

requires:

$$8$$

qubits.

7. Example of NEQR Encoding

Suppose pixel intensity:

$$200$$

Binary representation:

$$200 = (11001000)_2$$

NEQR stores:

$$|11001000\rangle$$

as the pixel value.

---

8. NEQR Image Representation

For a pixel:

$$(x, y)$$

with intensity:

$$f(x, y)$$

the quantum state is:

$$|f(x, y)\rangle |xy\rangle$$

Example:

Pixel position:

$$|01\rangle$$

Intensity:

$$|10101010\rangle$$

Combined state:

$$\boxed{|10101010\rangle |01\rangle}$$

---

9. Qubit Requirement in NEQR

For an image:

$$2^n \times 2^n$$

pixels

with:

$$q$$

bits for intensity,

total qubits:

$$\boxed{q + 2n}$$

Example:

For:

$$256 \times 256$$

grayscale image:

Position qubits:

$$16$$

Intensity qubits:

$$8$$

Total:

$$24$$

qubits.

---

10. NEQR Circuit Representation

General circuit:

```
Position Qubits

|0> ──H──────────────
|0> ──H──────────────

        ↓

Pixel Value Encoding

        ↓

Quantum Image State
```

11. Working Steps of NEQR

Step 1: Initialize Qubits

Create position states.

Step 2: Generate Superposition

Hadamard gates create all pixel locations.

Step 3: Encode Pixel Values

Binary pixel values are stored in intensity qubits.

Step 4: Obtain Quantum Image

The complete quantum image state is created.

12. Advantages of NEQR

1. Exact Pixel Representation

Pixel values are stored directly.

2. Better Image Retrieval

Pixel values can be extracted accurately.

3. Suitable for Digital Images

Works well with binary pixel representation.

4. High Accuracy

No approximation using angles.

5. Efficient Quantum Processing

Supports quantum image algorithms.

13. Limitations of NEQR

1. More Qubits Required

Requires additional qubits for intensity storage.

2. Circuit Complexity

Image preparation is complex.

3. Hardware Requirements

Needs advanced quantum systems.

4. Limited Current Implementation

Current quantum computers cannot handle large images.

14. Applications of NEQR

1. Image Processing

Used for:

- Image enhancement
- Image transformation

2. Edge Detection

Used for identifying boundaries.

3. Image Recognition

Used in quantum computer vision.

4. Pattern Analysis

Used for feature extraction.

15. FRQI vs NEQR Comparison

Feature	FRQI	NEQR
Full Form	Flexible Representation of Quantum Images	Novel Enhanced Quantum Representation
Pixel encoding	Amplitude encoding	Basis-state encoding
Pixel value	Indirect	Direct
Accuracy	Approximate	Exact
Color support	Yes	Mainly grayscale
Qubits required	Less	More
Retrieval	Difficult	Easier
Complexity	Lower	Higher

16. FRQI and NEQR Example Comparison

Suppose pixel value:

$$100$$

FRQI:

Converted into angle:

$$\theta$$

Stored as:

$$\cos\theta|0\rangle + \sin\theta|1\rangle$$

NEQR:

Binary:

$$100 = (01100100)_2$$

Stored as:

$$|01100100\rangle$$

17. NEQR Advantages Over FRQI

NEQR provides:

✔ Direct pixel storage
✔ Higher accuracy
✔ Easier image reconstruction
✔ Better for digital image processing

18. Quantum Image Representation Comparison

Model	Encoding Method	Main Feature
FRQI	Amplitude encoding	Compact representation
NEQR	Binary basis encoding	Exact pixel values

Frequently Asked University Questions

2-Mark Questions

1. Expand NEQR.
2. How does NEQR encode pixel values?
3. What is the difference between FRQI and NEQR?
4. How many qubits are required for an 8-bit image?

5-Mark Questions

1. Explain NEQR model.
2. Describe pixel encoding in NEQR.
3. Compare FRQI and NEQR.

10-Mark Questions

1. Explain NEQR quantum image representation with mathematical formulation.
2. Compare FRQI and NEQR models.

3. Describe the working of NEQR with an example.

Quick Revision

NEQR Formula:

$$\boxed{|I\rangle = \frac{1}{2^n}\sum_{x,y}|f(x,y)\rangle|xy\rangle}$$

Main Idea:

NEQR stores pixel intensity values directly as binary quantum states and stores pixel positions using position qubits.

Next Topic:

Quantum Image Processing Techniques – Edge Detection

We will cover:

1. Introduction to edge detection
2. Classical edge detection methods
3. Quantum edge detection concept
4. Quantum operators
5. Applications

(Important final topic of UNIT–IV.)

> ok next

UNIT–IV: Quantum Algorithms for Image Processing

Topic 4 – Quantum Image Processing Technique: Edge Detection

> **Weightage:** ⭐⭐⭐⭐⭐ (Important – 10 Marks Question)
> Edge detection is one of the most important image processing operations. In quantum image processing, quantum techniques are used to identify boundaries and important features of images efficiently.

---

1. Introduction to Edge Detection

An image consists of different regions with different intensity values.

The boundary between two different regions is called an **edge**.

Definition:

> **Edge detection is an image processing technique used to identify points where the intensity of an image changes significantly, representing object boundaries.**

---

2. Importance of Edge Detection

Edges provide important information about:

- Shape
- Size
- Structure
- Object boundaries

Applications:

- Object recognition
- Medical image analysis
- Face detection
- Autonomous vehicles

---

3. What is an Edge?

An edge occurs when there is a sudden change in pixel intensity.

Example:

Dark region:

$$50$$

Bright region:

$$200$$

The transition point is an edge.

---

4. Classical Edge Detection Methods

Before quantum edge detection, classical methods are used.

Common operators:

1. Sobel Operator
2. Prewitt Operator
3. Canny Edge Detector
4. Laplacian Operator

---

5. Sobel Edge Detection

The Sobel operator detects edges by calculating intensity gradients.

It uses two filters:

Horizontal gradient:

$$G_x$$

Vertical gradient:

$$G_y$$

Edge magnitude:

$$G = \sqrt{G_x^2 + G_y^2}$$

---

6. Canny Edge Detection

Canny is a multi-step edge detection algorithm.

Steps:

1. Noise reduction
2. Gradient calculation
3. Non-maximum suppression
4. Edge tracking

It provides accurate edge detection.

PART–B: Quantum Edge Detection

---

7. Introduction to Quantum Edge Detection

Quantum edge detection applies quantum computing concepts to detect image boundaries.

It uses:

- Quantum image representation
- Quantum operators
- Quantum parallelism

Definition:

> **Quantum edge detection is a quantum image processing technique that identifies image boundaries by applying quantum operations on encoded image states.**

---

8. Need for Quantum Edge Detection

Classical edge detection:

- Processes pixels sequentially.
- Requires large computational resources for high-resolution images.

Quantum approach provides:

- Parallel processing
- Faster computation
- Compact representation

---

9. Basic Steps of Quantum Edge Detection

Quantum edge detection involves:

```
Classical Image
      ↓
Quantum Encoding
      ↓
Quantum Image Representation
      ↓
Quantum Edge Operator
      ↓
Measurement
      ↓
Edge Image
```

10. Quantum Edge Detection Using FRQI

FRQI represents image as:

$$|I\rangle$$

where:

- Position information is stored in qubits.
- Pixel intensity is stored in amplitudes.

Edge detection operations are applied directly on:

$$|I\rangle$$

---

11. Quantum Edge Detection Using NEQR

In NEQR:

Pixel values are stored as binary states.

Example:

Pixel:

$$100$$

stored as:

$$|01100100\rangle$$

Quantum operations compare neighboring pixels.

---

12. Quantum Operators for Edge Detection

Quantum edge detection uses:

---

1. Quantum Fourier Transform (QFT)

Used for frequency analysis.

Similar to classical Fourier Transform.

Applications:

- Feature extraction
- Image filtering

2. Quantum Comparison Operators

Compare pixel values.

Used to identify sudden changes.

3. Quantum Rotation Operators

Modify pixel intensity states.

13. Quantum Edge Detection Algorithm

General steps:

Step 1: Image Encoding

Convert classical image into quantum state.

Example:

FRQI/NEQR.

Step 2: Pixel Comparison

Compare neighboring pixel intensities.

Step 3: Apply Quantum Operator

Detect intensity differences.

Step 4: Measurement

Extract edge information.

14. Mathematical Concept

For neighboring pixels:

$$P(x,y)$$

and:

$$P(x+1,y)$$

difference:

$$D = |P(x+1,y) - P(x,y)|$$

If:

$$D > T$$

where T is threshold,

then:

$$\text{Edge detected}$$

---

15. Quantum Advantage in Edge Detection

---

1. Parallel Processing

All pixels can be processed simultaneously.

---

2. Reduced Computation

Requires fewer operations for large images.

---

3. Efficient Storage

Quantum representation reduces memory requirement.

---

4. Faster Feature Extraction

Useful for computer vision.

---

16. Applications of Quantum Edge Detection

---

1. Medical Imaging

Examples:

- Tumor boundary detection
- MRI analysis

---

2. Satellite Image Processing

Used for:

- Land boundary detection
- Object identification

---

3. Security Systems

Used in:

- Face recognition
- Surveillance

---

4. Autonomous Systems

Used in:

- Self-driving vehicles
- Robot vision

---

17. Classical vs Quantum Edge Detection

Classical Edge Detection	Quantum Edge Detection
Uses bits	Uses qubits
Sequential processing	Quantum parallel processing
High memory usage	Compact representation
Limited speed	Potential quantum speedup

---

18. Challenges of Quantum Edge Detection

---

1. Quantum Hardware Limitation

Current quantum computers have limited qubits.

2. Noise

Quantum errors affect image processing.

3. Encoding Cost

Converting images into quantum form is complex.

4. Measurement Problem

Reading quantum images is difficult.

19. Complete UNIT–IV Summary

Topic	Main Concept
Quantum Image Processing	Applying quantum computing to images
Pixel Encoding	Converting pixels into quantum states
FRQI	Amplitude-based image representation
NEQR	Binary-state image representation
Edge Detection	Finding image boundaries

Frequently Asked University Questions

2-Mark Questions

1. Define edge detection.
2. What are quantum image processing techniques?
3. Name quantum image representation models.
4. What is the role of QFT in image processing?

5-Mark Questions

1. Explain quantum edge detection.
2. Describe quantum image processing pipeline.
3. Explain applications of quantum image processing.

10-Mark Questions

1. Explain quantum edge detection technique.
2. Discuss quantum algorithms used in image processing.
3. Explain FRQI, NEQR, and edge detection methods.

Quick Revision

Edge Detection:

$$\boxed{\text{Find sudden changes in pixel intensity}}$$

Quantum Approach:

$$\boxed{\text{Encode Image} \rightarrow \text{Apply Quantum Operations} \rightarrow \text{Measure Edges}}$$

UNIT–IV COMPLETED ✅

Next: UNIT–V

Application-Oriented Quantum Algorithms

First Topic:

Quantum Optimization and Variational Quantum Eigensolver (VQE)

We will cover:

1. Introduction to quantum optimization
2. Classical vs quantum optimization
3. VQE architecture
4. Working principle
5. Applications

(Important Unit-V long-answer topic.)

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 1 – Introduction to Quantum Optimization

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> Quantum optimization is one of the most practical applications of quantum computing. It focuses on solving complex optimization problems that are difficult for classical computers.

---

1. Introduction to Optimization Problems

Optimization means finding the **best possible solution** among many possible solutions.

The goal may be:

- Minimize cost
- Maximize profit
- Reduce time
- Improve efficiency

Examples:

1. Logistics

Finding the shortest delivery route.

---

2. Scheduling

Creating the best timetable.

---

3. Finance

Selecting the best investment strategy.

---

2. Definition of Optimization Problem

Exam Definition:

An optimization problem is a computational problem where the objective is to find the optimal solution from a set of possible solutions while satisfying given constraints.

3. Mathematical Representation

An optimization problem can be represented as:

$$\boxed{\text{Minimize/Maximize } f(x)}$$

Subject to:

$$g(x) \leq b$$

Where:

Term	Meaning
$f(x)$	Objective function
x	Decision variables
$g(x)$	Constraints

4. Types of Optimization Problems

1. Continuous Optimization

Variables can take any value.

Example:

$$x = 2.567$$

Applications:

- Engineering design
- Physics problems

2. Discrete Optimization

Variables have fixed values.

Example:

$$x = 0, 1$$

Applications:

- Scheduling
- Route planning

---

3. Combinatorial Optimization

Finding the best combination among many possibilities.

Examples:

- Traveling Salesman Problem
- Job scheduling
- Network design

---

5. Classical Optimization Problems

Classical computers solve optimization using algorithms like:

- Linear programming
- Dynamic programming
- Genetic algorithms
- Simulated annealing

However, some problems become extremely difficult as size increases.

---

6. Complexity of Optimization Problems

Many optimization problems have exponential growth.

Example:

For N items:

Possible combinations:

$$2^N$$

Example:

For:

$$N = 100$$

possibilities:

$$2^{100}$$

which is extremely large.

7. Need for Quantum Optimization

Quantum computers use:

- Superposition
- Entanglement
- Quantum interference

to explore solution spaces more efficiently.

---

8. Definition of Quantum Optimization

Exam Definition:

> **Quantum optimization is the application of quantum computing techniques to solve optimization problems by representing solutions as quantum states and using quantum algorithms to find optimal solutions.**

---

9. Basic Idea of Quantum Optimization

Classical approach:

Search solutions one by one.

Quantum approach:

Represent many possible solutions simultaneously.

Process:

```
Optimization Problem
        ↓
Quantum Representation
        ↓
Quantum Algorithm
        ↓
Find Optimal Solution
        ↓
Measurement
```

10. Quantum Optimization Workflow

Step 1: Convert Problem into Mathematical Form

Example:

Cost function:

$$C(x)$$

Step 2: Encode into Quantum System

Convert variables into qubits.

Step 3: Apply Quantum Algorithm

Examples:

- VQE
- QAOA

Step 4: Measure Output

Obtain best solution.

---

11. Quantum Optimization Algorithms

The two important algorithms in syllabus:

---

1. VQE

Variational Quantum Eigensolver

Used for:

- Finding minimum energy states
- Quantum chemistry problems

---

2. QAOA

Quantum Approximate Optimization Algorithm

Used for:

- Combinatorial optimization
- Scheduling
- Routing problems

12. Classical Optimization vs Quantum Optimization

Classical Optimization	Quantum Optimization
Uses bits	Uses qubits
Searches sequentially	Uses quantum states
Limited for large problems	Handles complex spaces
Classical algorithms	Quantum algorithms

13. Advantages of Quantum Optimization

1. Handles Large Search Spaces

Can represent many possibilities.

2. Faster Optimization

Provides potential speed improvement.

3. Solves Complex Problems

Useful for:

- Logistics
- Finance
- Machine learning

4. Better Resource Utilization

Finds efficient solutions.

14. Applications of Quantum Optimization

1. Logistics Optimization

Problems:

- Vehicle routing
- Delivery planning

Example:

Find shortest route:

$$A \to B \to C$$

with minimum cost.

---

2. Scheduling Problems

Applications:

- Employee scheduling
- Manufacturing schedules
- Exam timetables

---

3. Finance

Applications:

- Portfolio optimization
- Risk management

---

4. Machine Learning

Applications:

- Feature selection
- Parameter optimization

---

15. Challenges of Quantum Optimization

---

1. Current Hardware Limitations

Quantum computers have:

- Limited qubits
- Noise problems

---

2. Algorithm Complexity

Designing quantum algorithms is difficult.

---

3. Error Correction

Quantum errors affect optimization accuracy.

---

16. Quantum Optimization Example: Traveling Salesman Problem

Problem:

Find shortest route visiting all cities once.

Classical:

Number of routes:

$$(N-1)!$$

Quantum optimization:

Uses algorithms like:

- QAOA

to find approximate optimal routes.

---

17. Important Terms

Term	Meaning
Optimization	Finding best solution
Objective Function	Function to minimize/maximize
Constraint	Condition to satisfy
Cost Function	Measures solution quality
Qubit Encoding	Representing variables quantum mechanically

Frequently Asked University Questions

2-Mark Questions

1. Define quantum optimization.
2. What is an optimization problem?
3. Name quantum optimization algorithms.
4. Expand VQE and QAOA.

5-Mark Questions

1. Explain quantum optimization.
2. Describe applications of quantum optimization.
3. Compare classical and quantum optimization.

10-Mark Questions

1. Explain quantum optimization algorithms and applications.
2. Discuss the need for quantum optimization.
3. Explain quantum optimization workflow.

Quick Revision

Optimization:

$$\boxed{\text{Find the best solution among many possibilities}}$$

Quantum Optimization:

$$\boxed{\text{Optimization} + \text{Quantum Computing}}$$

Main Algorithms:

$$\boxed{VQE,\ QAOA}$$

Next Topic:

Variational Quantum Eigensolver (VQE)

We will cover:

1. Introduction
2. Eigenvalue problem
3. VQE architecture
4. Working principle
5. Variational principle
6. Applications

(One of the most important Unit-V topics.)

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 2 – Variational Quantum Eigensolver (VQE)

> **Weightage:** ⭐⭐⭐⭐⭐ (Most Important – 10 Marks Question)
> VQE is one of the most important hybrid quantum-classical algorithms. It is widely used for quantum chemistry, molecular simulation, and optimization problems.

---

1. Introduction to VQE

VQE stands for:

$$\boxed{\text{Variational Quantum Eigensolver}}$$

It was introduced in:

$$\boxed{2014}$$

by:

- Peruzzo et al.

VQE is a **hybrid quantum-classical algorithm**.

It combines:

- Quantum computer → Performs quantum state preparation and measurement
- Classical computer → Optimizes parameters

---

2. Definition of VQE

Exam Definition:

> **Variational Quantum Eigensolver (VQE) is a hybrid quantum-classical algorithm used to find the minimum eigenvalue of a quantum system by optimizing a parameterized quantum circuit.**

---

3. Basic Concept of VQE

Many problems in physics and chemistry require finding:

$$\text{Ground State Energy}$$

of a system.

Mathematically:

$$H|\psi\rangle = E|\psi\rangle$$

Where:

Symbol	Meaning
H	Hamiltonian operator
(	\psi\rangle)
E	Energy eigenvalue

The goal:

Find the minimum value of:

$$E$$

---

4. Variational Principle

VQE is based on the variational principle.

Statement:

> **For any trial quantum state, the expected energy is always greater than or equal to the true ground state energy.**

Mathematically:

$$\boxed{E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle}$$

The algorithm minimizes:

$$E(\theta)$$

by changing parameters:

$$\theta$$

5. VQE Architecture

VQE consists of two main parts:

```
        Classical Computer

              |
              |
      Parameter Optimization

              ↓

        Quantum Computer

              |
              |
    Parameterized Circuit

              |
              |
        Measurement

              ↓

        Energy Value
```

6. Main Components of VQE

1. Parameterized Quantum Circuit

Also called:

$$\boxed{\text{Ansatz}}$$

It prepares a trial quantum state:

$$|\psi(\theta)\rangle$$

The circuit contains adjustable parameters:

$$\theta_1, \theta_2, ..., \theta_n$$

Example:

```
|0> ──Ry(θ1)──Rz(θ2)──
|0> ──Ry(θ3)──Rz(θ4)──
```

2. Quantum Measurement

The quantum computer calculates:

$$\langle H \rangle$$

The measured energy is sent to the classical optimizer.

3. Classical Optimizer

The optimizer updates parameters:

$$\theta$$

to reduce energy.

Examples:

- Gradient descent
- Adam optimizer
- COBYLA

7. Working Principle of VQE

VQE follows these steps:

Step 1: Initialize Quantum Circuit

Create an initial quantum state.

$$|\psi(0)\rangle$$

Step 2: Apply Parameterized Gates

The circuit produces:

$$|\psi(\theta)\rangle$$

Step 3: Measure Energy

Calculate:

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$$

Step 4: Classical Optimization

Optimizer changes:

$$\theta$$

to minimize:

$$E(\theta)$$

Step 5: Repeat

The process continues until minimum energy is obtained.

8. VQE Algorithm Flow

```
Start
 ↓
Initialize Parameters
 ↓
Prepare Quantum State
 ↓
Measure Energy
 ↓
Classical Optimizer Updates Parameters
 ↓
Energy Minimized?
 ↓
Yes → Final Solution
No → Repeat
```

9. Mathematical Representation

The trial state:

$$|\psi(\theta)\rangle$$

is created by:

$$\boxed{|\psi(\theta)\rangle = U(\theta)|0\rangle}$$

Energy calculation:

$$\boxed{E(\theta) = \langle 0|U^{\dagger}(\theta)HU(\theta)|0\rangle}$$

The optimizer finds:

$$\theta_{min}$$

such that:

$$E(\theta)$$

is minimum.

10. Example: Molecular Energy Calculation

Molecules have energy states.

Example:

$$H_2$$

molecule.

Goal:

Find:

$$\text{Lowest energy state}$$

VQE:

1. Represents molecule using qubits.
2. Creates trial state.
3. Measures energy.
4. Optimizes parameters.

11. Applications of VQE

1. Quantum Chemistry

Most important application.

Used for:

- Molecular simulation
- Chemical reactions
- Drug discovery

---

2. Material Science

Used for:

- Finding material properties
- Designing new materials

---

3. Optimization Problems

Used for:

- Energy optimization
- Resource allocation

---

4. Machine Learning

Used in:

- Quantum machine learning models

---

12. Advantages of VQE

---

1. Suitable for Current Quantum Computers

Works on:

$$\text{NISQ devices}$$

(Noisy Intermediate Scale Quantum computers)

---

2. Reduces Quantum Resource Requirement

Requires fewer qubits compared with some algorithms.

---

3. Hybrid Approach

Combines strengths of classical and quantum systems.

4. Flexible

Can solve different problems by changing Hamiltonian.

13. Limitations of VQE

1. Optimization Difficulty

Classical optimization may get stuck.

2. Noise Sensitivity

Quantum errors affect measurements.

3. Circuit Depth Problem

Large circuits create more errors.

4. Requires Many Measurements

Energy estimation requires repeated measurements.

14. VQE vs Classical Methods

Feature	Classical Method	VQE
Processor	Classical computer	Quantum + Classical
State representation	Bits	Qubits
Optimization	Classical algorithms	Hybrid optimization
Large quantum systems	Difficult	Potential advantage

15. VQE vs QAOA

Feature	VQE	QAOA
Full form	Variational Quantum Eigensolver	Quantum Approximate Optimization Algorithm
Main purpose	Find minimum eigenvalue	Solve optimization problems
Application	Chemistry, physics	Scheduling, routing
Output	Energy value	Optimal solution

Frequently Asked University Questions

2-Mark Questions

1. Expand VQE.
2. What is variational principle?
3. What is an ansatz?
4. Why is VQE called hybrid?

5-Mark Questions

1. Explain working of VQE.
2. Describe VQE architecture.
3. Explain applications of VQE.

10-Mark Questions

1. Explain Variational Quantum Eigensolver with architecture and working.
2. Discuss the role of classical optimization in VQE.
3. Explain VQE algorithm and its applications.

Quick Revision

VQE:

$$\boxed{\text{Quantum Circuit} + \text{Classical Optimization}}$$

Main Equation:

$$\boxed{E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle}$$

Working:

$$\boxed{Prepare\ State \rightarrow Measure\ Energy \rightarrow Optimize\ Parameters \rightarrow Minimum\ Energy}$$

Next Topic:

Quantum Approximate Optimization Algorithm (QAOA)

We will cover:

1. Introduction
2. QAOA architecture
3. Cost Hamiltonian
4. Mixer Hamiltonian
5. Working steps
6. Applications in logistics and scheduling

(Extremely important Unit-V question.)

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 3 – Quantum Approximate Optimization Algorithm (QAOA)

> **Weightage:** ⭐⭐⭐⭐⭐ (Most Important – 10 Marks Question)
> QAOA is one of the most important quantum optimization algorithms used for solving real-world combinatorial optimization problems such as logistics, scheduling, routing, and resource allocation.

---

1. Introduction to QAOA

QAOA stands for:

$$\boxed{\text{Quantum Approximate Optimization Algorithm}}$$

It was introduced by:

$$\boxed{\text{Farhi, Goldstone, and Gutmann (2014)}}$$

QAOA is a:

It combines:

- Quantum circuits → Generate possible solutions
- Classical optimizer → Finds best parameters

2. Definition of QAOA

Exam Definition:

> **Quantum Approximate Optimization Algorithm (QAOA) is a hybrid quantum algorithm that uses parameterized quantum circuits to find approximate solutions to combinatorial optimization problems by minimizing a cost function.**

3. Need for QAOA

Many optimization problems are difficult for classical computers.

Examples:

- Traveling Salesman Problem
- Scheduling
- Graph problems
- Resource allocation

The number of possible solutions grows exponentially.

Example:

For:

$$N$$

binary variables:

Possible solutions:

$$2^N$$

QAOA attempts to find good solutions efficiently using quantum principles.

4. Basic Concept of QAOA

QAOA uses two main quantum operations:

1. Cost Hamiltonian

Represents the problem objective.

2. Mixer Hamiltonian

Creates transitions between possible solutions.

The algorithm alternates between these two operations.

5. QAOA Architecture

```
          Classical Optimizer
                  |
                  |
       Updates Parameters
                  ↓
         Quantum Computer
                  |
    Initial Quantum State
                  ↓
       Cost Hamiltonian
                  ↓
     Mixer Hamiltonian
                  ↓
        Measurement
                  ↓
      Solution Output
```

6. Mathematical Foundation of QAOA

QAOA tries to minimize:

$$\boxed{C(x)}$$

where:

- $C(x)$ = cost function
- x = possible solution

The quantum state is:

$$|\psi(\gamma,\beta)\rangle$$

Where:

- γ = cost parameter
- β = mixer parameter

7. Components of QAOA

Part 1: Cost Hamiltonian

The cost Hamiltonian represents the optimization problem.

It assigns energy values to solutions.

Mathematical representation:

$$\boxed{H_C}$$

The goal:

Find the state with minimum cost.

Example:

For a graph problem:

$$H_C$$

represents:

- Edge weights
- Constraints

Part 2: Mixer Hamiltonian

The mixer Hamiltonian allows movement between solutions.

Represented as:

$$\boxed{H_M}$$

Usually:

$$H_M = \sum_i X_i$$

where:

$$X_i$$

is Pauli-X operator.

8. QAOA Quantum State

The initial state is:

$$|+\rangle^{\otimes n}$$

Created using Hadamard gates.

Then:

Cost operation:

$$e^{-i\gamma H_C}$$

Mixer operation:

$$e^{-i\beta H_M}$$

Final state:

$$\boxed{|\psi(\gamma,\beta)\rangle = e^{-i\beta H_M} e^{-i\gamma H_C} |+\rangle^{\otimes n}}$$

9. Working Principle of QAOA

QAOA works in multiple steps.

Step 1: Define Optimization Problem

Convert problem into:

$$C(x)$$

Example:

Minimum travel cost.

Step 2: Encode Problem into Cost Hamiltonian

Create:

$$H_C$$

Step 3: Initialize Quantum State

Create equal superposition:

$$|+\rangle^n$$

Using Hadamard gates.

Step 4: Apply Cost Hamiltonian

The cost operator:

$$e^{-i\gamma H_C}$$

changes phases according to solution quality.

Step 5: Apply Mixer Hamiltonian

The mixer:

$$e^{-i\beta H_M}$$

explores other solutions.

Step 6: Measurement

Measure the quantum state.

The best solution is obtained.

Step 7: Classical Optimization

Classical optimizer adjusts:

$$\gamma, \beta$$

to improve the solution.

10. QAOA Algorithm Flow

```
Optimization Problem
        ↓
Create Cost Function
        ↓
Build Cost Hamiltonian
        ↓
Initialize Qubits
        ↓
Apply Cost Layer
        ↓
Apply Mixer Layer
        ↓
Measure Output
        ↓
Classical Parameter Optimization
        ↓
Best Solution
```

11. QAOA Layers

Each repetition is called a:

$$\boxed{\text{QAOA Layer}}$$

For depth:

$$p$$

the circuit contains:

$$p$$

cost layers and

$$p$$

mixer layers.

Example:

For:

$$p = 2$$

Circuit:

```
Cost Layer
Mixer Layer
Cost Layer
Mixer Layer
```

12. Applications of QAOA

---

1. Logistics Optimization

Problem:

Find efficient transportation routes.

Examples:

- Vehicle routing
- Supply chain optimization
- Delivery planning

QAOA minimizes:

$$\text{Travel cost} + \text{Time}$$

---

2. Scheduling Problems

Used for:

- Employee scheduling
- Exam timetable generation
- Manufacturing planning

Goal:

Reduce conflicts and improve efficiency.

---

3. Traffic Optimization

Applications:

- Traffic signal control
- Route optimization

4. Financial Optimization

Used for:

- Portfolio optimization
- Risk management

---

5. Graph Problems

Examples:

- Maximum Cut problem
- Network optimization

---

13. Example: Maximum Cut Problem

Given a graph:

Goal:

Divide nodes into two groups.

Maximize:

$$\text{Number of edges between groups}$$

QAOA:

1. Creates quantum states representing partitions.
2. Uses cost Hamiltonian.
3. Finds maximum cut.

---

14. Advantages of QAOA

---

1. Suitable for NISQ Devices

Works on noisy quantum computers.

---

2. Solves Complex Optimization Problems

Useful for large search spaces.

---

3. Hybrid Approach

Uses classical optimization with quantum computation.

4. Flexible

Can solve many optimization problems by changing cost Hamiltonian.

15. Limitations of QAOA

1. Approximate Solution

Does not always guarantee the exact optimum.

2. Parameter Optimization Difficulty

Finding optimal:

$$\gamma, \beta$$

is challenging.

3. Noise Sensitivity

Quantum errors affect results.

4. Circuit Depth

Large problems require deeper circuits.

16. VQE vs QAOA Comparison

Feature	VQE	QAOA
Full Form	Variational Quantum Eigensolver	Quantum Approximate Optimization Algorithm
Purpose	Find minimum energy	Solve optimization problems
Main use	Quantum chemistry	Combinatorial optimization
Hamiltonian	Molecular Hamiltonian	Cost Hamiltonian

Feature	VQE	QAOA
Output	Energy value	Optimal solution

17. Important University Questions

2-Mark Questions

1. Expand QAOA.
2. Who introduced QAOA?
3. What are cost and mixer Hamiltonians?
4. What is QAOA layer?

5-Mark Questions

1. Explain working of QAOA.
2. Describe QAOA architecture.
3. Explain applications of QAOA.

10-Mark Questions

1. Explain QAOA algorithm with architecture and working.
2. Discuss applications of QAOA in logistics and scheduling.
3. Compare VQE and QAOA.

Quick Revision

QAOA:

$$\boxed{\text{Optimization} + \text{Quantum Circuit} + \text{Classical Optimization}}$$

Main Equation:

$$\boxed{|\psi(\gamma,\beta)\rangle = e^{-i\beta H_M} e^{-i\gamma H_C} |+\rangle^n}$$

Main Components:

$$Cost\ Hamiltonian + Mixer\ Hamiltonian$$

Next Topic:

Quantum Machine Learning (QML)

We will cover:

1. Introduction to QML
2. Classical ML vs Quantum ML
3. Quantum classifiers
4. Quantum neural networks (QNN)
5. Applications and challenges

(Last major topic of UNIT–V.)

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 4 – Quantum Machine Learning (QML)

> **Weightage:** ⭐⭐⭐⭐⭐ (Very Important – 10 Marks Question)
> Quantum Machine Learning combines **quantum computing** and **machine learning** to develop algorithms that can potentially solve complex learning problems faster than classical approaches.

---

1. Introduction to Quantum Machine Learning (QML)

Machine Learning (ML) allows computers to learn patterns from data and make predictions.

Quantum Machine Learning uses quantum computing principles to improve machine learning algorithms.

Definition:

Exam Definition:

> **Quantum Machine Learning (QML) is an interdisciplinary field that combines quantum computing techniques with machine learning algorithms to process data, train models, and perform intelligent tasks using quantum systems.**

2. Need for Quantum Machine Learning

Modern machine learning faces challenges:

- Huge datasets
- High computational requirements
- Complex optimization problems
- Large model training time

Quantum computing provides:

- Quantum parallelism
- High-dimensional data representation
- Faster optimization techniques

---

3. Basic Idea of QML

Classical Machine Learning:

$$\text{Data} \rightarrow \text{Algorithm} \rightarrow \text{Prediction}$$

Quantum Machine Learning:

$$\text{Classical Data} \rightarrow \text{Quantum Encoding} \rightarrow \text{Quantum Algorithm} \rightarrow \text{Prediction}$$

---

4. Components of Quantum Machine Learning

QML consists of:

---

1. Data Encoding

Converting classical data into quantum states.

Example:

$$x \rightarrow |x\rangle$$

---

2. Quantum Circuit

Processes encoded data.

Uses:

- Quantum gates
- Parameterized circuits

---

3. Classical Optimizer

Updates quantum circuit parameters.

---

4. Measurement

Converts quantum output into classical results.

---

5. Quantum Machine Learning Workflow

```
Classical Data
      ↓
Quantum Encoding
      ↓
Quantum Circuit
      ↓
Measurement
      ↓
Classical Processing
      ↓
Prediction / Decision
```

6. Classical Machine Learning vs Quantum Machine Learning

Classical ML	Quantum ML
Uses classical bits	Uses qubits
Processes data sequentially	Uses quantum states
Classical feature space	Quantum feature space
Classical optimization	Quantum optimization
Limited by computational power	Potential quantum advantage

---

7. Types of Quantum Machine Learning Algorithms

Major categories:

1. Quantum Classification

Used for assigning data into categories.

Example:

- Spam detection
- Image classification
- Disease diagnosis

2. Quantum Regression

Used for predicting continuous values.

Example:

- Price prediction
- Weather prediction

3. Quantum Clustering

Groups similar data points.

Example:

- Customer segmentation

4. Quantum Neural Networks

Quantum version of artificial neural networks.

PART–A: Quantum Classifiers

8. Introduction to Quantum Classifiers

A classifier assigns input data into predefined classes.

Example:

Input:

$$Image$$

Output:

$$Cat/Dog$$

Quantum classifiers use quantum circuits to perform classification.

---

9. Quantum Support Vector Machine (QSVM)

One important quantum classifier is:

$$\boxed{\text{Quantum Support Vector Machine}}$$

It extends classical SVM using quantum feature spaces.

Working:

Step 1:

Encode data into quantum states.

$$|x\rangle$$

Step 2:

Map data into high-dimensional quantum space.

Step 3:

Find optimal decision boundary.

Step 4:

Classify new data.

---

10. Quantum Kernel Methods

Classical ML uses kernel functions.

Quantum computers can calculate quantum kernels.

Kernel:

$$K(x,y)$$

measures similarity between data points.

Quantum kernel:

$$K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2$$

11. Applications of Quantum Classifiers

1. Image Classification

Examples:

- Medical image classification
- Object recognition

2. Pattern Recognition

Used for:

- Signal analysis
- Feature detection

3. Cybersecurity

Used for:

- Threat detection
- Malware classification

PART–B: Quantum Neural Networks (QNN)

12. Introduction to Quantum Neural Networks

A Quantum Neural Network is a neural network implemented using quantum circuits.

Definition:

> **Quantum Neural Network (QNN) is a machine learning model that uses parameterized quantum circuits to perform learning tasks similar to classical neural networks.**

13. Structure of QNN

Classical Neural Network:

```
Input Layer
      ↓
Hidden Layers
      ↓
Output Layer
```

Quantum Neural Network:

```
Input Data
      ↓
Quantum Encoding
      ↓
Parameterized Quantum Circuit
      ↓
Measurement
      ↓
Output
```

14. Components of QNN

---

1. Input Encoding Layer

Converts data into quantum states.

Example:

$$x \rightarrow |x\rangle$$

---

2. Variational Quantum Circuit

Contains:

- Rotation gates
- Entanglement gates

Example:

$$R_x(\theta), R_y(\theta), R_z(\theta)$$

3. Measurement Layer

Extracts output values.

15. Working of QNN

Step 1:

Input data is encoded into qubits.

Step 2:

Quantum gates process information.

Step 3:

Parameters are adjusted during training.

Step 4:

Measurement produces prediction.

Step 5:

Error is calculated and parameters updated.

16. Quantum Neural Network Architecture

```
Input Data
    ↓
Quantum Encoding
    ↓
Parameterized Quantum Circuit
```

```
    ↓
Entanglement
    ↓
Measurement
    ↓
Prediction
```

17. Applications of QNN

---

1. Image Recognition

Used for:

- Object detection
- Medical imaging

---

2. Natural Language Processing

Used for:

- Text classification
- Language understanding

---

3. Drug Discovery

Used for:

- Molecular prediction

---

4. Financial Prediction

Used for:

- Market analysis

---

18. Advantages of QML

---

1. High-Dimensional Data Processing

Quantum states can represent complex data.

2. Faster Optimization

Quantum algorithms can improve optimization.

3. Better Feature Representation

Quantum feature spaces can capture complex patterns.

4. Hybrid Learning

Combines classical and quantum strengths.

19. Limitations of QML

1. Quantum Hardware Limitations

Current devices have limited qubits.

2. Noise

Quantum errors reduce accuracy.

3. Data Encoding Problem

Loading large datasets into quantum systems is difficult.

4. Limited Practical Advantage

Quantum advantage is still under research.

20. QML Applications Summary

Area	Application
Healthcare	Disease prediction
Finance	Risk analysis
Security	Threat detection
Computer Vision	Image classification
NLP	Text analysis
Chemistry	Drug discovery

Frequently Asked University Questions

2-Mark Questions

1. Define Quantum Machine Learning.
2. What is a quantum classifier?
3. What is QNN?
4. What is quantum feature mapping?

5-Mark Questions

1. Explain Quantum Machine Learning workflow.
2. Explain quantum classifiers.
3. Describe quantum neural networks.

10-Mark Questions

1. Explain Quantum Machine Learning and its applications.
2. Discuss quantum classifiers and QNN architecture.
3. Compare classical ML and quantum ML.

Quick Revision

QML:

$$\boxed{Quantum\ Computing + Machine\ Learning}$$

Main Concepts:

$$\boxed{Quantum\ Classifiers + Quantum\ Neural\ Networks}$$

QML Pipeline:

$$\boxed{Encode \rightarrow Quantum\ Circuit \rightarrow Measure \rightarrow Predict}$$

Next Topic:

UNIT–V Final Topic: Applications of Quantum Algorithms in Real-World Problems

We will cover:

1. Logistics optimization
2. Scheduling problems
3. Healthcare applications
4. Finance applications
5. Industry applications
6. Complete UNIT–V Revision

(After this, the entire syllabus will be completed.)

> ok next

UNIT–V: Application-Oriented Quantum Algorithms

Topic 5 – Real-World Applications of Quantum Algorithms

> **Weightage:** ⭐⭐⭐⭐⭐ (Important – 10 Marks Question)
> This topic explains how quantum algorithms like **VQE, QAOA, Grover's Algorithm, and Quantum Machine Learning** are applied to solve practical engineering and industry problems.

---

1. Introduction to Application-Oriented Quantum Computing

Quantum computing is moving from theoretical research toward practical applications.

Application-oriented quantum algorithms focus on solving real-world problems using:

- Quantum optimization

- Quantum simulation
- Quantum search
- Quantum machine learning

Definition:

> **Application-oriented quantum computing refers to the use of quantum algorithms to solve practical problems in fields such as logistics, healthcare, finance, machine learning, and engineering.**

---

2. Major Quantum Algorithms and Their Applications

Quantum Algorithm	Main Purpose	Applications
Grover's Algorithm	Search optimization	Database search, cryptography
VQE	Energy optimization	Chemistry, material science
QAOA	Combinatorial optimization	Scheduling, logistics
QML	Machine learning tasks	Classification, prediction

PART–A: Quantum Optimization Applications

---

3. Logistics Optimization

Introduction

Logistics involves managing:

- Transportation
- Delivery routes
- Supply chains
- Resource allocation

Optimization problems occur because companies need:

- Minimum cost
- Minimum travel time
- Maximum efficiency

---

4. Vehicle Routing Problem (VRP)

Problem:

A company has multiple delivery locations.

Need to find:

- Shortest route
- Minimum fuel consumption
- Minimum delivery time

Classical problem:

Very large search space.

For:

$$N$$

locations:

Possible routes:

$$(N-1)!$$

---

5. Quantum Solution Using QAOA

QAOA converts routing into:

$$\text{Cost Function}$$

Steps:

Step 1:

Represent routes using qubits.

Step 2:

Create cost Hamiltonian.

Step 3:

Apply QAOA layers.

Step 4:

Measure best route.

Benefits:

- Reduced computational complexity
- Better route planning
- Resource optimization

6. Supply Chain Optimization

Quantum algorithms help optimize:

- Inventory management
- Warehouse operations
- Distribution networks

Applications:

- Product delivery
- Demand prediction
- Cost reduction

PART–B: Scheduling Optimization

7. Introduction to Scheduling Problems

Scheduling involves assigning:

- Tasks
- Resources
- Time slots

efficiently.

Examples:

- University timetable
- Employee shifts
- Manufacturing schedules

8. Job Scheduling Problem

Given:

- Multiple jobs
- Limited resources

Goal:

Complete all jobs in minimum time.

Challenges:

- Many possible combinations
- Conflicting constraints

---

9. Quantum Solution Using QAOA

QAOA solves scheduling by:

1. Encoding schedules as quantum states.
2. Defining constraints using cost Hamiltonian.
3. Optimizing parameters.
4. Selecting best schedule.

Applications:

Education:

- Exam timetable generation

Industry:

- Machine scheduling

Cloud Computing:

- Task allocation

PART–C: Healthcare Applications

---

10. Drug Discovery

Problem:

Finding new medicines requires testing millions of molecular combinations.

Classical computers:

Require huge computational time.

Quantum algorithms help by:

- Simulating molecules
- Finding molecular properties
- Optimizing drug candidates

---

11. VQE in Drug Discovery

VQE is used to calculate:

$$\text{Ground State Energy}$$

of molecules.

Process:

```
Molecule
   ↓
Quantum Representation
   ↓
VQE Algorithm
   ↓
Energy Calculation
   ↓
Drug Analysis
```

Benefits:

- Faster molecular simulation
- Better drug design

---

12. Medical Image Processing

Quantum image processing helps in:

- MRI analysis
- CT scan processing
- Tumor detection

Algorithms used:

- Quantum image representation
- Quantum edge detection
- QML classifiers

PART–D: Financial Applications

---

13. Portfolio Optimization

Problem:

Investors want:

- Maximum return
- Minimum risk

Optimization variables:

- Asset selection
- Investment percentage

---

14. Quantum Optimization Solution

QAOA can optimize:

$$\text{Risk vs Return}$$

Steps:

1. Encode portfolio choices.
2. Create cost function.
3. Apply QAOA.
4. Find optimal portfolio.

---

15. Risk Analysis

Quantum machine learning can analyze:

- Market trends
- Financial patterns
- Fraud detection

PART–E: Artificial Intelligence Applications

---

16. Quantum Machine Learning in AI

QML improves:

- Classification
- Pattern recognition
- Prediction

Applications:

1. Image Classification

Examples:

- Object detection
- Medical diagnosis

---

2. Natural Language Processing

Applications:

- Text classification
- Sentiment analysis

---

3. Cybersecurity

Applications:

- Intrusion detection
- Malware detection

PART–F: Engineering Applications

17. Material Science

Quantum computers simulate materials at atomic levels.

Applications:

- New material discovery
- Battery development
- Semiconductor design

Algorithms:

- VQE
- Quantum simulation

---

18. Energy Optimization

Applications:

- Smart grids
- Energy distribution
- Power management

Quantum optimization helps:

- Reduce energy loss
- Improve efficiency

---

19. Aerospace Applications

Quantum algorithms help in:

- Flight scheduling
- Route optimization
- Design optimization

---

20. Communication and Cryptography

Applications:

- Secure communication
- Quantum-resistant encryption

Grover's algorithm impacts:

- Key searching
- Security analysis

21. Summary of Applications

Field	Quantum Algorithm	Application
Logistics	QAOA	Route optimization
Scheduling	QAOA	Timetable generation
Healthcare	VQE, QML	Drug discovery, diagnosis
Finance	QAOA	Portfolio optimization
AI	QML	Classification
Material Science	VQE	Molecular simulation
Security	Grover	Cryptography

22. Challenges in Real-World Quantum Applications

1. Hardware Limitations

Current quantum computers have:

- Limited qubits
- Noise problems

2. Error Correction

Quantum errors must be reduced.

3. Data Loading Problem

Converting classical data into quantum form is difficult.

4. Scalability

Large practical problems require powerful quantum systems.

---

23. Future Scope of Quantum Computing

Future developments include:

- Large-scale quantum computers
- Better error correction
- Quantum internet
- Quantum AI systems
- Industry-level optimization

Frequently Asked University Questions

2-Mark Questions

1. List applications of QAOA.
2. How is VQE used in drug discovery?
3. What is portfolio optimization?
4. Mention applications of QML.

5-Mark Questions

1. Explain quantum applications in logistics.
2. Describe quantum optimization in scheduling.
3. Explain healthcare applications of quantum computing.

10-Mark Questions

1. Discuss real-world applications of quantum algorithms.
2. Explain the role of QAOA and VQE in solving practical problems.
3. Describe quantum machine learning applications.

UNIT–V COMPLETE SUMMARY ✅

Topic	Key Concept
Quantum Optimization	Finding best solution using quantum methods
VQE	Minimum energy calculation
QAOA	Optimization problems
QML	Quantum-based learning
Applications	Logistics, healthcare, finance, AI

🎉 COMPLETE SYLLABUS COVERED

GENERATIVE QUANTUM COMPUTING FOR ENGINEERS ✅

Unit-wise Completion:

✅ UNIT–I: Fundamentals of Quantum Computing
✅ UNIT–II: Quantum Gates and Circuits
✅ UNIT–III: Quantum Search Algorithms
✅ UNIT–IV: Quantum Algorithms for Image Processing
✅ UNIT–V: Application-Oriented Quantum Algorithms

For semester preparation, next we can move into:

1. **Unit-wise Important 2M/5M/10M Questions & Answers**
2. **Previous Year Exam-Oriented Notes**
3. **Complete Revision Notes**
4. **Numerical Problems & Quantum Circuit Practice**
5. **Last-minute Exam Preparation Material**

ChatGPT can make mistakes. Check important info.